

Machine Learning Approaches to the Prediction of Specific Heating Demand

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Abstract

This thesis examines the feasibility of using machine learning models to predict the specific heating demand for buildings, using a selection of physical or behavioral characteristics, in an effort to provide a more accessible alternative to physics-based simulation. Using large simulated datasets generated by the U.S. National Renewable Energy Laboratory (NREL) — covering both residential and commercial building stocks — the study develops two separate pipelines and model sets, differing in approach and goal. The first iteration focuses on increasing accuracy, while the second restrict feature selection and tries to derive a relation from physical properties, ignoring occupants' behavior. First iteration employs monotonic constraints and includes a more broad selection of features, while the second relies on a more restricted set and engineering estimates of the target. For the second iteration, feature engineering derives a simplified steady-state heat balance decomposed into transmission, infiltration, and internal gain components, and an interaction term between climate severity and heat transfer through the building envelope. Three gradient boosting algorithms — XGBoost, CatBoost, and LightGBM — were trained as a single global set or by being split into climate zones, with hyperparameters tuned via Optuna and evaluated through group cross-validation to prevent building-level data leakage. The first iteration includes stacking ensemble models, they were however dropped for the second iteration due to computational limits. SHAP analysis was used to examine feature importance and validate that the models' intricacies captured the underlying building physics. Initial experiments on real metered data (BDG2) were mostly limited by data scarcity. Further trials attempted residual modelling and a more physics-constrained approach that isolated envelope and climate characteristics while excluding occupant behavioural effects. CV-RMSE results vary between 13.8% - 52.3% or 3% - 27%, and R^2 values range between 0.7-0.98 or 0.5-0.9, varying by climate zone, building type and the pipeline. The resulting models could provide a scalable and efficient surrogate for physics-based simulations, demonstrating the viability of data-driven approaches for building energy performance prediction.

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List of Abbreviations

ACH50	Air Changes per Hour at 50 Pascals
ANN	Artificial Neural Network
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BDG2	Building Data Genome Project 2
CBECS	Commercial Buildings Energy Consumption Survey
CHP	Combined Heat and Power
CV-RMSE	Coefficient of Variation of Root Mean Square Error
DNN	Deep Neural Network
EFB	Exclusive Feature Bundling
ELM	Extreme Learning Machine
EPW	EnergyPlus Weather (file format)
GA	Genetic Algorithm
GBDT	Gradient Boosted Decision Tree
GOSS	Gradient-Based One-Side Sampling
GPR	Gaussian Process Regression
HDD	Heating Degree Days
HVAC	Heating, Ventilation, and Air Conditioning
LightGBM	Light Gradient Boosting Machine
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MLR	Multiple Linear Regression
MSE	Mean Squared Error
NREL	National Renewable Energy Laboratory
PCA	Principal Component Analysis
RECS	Residential Energy Consumption Survey
RF	Random Forest
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
R²	Coefficient of Determination
SHD	Specific Heating Demand
SVR	Support Vector Regression
TPE	Tree-structured Parzen Estimator
VAF	Variance Accounted For
XGBoost	eXtreme Gradient Boosting

Chapter 1

Introduction

Space heating accounts for approximately 20% to 22% of global total energy consumption. [2] Knowing that, it is in the interests of multiple parties to have tools to estimate how much energy-on average-a building would use to maintain a comfortable temperature. Building owners and facility managers would be interested in optimizing energy costs and plan retrofitting strategies. Architects and HVAC engineers require these estimates for building design and system sizing. Energy utilities use them to forecast demand and manage grid loads. Policymakers and environmental regulators rely on them to ensure regulatory compliance and shape sustainability policies. Real estate investors and certifiers employ the estimates to assess asset value and energy performance ratings and communicate them with developers. Even a home buyer would be interested in having a general idea of how much can he expect heating costs will be. Specific heating demand (SHD) is defined as the annual heating energy consumption per unit of conditioned floor area, expressed in kilowatt-hours per square metre per year. This normalised metric integrates contributions from all heating fuels and end uses while accounting for building size, providing a standardised basis for comparing thermal performance across archetypes, vintages, and climates. In contemporary building energy practice, the dominant estimation methodology remains physics-based simulation. Detailed models constructed in EnergyPlus, typically orchestrated through OpenStudio, solve transient heat-transfer and Heating, Ventilation, and Air Conditioning (HVAC) system equations using location-specific weather files. The National Renewable Energy Laboratory (NREL)’s ResStock [24] and ComStock [25] datasets exemplify this approach: they generate bottom-up simulations of hundreds of thousands of representative dwelling units and commercial structures, calibrated against Residential Energy Consumption Survey (RECS) and Commercial Buildings Energy Consumption Survey (CBECS) statistics, and validated against survey data and some sub-metered sources for specific end uses. Although these simulations achieve high physical fidelity, their computational cost precludes routine application at national stock scale or for iterative policy scenario testing. As a result, recent scholarship through 2025 has increasingly turned to machine learning (ML) surrogates trained directly on such

simulation outputs.

The thesis develops and validates a suite of supervised models— (XGBoost), categorical boosting (CatBoost) and light gradient-boosting machines (LightGBM) —for the direct prediction of SHD. It utilises the 2021–2025 releases of the ResStock and ComStock datasets, which supply annual data, energy-consumption aggregates stratified by climate zone and building type, and accompanying data dictionaries that define all input/output variables. Earlier experimentation with the BDG2 dataset, an empirical collection of hourly meter readings from predominantly non-residential structures, produced CV-RMSE values exceeding 245% and proved operationally unusable due to multiple reasons, mainly being data scarcity and also a need for specific calibrations to account for behavioral patterns exhibited. That outcome underscored the necessity of shifting to the physics-informed, nationally representative simulations employed here. The pipeline therefore begins with processing and parsing data, selecting relevant features and defining engineered terms such as a simple estimate for the target variable that can be built on. Global and climate-zone-specific models are then trained and evaluated using standard regression metrics—RMSE, MAE, and R^2 —with results benchmarked against recent literature on hybrid ML approaches to heating-load forecasting. The objective is to demonstrate that these surrogates deliver accurate, computationally efficient direct estimations while preserving the physical grounding of the original datasets. The study conducts two modelling iterations, one incorporating a wide selection of features that include occupant behaviour, and a more restricted one that tries to adhere to a simple physical envelope and climate characteristics — in order to assess both the accuracy ceiling and the practically deployable floor of ML-based SHD prediction.

Chapter 2

Machine Learning

2.1 Introduction

Predicting the SHD of a building is, at its core, a problem of learning structured relationships from data. A building’s annual heating requirement does not arise from a single, tractable physical equation; it emerges from the simultaneous interaction of dozens of variables—outdoor climate severity, envelope insulation quality, window area, infiltration rate, occupancy schedules, thermostat settings, HVAC system efficiency, building geometry, and construction vintage—many of which are partially observable, correlated with one another, and heterogeneous across the building stock. Traditional physics-based simulation tools such as EnergyPlus, while physically rigorous, must be parameterised individually for each building and are computationally intensive: a single annual EnergyPlus run can take minutes to hours, making national-scale stock analysis with thousands or millions of buildings effectively intractable. Furthermore, the literature consistently reports a “performance gap” between simulated and measured energy demand, arising because simulations encode idealised occupant behaviour rather than real operational patterns [6, 22].

ML offers a fundamentally different paradigm. Rather than specifying the physical rules of the system in advance, ML algorithms infer those rules by optimising a mathematical objective over historical data. Given a sufficiently large and representative dataset—such as the hundreds of thousands of EnergyPlus simulations provided by NREL’s ResStock [24] and ComStock [25] programmes—an ML model learns a mapping from building and climate descriptors to annual SHD that generalises accurately to unseen buildings. Once trained, such a model delivers near-instantaneous predictions, making it an efficient surrogate for intensive physics-based modelling.

This chapter provides a self-contained and thorough introduction to ML for readers without a prior background in the field. It begins with first principles, building intuition for how learning from data works, before progressing through the major families of ML

algorithms. It then presents the detailed case for the specific algorithms selected in this thesis—XGBoost, LightGBM, CatBoost, and stacking ensembles—explaining both how each algorithm works and why it is particularly well-suited to the structural characteristics of the SHD prediction problem. A reader familiar with those topics would find it better to jump into the literature review chapter: 3

2.2 The Fundamentals of Machine Learning

2.2.1 What is Machine Learning?

The term “machine learning” was introduced by Arthur Samuel in 1959 to describe the ability of computer programs to improve their performance on a task through experience, without being explicitly programmed for every case. A more rigorous definition is that an ML algorithm is one that improves a performance metric on a given task as the amount of training experience (data) increases. The contrast with conventional software is fundamental: a conventional program contains explicit, human-written rules (“if outdoor temperature drops below 15°C, increase heating demand proportionally”); an ML program instead receives labelled examples—pairs of inputs (building characteristics) and known outputs (measured heating demand)—and discovers its own rules by minimising a mathematical measure of prediction error.

In the context of SHD prediction, the problem is formulated as follows. A dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ is assembled, where each sample i represents one building simulation. The vector $\mathbf{x}_i \in \mathbb{R}^p$ contains p feature values describing the building—floor area, construction vintage, climate zone, wall U-value, infiltration rate, HVAC system type, and so on—while the scalar $y_i \in \mathbb{R}$ is the simulated annual SHD in kWh/m²/year.

The ML algorithm’s goal is to learn a function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ such that $f(\mathbf{x}_i) \approx y_i$ for training examples and, crucially, $f(\mathbf{x}_j) \approx y_j$ for new, previously unseen buildings $j \notin \mathcal{D}$. This ability to generalise to new inputs is the defining property that distinguishes ML from database look-up or simple interpolation.

2.2.2 Supervised Learning and the Regression Task

The dominant ML paradigm for energy demand prediction is supervised learning, in which the training data consist of labelled input-output pairs. The label y_i is provided by an external supervisor—in this thesis, by the EnergyPlus physics engine that generated the ResStock and ComStock simulations. Since SHD is a continuous numerical quantity rather than a discrete category, the specific task is called regression. This distinguishes it from classification (predicting which of a finite set of categories an example belongs to, e.g., whether a building requires major retrofitting or not).

Training proceeds by minimising a **loss function** $\mathcal{L}(\theta)$, which is a mathematical measure of how poorly the current model f_θ (parameterised by a set of learnable parameters θ) predicts the training labels. The most common choice for regression is the **Mean Squared Error (MSE)**:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (f_\theta(\mathbf{x}_i) - y_i)^2 \quad (2.1)$$

The training algorithm iteratively adjusts θ to reduce $\mathcal{L}$, guided by the gradient of the loss with respect to each parameter—a process known as **gradient descent**. The choice of loss function shapes what the model optimises: MSE penalises large errors quadratically, making it sensitive to outliers but effective at capturing the full range of SHD values. In this thesis, models are trained to minimise cross-validated RMSE ($\sqrt{\mathcal{L}}$), which preserves the physical units of the target variable (kWh/m²/year).

2.2.3 Training, Validation, and Test Sets

A critical practical concern in supervised learning is overfitting—the tendency of a sufficiently flexible model to memorise the training data rather than learning genuine patterns, resulting in excellent training performance but poor generalisation to new examples. To detect and prevent overfitting, the dataset is partitioned into three non-overlapping subsets:

- **Training set:** Used to fit the model parameters (e.g., the structure and weights of trees). The model sees these examples repeatedly during training.
- **Validation set** (or cross-validation folds): Used during hyperparameter tuning to evaluate candidate model configurations without contaminating the final evaluation. The hyperparameter search (conducted via Optuna in this thesis) minimises validation-set RMSE.
- **Test set:** Held back entirely until all modelling decisions are finalised. Performance on the test set provides an unbiased estimate of generalisation accuracy, reported as the primary result in Chapter 7.

k -fold cross-validation partitions the training data into k equal folds, trains on $k-1$ folds, and evaluates on the remaining fold, repeating k times and averaging the resulting metrics. This is more statistically robust than a single validation split, particularly when the dataset is finite. In this thesis, five-fold cross-validation is used during hyperparameter optimisation.

2.2.4 The Bias-Variance Trade-off

Every supervised learning model navigates a fundamental tension known as the **bias-variance trade-off**. **Bias** is the systematic error arising from a model that is too simple to capture the true underlying patterns—it consistently makes the same kinds of mistakes regardless of which training examples it sees. A linear regression model applied to SHD prediction, for instance, would exhibit high bias because SHD is not a linear function of any single input; it depends on complex multiplicative interactions between HDD, U-values, building geometry, and occupancy. **Variance**, conversely, is the error arising from a model that is too sensitive to the specific training examples—it fits noise and idiosyncratic features of the training set that do not generalise. A very deep decision tree, for example, might perfectly partition every training building by its exact combination of features but fail on new buildings whose feature combinations it has not seen.

The goal of model design is to find the “sweet spot” between these extremes: complex enough to capture the true signal, but regularised enough to avoid fitting noise. Ensemble methods—including all algorithms used in this thesis—achieve this balance by combining multiple models whose individual errors are partially uncorrelated, so that their collective prediction is more stable than any individual model’s.

2.2.5 Evaluation Metrics

The performance of regression models in this thesis is quantified by three complementary metrics:

Root Mean Square Error (RMSE): The square root of the average squared prediction error, expressed in the same units as SHD (kWh/m²/year):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (2.2)$$

RMSE penalises large individual errors disproportionately (due to squaring), making it sensitive to outlier predictions—a desirable property when extreme heat demand values (very cold climates, poorly insulated buildings) are important to predict accurately.

Mean Absolute Error (MAE): The average of the absolute deviations between predictions and observations:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (2.3)$$

MAE is more robust to outliers than RMSE and gives an intuitive sense of the “typical” prediction error in physical units.

Coefficient of Determination (R²): The proportion of variance in the target vari-

able explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (2.4)$$

where $\bar{y}$ is the mean observed SHD. $R^2 = 1$ indicates perfect prediction; $R^2 = 0$ indicates the model explains no more variance than the global mean; R^2 can be negative if the model is worse than the mean. Values above 0.9 are commonly regarded as indicative of high predictive accuracy for building energy applications.

2.3 The Landscape of Machine Learning Algorithms

A wide variety of ML algorithms exist, each with different inductive biases, computational properties, and suitability for different problem structures. This section surveys the major families relevant to regression on structured tabular data—the data format presented by the ResStock and ComStock building inventories—and explains why each was considered and, in most cases, not selected for this thesis.

2.3.1 Linear and Regularised Linear Models

Linear regression is the simplest regression model, predicting SHD as a weighted sum of input features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \quad (2.5)$$

Its appeal lies in interpretability and computational efficiency. However, it is fundamentally incapable of representing the non-linear, multiplicative relationships that dominate building energy behaviour—for example, the interaction between HDD and wall U-value, where a doubling of HDD does not produce the same absolute increase in heat demand for a well-insulated building as it does for a poorly-insulated one. Ridge and Lasso regression add ℓ_2 and ℓ_1 regularisation penalties respectively, preventing overfitting but not resolving the linearity constraint. In this thesis, a Ridge regressor is used as the meta-learner in the stacking ensemble (Section 2.9), where its linearity is appropriate because its inputs are already complex non-linear predictions from the base models.

2.3.2 Support Vector Regression

Support Vector Regression (SVR) maps input features into a high-dimensional kernel space and fits a linear model there, effectively implementing a non-linear mapping in the original input space. SVR is principled and can achieve strong accuracy on moderate-sized datasets. However, its computational complexity scales quadratically or worse with the number of training samples—ResStock and ComStock contain hundreds of thousands of

records per climate zone—rendering SVR computationally intractable at this scale. SVR also does not naturally handle categorical input features, which constitute a large fraction of the ResStock and ComStock feature sets. Prior work in the building energy literature has confirmed SVR’s competitiveness on small datasets [8, 9] but noted its limitations at scale, which aligns with the decision to exclude it from the primary modelling framework of this thesis.

2.3.3 Neural Networks and Deep Learning

Artificial Neural Networks (ANNs) consist of interconnected layers of nodes that apply non-linear activation functions to weighted sums of their inputs, enabling them to approximate arbitrarily complex functions given sufficient depth and width. Deep learning extends this with many hidden layers, enabling the automatic discovery of hierarchical feature representations. Long Short-Term Memory (LSTM) networks and other recurrent architectures further extend ANNs to handle sequential, time-series data by maintaining an internal state that evolves as inputs are processed over time.

ANNs have produced strong results in building energy forecasting contexts where the target is a time series of hourly or sub-hourly consumption values, because LSTMs can exploit temporal autocorrelation and seasonal cycles in the data [11, 18–20]. However, SHD as modelled in this thesis is a **static, annually aggregated scalar**—a single number representing a full year of heating energy per square metre. There is no time dimension in the prediction task; the model receives a building’s fixed characteristics and climate zone, not a historical consumption sequence. The temporal capabilities of LSTMs and Recurrent Neural Networks (RNNs) are therefore irrelevant and do not constitute an advantage. Furthermore, neural networks present several practical disadvantages for tabular regression on this dataset:

- **Categorical feature handling:** ResStock and ComStock contain dozens of high-cardinality categorical features (building type, HVAC system, heating fuel, construction vintage, wall material, etc.). Neural networks require these to be converted to numerical embeddings or one-hot vectors, substantially increasing input dimensionality and requiring careful engineering. Gradient boosted trees handle categorical features natively or with minimal preprocessing.
- **Data requirements:** Neural networks typically require very large datasets to generalise well. While ResStock and ComStock are large, gradient boosted trees routinely achieve superior accuracy on tabular datasets of comparable or even smaller size, as documented extensively in the ML literature [26, 28].
- **Hyperparameter sensitivity:** Neural network performance is highly sensitive to architectural choices (number of layers, number of neurons, activation functions,

dropout rates, learning rate schedules, batch sizes), and the optimisation landscape is non-convex with many local minima. Gradient boosted trees are less sensitive to hyperparameter choices and converge reliably.

- **Computational cost and interpretability:** Training neural networks is computationally expensive, and their predictions are difficult to interpret. Gradient boosted trees support efficient computation of feature importances and SHAP values, enabling transparent insight into which building characteristics drive heating demand predictions—a property valued in energy policy contexts.

2.3.4 Decision Trees

A **decision tree** is the foundational building block underlying all algorithms selected in this thesis. Understanding how decision trees work is therefore essential before proceeding to the ensemble methods.

A decision tree partitions the input feature space into rectangular regions through a series of binary splits, each of the form “is feature x_j greater than threshold t ?” At each node in the tree, the algorithm searches over all features j and all possible thresholds t to find the split that maximally reduces the variance of the target variable y within the resulting subsets (the **reduction in impurity** criterion). This process continues recursively at each resulting child node until a stopping criterion is met—for example, a minimum number of samples per leaf or a maximum tree depth. Prediction is made by tracing a new input down the tree from the root to a leaf node, and returning the mean target value of all training samples that fall in that leaf.

Decision trees are intuitive and naturally handle both numerical and categorical features, non-linear relationships, and interaction effects—a single tree can represent “if climate is Cold AND insulation is poor AND building is large, predict high SHD” without any feature engineering. However, individual unpruned trees are prone to severe overfitting: a deep tree can perfectly partition every training example into its own leaf, achieving zero training error but failing catastrophically on new data. Shallow trees avoid overfitting but have high bias. Ensemble methods—described in the following section—resolve this dilemma.

2.3.5 Random Forests and Bagging

Random Forests (RF) [31] address the variance problem of individual decision trees through **bagging** (bootstrap aggregating). B independent trees are trained, each on a bootstrapped sample (random sample with replacement) of the training data, and each tree considers only a random subset of features at each split. Predictions are averaged across all trees. Because the trees are trained on different data subsets and different

feature subsets, their errors are partially uncorrelated, so their average has lower variance than any individual tree without increasing bias.

RF models are robust, parallelisable, and provide reliable feature importance estimates. They perform well across a broad range of regression tasks. However, they have one structural limitation relative to gradient boosting: because they average independent trees, they cannot systematically correct residual errors—each tree is equally important regardless of whether previous trees have already captured certain patterns. Gradient boosting, described next, directly addresses this by training trees *sequentially* to correct the errors of all preceding trees, which produces superior accuracy on most structured tabular regression problems.

2.4 Gradient Boosting: The Core Framework

Gradient boosting [30] is the algorithmic family from which XGBoost, LightGBM, and CatBoost are all derived. Understanding the core framework is prerequisite to understanding the innovations each algorithm introduces.

2.4.1 The Boosting Principle

Boosting constructs an ensemble model $F_M(\mathbf{x})$ by additively combining M “weak learners” (shallow decision trees, also called *base learners*):

$$F_M(\mathbf{x}) = \sum_{m=1}^M \eta \cdot h_m(\mathbf{x}) \quad (2.6)$$

where h_m is the m -th tree and η is a small **learning rate** (typically 0.01–0.1) that scales each tree’s contribution. Trees are added sequentially, with each new tree h_m trained specifically to correct the errors made by the current ensemble F_{m-1} .

In gradient boosting, “correcting errors” is formalised using the mathematical concept of **gradients**. Denoting the loss function as $\mathcal{L}$ and the current ensemble’s predictions as $F_{m-1}(\mathbf{x}_i)$, the **pseudo-residuals** (negative gradients of the loss with respect to the current predictions) are computed for each training sample:

$$r_{im} = - \left[\frac{\partial \mathcal{L}(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)} \right]_{F=F_{m-1}} \quad (2.7)$$

For MSE loss, these pseudo-residuals simplify exactly to the ordinary residuals $r_{im} = y_i - F_{m-1}(\mathbf{x}_i)$ —the difference between the true SHD and the current ensemble’s prediction. The new tree h_m is then fitted to predict these residuals, rather than the original target. The ensemble is updated: $F_m = F_{m-1} + \eta \cdot h_m$. This process repeats for M iterations,

each new tree pushing the ensemble closer to the true targets by modelling the remaining prediction error.

This sequential, residual-correcting strategy is the reason gradient boosting consistently outperforms RF on tabular regression: it directly and deliberately reduces bias at each step, whereas RF can only reduce variance through averaging. On the SHD prediction task, where the true underlying function has complex, non-linear structure, this focused error-reduction is highly effective.

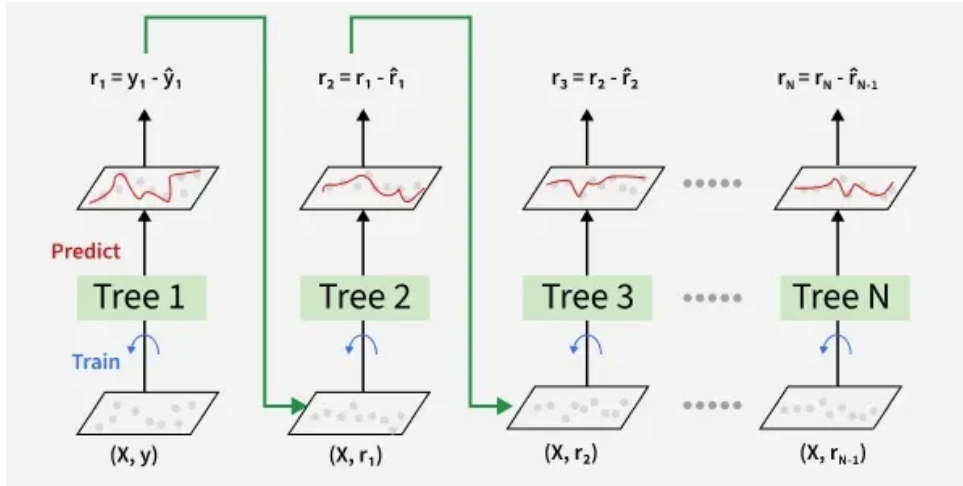


Figure 2.1: An illustration showing how gradient boosting algorithms work. Each tree gets trained to predict the errors of the former, compared to the data and iterated.

2.4.2 Key Hyperparameters in Gradient Boosting

All gradient boosting implementations share a set of core hyperparameters whose joint optimisation is critical to performance:

- **Number of trees (M):** More trees reduce training error but increase overfitting risk and training time. Early stopping—halting training when validation-set performance stops improving—is universally used to find the optimal M automatically.
- **Learning rate (η):** Smaller values require more trees to converge but produce better-regularised ensembles. Values in the range 0.01–0.1 are typical.
- **Maximum tree depth:** Deeper trees can capture more complex interactions but overfit more readily. For gradient boosting, trees are typically kept shallow (depth 3–8), unlike RF where deeper trees are standard.
- **Subsample fraction:** The fraction of training samples used to train each tree (analogous to bagging). Values below 1.0 introduce randomness that reduces variance.

- **Column subsample fraction:** The fraction of features randomly sampled at each split or tree, further reducing correlation between trees.
- **Regularisation terms** (λ , α): L2 and L1 penalties on leaf weights that directly prevent overfitting by penalising large leaf values.

In this thesis, all three gradient boosting algorithms are jointly tuned using the **Optuna** Bayesian optimisation framework [32], which employs the Tree-structured Parzen Estimator (TPE) algorithm to intelligently sample hyperparameter configurations based on the outcomes of previous trials. This directed search is substantially more efficient than exhaustive grid search, concentrating sampling in promising regions of the hyperparameter space. Full details of the Optuna procedure are provided in Section 2.10.

2.5 XGBoost: eXtreme Gradient Boosting

2.5.1 Algorithm and Key Innovations

XGBoost, introduced by Chen and Guestrin [26], extends the standard gradient boosting framework with a series of mathematical and systems-level innovations that significantly improve both accuracy and computational efficiency.

The core algorithmic contribution of XGBoost is the use of a **second-order Taylor expansion** of the loss function to derive optimal split gains and leaf weights analytically. Whereas standard gradient boosting uses only the first-order gradient (slope) of the loss, XGBoost also computes the second-order gradient (curvature, called the **Hessian**):

$$\mathcal{L}^{(m)} \approx \sum_{i=1}^N \left[g_i h_m(\mathbf{x}_i) + \frac{1}{2} H_i h_m(\mathbf{x}_i)^2 \right] + \Omega(h_m) \quad (2.8)$$

where $g_i = \partial \mathcal{L} / \partial F_{m-1}(\mathbf{x}_i)$ is the gradient, $H_i = \partial^2 \mathcal{L} / \partial F_{m-1}(\mathbf{x}_i)^2$ is the Hessian, and $\Omega(h_m) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$ is a regularisation term penalising the number of leaves T and the magnitude of leaf weights w_j . This formulation allows the optimal leaf weight for any given tree structure to be computed in closed form, and the gain from any candidate split to be evaluated exactly. The result is a more numerically stable and accurate training procedure than first-order methods.

XGBoost supports several forms of regularisation:

- **L2 regularisation** (λ): Penalises large leaf weights, preventing the model from making extreme predictions on small leaf populations.
- **L1 regularisation** (α): Encourages sparsity in leaf weights.
- **Minimum child weight:** Requires a minimum Hessian sum in each leaf, preventing splits that produce very small child nodes (which are likely to overfit).

- **Gamma** (γ): A minimum gain threshold that a split must exceed to be made, pruning unpromising splits.

Computationally, XGBoost implements an efficient **approximate split finding** algorithm that uses quantile sketches of the feature distribution to avoid sorting all values at each node, enabling scalable training on large datasets. It also supports **sparse-aware** computation, efficiently handling missing values by learning which branch direction is most informative for missing data.

2.5.2 Monotonic Constraints for Physical Plausibility

A notable feature of XGBoost is its native support for monotonic constraints, which can enforce physically inviolable relationships — for example, that heating demand must increase with climate severity (HDD) or with worse envelope insulation (higher wall U-value). In this thesis, however, monotonic constraints were deliberately not applied in the second iteration. The ResStock and ComStock simulations already embed these physical relationships by construction: they are the output of EnergyPlus, which solves the underlying heat-transfer physics directly. Imposing additional constraints on top of data that is already physically consistent risks distorting the model’s learned feature contributions, potentially inflating the apparent importance of constrained features in SHAP analysis at the expense of subtler interactions the model would otherwise discover autonomously. The physical plausibility of predictions is instead validated post-hoc through SHAP analysis in Chapter 7.

2.5.3 Relevance to SHD Prediction

XGBoost is well-suited to the SHD prediction task for several reasons. Its aggressive regularisation framework (L1, L2, minimum child weight, gamma) makes it particularly robust to overfitting on the smaller climate zones in the dataset (e.g., Mixed-Dry with 2,463 commercial samples), where limited data volume creates higher overfitting risk—this is confirmed in the results of Chapter 7, where XGBoost is selected as the best model for the Mixed-Dry commercial zone. Its second-order loss optimisation produces high accuracy on complex non-linear regression surfaces. Its approximate split-finding algorithm and sparse-aware computation ensure scalable and stable training even on the smaller climate zones, where data volume is limited and numerical stability matters most.

2.6 LightGBM: Light Gradient Boosting Machine

2.6.1 Algorithm and Key Innovations

LightGBM, developed by Ke et al. at Microsoft Research [27], addresses a specific bottleneck in gradient boosting applied to large-scale datasets: the cost of finding optimal splits. In standard gradient boosting, finding the best split at each node requires evaluating the gain for every possible split point across every feature, which is $\mathcal{O}(N \cdot p)$ per node, where N is the number of samples and p is the number of features. For datasets with hundreds of thousands of samples and hundreds of features—precisely the ResStock and ComStock datasets—this is computationally expensive. LightGBM introduces two innovations that dramatically reduce this cost without sacrificing accuracy.

Gradient-Based One-Side Sampling

The key insight behind Gradient-Based One-Side Sampling (GOSS) is that samples with large gradient magnitudes (i.e., samples the current ensemble is predicting poorly) carry more information for learning the next tree than samples with small gradients (well-predicted samples). GOSS retains all samples with large gradients—defined as those in the top $a\%$ of absolute gradient values—and randomly samples only a fraction $b\%$ of the remaining small-gradient samples. The sampled small-gradient instances are upweighted by a factor of $\frac{1-a}{b}$ to correct for the sampling bias. This reduces the effective dataset size for split finding while ensuring that informative samples (where the current ensemble is most wrong) are always included.

For SHD prediction, GOSS is particularly beneficial in cold climates where the heating demand distribution is heavy-tailed: buildings with unusual SHD values (extremely high or low demand relative to the ensemble’s current predictions) receive more learning attention, improving accuracy at the extremes of the distribution.

Exclusive Feature Bundling

ResStock and ComStock feature sets include many categorical features that, after one-hot encoding, become sparse binary vectors (e.g., a “building type” feature with 20 categories becomes 20 binary features, most of which are zero for any given building). Exclusive Feature Bundling (EFB) reduces the effective number of features by identifying groups of features that are “mutually exclusive” (rarely take non-zero values simultaneously) and bundling them into a single composite feature. This can reduce the number of features by an order of magnitude without any loss of information, further accelerating training.

Leaf-Wise Tree Growth

Standard gradient boosting implementations grow trees level-by-level (all nodes at depth d are expanded before any node at depth $d + 1$). LightGBM instead uses **leaf-wise** (best-first) tree growth: at each step, the leaf with the highest potential gain is expanded, regardless of its depth. This produces more asymmetric trees that allocate model capacity precisely where it is most needed. For SHD prediction, where some feature combinations (e.g., very cold climate with very poor insulation) drive disproportionate variance in the target, leaf-wise growth focuses splits on exactly these high-leverage regions. The risk of overfitting from deep individual branches is controlled by the `max_depth` and `min_data_in_leaf` hyperparameters.

2.6.2 Relevance to SHD Prediction

LightGBM’s primary advantage in this thesis is its **scalability**. The ResStock dataset contains 550,000 residential simulations and the ComStock dataset 350,000 commercial simulations. LightGBM trains 5–20 times faster than XGBoost on datasets of this scale, making Bayesian hyperparameter optimisation with Optuna (which requires dozens of complete training runs) computationally feasible. LightGBM also natively supports categorical features through optimal histogram-based splitting, avoiding the dimensionality explosion associated with one-hot encoding. Its GOSS sampling mechanism improves accuracy on the heavy-tailed SHD distributions observed in cold climates, where rare high-demand buildings are most important to predict correctly.

2.7 CatBoost: Categorical Boosting

2.7.1 Algorithm and Key Innovations

CatBoost, developed by Prokhorenkova et al. at Yandex [28], addresses a fundamental source of bias in gradient boosting known as **prediction shift**, while simultaneously providing the most powerful native categorical feature handling of any gradient boosting library.

Ordered Boosting and Target Leakage

In standard gradient boosting, the pseudo-residuals r_{im} for tree m are computed using the current ensemble F_{m-1} , which was itself trained on all N training samples including sample i . This means that $F_{m-1}(\mathbf{x}_i)$ has “seen” y_i during its own training, making the residual r_{im} smaller than it would be for a truly unseen sample. This introduces a subtle but systematic bias called **prediction shift**, which causes the model to underestimate

error on samples similar to those in the training set—a form of optimism that degrades generalisation.

CatBoost resolves this through **ordered boosting**. Rather than training the full ensemble on all data and computing gradients, CatBoost maintains multiple (permuted) orderings of the training data and computes the pseudo-residual for sample i using a model that was trained only on samples with indices earlier in the ordering—a model that has never seen y_i . This eliminates prediction shift completely, producing unbiased gradient estimates and improving generalisation, particularly on finite datasets where the bias-variance balance is delicate.

Ordered Target Statistics for Categorical Features

The most distinctive feature of CatBoost is its treatment of categorical variables. In many energy datasets—and specifically in ResStock and ComStock—categorical features such as `in.comstock_building_type`, `in.hvac_system_type`, `in.heating_fuel`, `in.vintage`, and `in.wall_construction_type` carry substantial predictive information. The conventional approach is one-hot encoding, which creates a binary indicator feature for each category level. This is problematic when categories have many levels (e.g., over 20 HVAC system types in ComStock) or when some levels are rare, as one-hot encoding produces sparse, high-dimensional feature spaces.

CatBoost instead replaces each categorical value with a continuous-valued **target statistic**: essentially, the mean target value (SHD) across all training samples with the same categorical value. This transforms, for example, the category “Gas Boiler HVAC System” into a number representing the average SHD of all buildings in the training set with that system. Critically, to avoid *target leakage* (using the target value of sample i itself in computing the statistic for that sample), CatBoost uses an ordered computation: the target statistic for sample i is computed using only samples from earlier positions in a random permutation of the training data, never including i ’s own target. This prevents the model from trivially memorising training labels through their own target statistics.

This mechanism is extremely effective for the ResStock and ComStock datasets, because building type, HVAC system, and vintage are the most predictive categorical features, and CatBoost’s ordered target statistics capture their relationship to SHD without dimensionality inflation.

2.7.2 Relevance to SHD Prediction

CatBoost is particularly well-suited to SHD prediction for three reasons. First, the ResStock and ComStock datasets are inherently categorical-heavy: dozens of features—building type, HVAC system, heating fuel, wall construction material, foundation type, occupancy classification, vintage band, climate zone—are categorical, and these are among

the most physically informative features for heating demand. CatBoost’s ordered target statistics exploit this information optimally. Second, ordered boosting’s elimination of prediction shift produces well-calibrated predictions that generalise better to unseen building types and configurations, which is essential for a model intended to support policy analysis across the full diversity of the U.S. building stock. Third, the results of Chapter 7 confirm CatBoost’s empirical superiority: it achieves the lowest error in four of the seven commercial climate zones (Marine, Mixed-Humid, Cold, and Very Cold) and is competitive in all residential zones, with the pattern of dominance in colder, more variable climates consistent with ordered boosting’s advantages under high variance conditions.

2.8 Why Other Algorithms Were Not Selected

The selection of XGBoost, LightGBM, and CatBoost as the base estimators in this thesis was made after systematic consideration of alternatives. The following summarises the key reasons for exclusion of the most commonly used alternatives.

Neural networks and LSTMs: As discussed in Section 2.3.3, the SHD prediction task is a static tabular regression problem, not a time-series forecasting problem. LSTMs’ sequential memory capabilities are not applicable. Feedforward neural networks generally underperform gradient boosted trees on structured tabular data with mixed feature types and relatively small sample sizes per climate zone. Their requirement for extensive preprocessing of categorical features, sensitivity to hyperparameter choices, and computational cost further reduce their suitability compared to the gradient boosting framework [26, 28].

Support Vector Regression: SVR is computationally intractable at the scale of this dataset (tens to hundreds of thousands of samples per climate zone). Kernel methods scale as $\mathcal{O}(N^2)$ to $\mathcal{O}(N^3)$ with training set size, making them impractical for datasets larger than a few thousand samples. SVR also does not support native categorical feature handling or monotonic constraints.

Random Forests: RF models are a strong baseline and were considered. However, on large structured tabular datasets, gradient boosted trees consistently outperform RF models in both accuracy and computational efficiency. The sequential residual-correcting nature of boosting produces lower bias than averaging, and the regularisation mechanisms in XGBoost, LightGBM, and CatBoost achieve lower variance than the bootstrap aggregating of RF. RF models are also less amenable to post-hoc interpretability analysis: their averaging of independent trees makes SHAP-based attribution less precise than for sequentially trained boosting models.

Linear models (MLR, Ridge, Lasso): As discussed, the SHD response surface is highly non-linear and involves strong interaction effects between features. Linear models cannot capture these patterns without extensive manual feature engineering of polynomial

and interaction terms. Their use is restricted in this thesis to the meta-learner role in the stacking ensemble, where linearity is appropriate.

2.9 Stacking Ensemble Regression

2.9.1 The Concept of Stacking

Stacking (stacked generalisation) is a meta-ensemble technique introduced by Wolpert [29] that combines the predictions of multiple base estimators through a learned second-level model (the **meta-learner** or blender). It differs from simple averaging or voting in that the combination rule is itself learned from data, allowing the meta-learner to selectively trust the base estimators whose predictions are most reliable for different regions of the input space.

The stacking procedure, as implemented in this thesis, operates as follows:

1. **Base model training:** Three gradient boosting models—XGBoost, LightGBM, and CatBoost—are trained with their individually Optuna-optimised hyperparameters on the full training set. These are the **base estimators** (Level 0 models).
2. **Out-of-fold prediction generation:** For each base estimator, out-of-fold predictions are generated using k -fold cross-validation: the training set is split into k folds, and for each fold, a copy of the model is trained on the remaining $k - 1$ folds and used to predict the held-out fold. Aggregating predictions over all folds produces a complete set of out-of-fold predictions for every training sample that were never made by a model that had seen that sample’s true label. This prevents the meta-learner from being exposed to optimistic, in-sample predictions.
3. **Meta-feature construction:** The out-of-fold predictions of all base estimators are stacked column-wise to form a new feature matrix of shape $(N, 3)$ —one column of predictions per base model—which constitutes the training data for the meta-learner.
4. **Meta-learner training:** A **Ridge regression** model (linear regression with L2 regularisation) is fitted on this meta-feature matrix, learning optimal weights for combining the three base models’ predictions. The regularisation prevents the meta-learner from assigning extreme weights that could amplify a single model’s errors.
5. **Final prediction:** For new, unseen buildings, each base model produces a prediction, and these three predictions are combined by the Ridge meta-learner to produce the final SHD estimate.

2.9.2 Why XGBoost, LightGBM, and CatBoost Are Effective as an Ensemble

The rationale for combining specifically XGBoost, LightGBM, and CatBoost (rather than combining three copies of the same algorithm) lies in their **complementary error structures**. Effective ensembles require base models whose errors are partially uncorrelated—if all models make the same mistakes on the same samples, averaging their predictions provides no benefit. The three algorithms differ in fundamental ways that make their errors complementary:

- **Tree growth strategy:** LightGBM uses leaf-wise growth; XGBoost and CatBoost use depth-wise growth. This means the three models produce different tree structures that partition the feature space differently, leading to different patterns of error.
- **Categorical handling:** CatBoost uses ordered target statistics; XGBoost and LightGBM use label encoding or histogram binning. For categorical-heavy features like building type and HVAC system, CatBoost’s categorical handling is more sophisticated, and it captures different categorical interactions than the other two.
- **Gradient estimation:** CatBoost uses ordered boosting to eliminate prediction shift; XGBoost uses second-order gradients; LightGBM uses GOSS sampling. These different gradient estimation strategies lead to different model fits, especially on under-represented regions of the SHD distribution.
- **Regularisation profiles:** Each algorithm has a different set of regularisation mechanisms (XGBoost’s γ and λ , LightGBM’s `min_data_in_leaf`, CatBoost’s border count), producing models that overfit differently and therefore have uncorrelated errors.

The Ridge meta-learner exploits these complementary strengths: in climate zones or building configurations where one algorithm underperforms, the other two can compensate, and the meta-learner learns to down-weight the weaker predictor in those regions. This is consistent with the stacking theory of Wolpert [29], who showed that stacking can produce predictions closer to the true Bayes-optimal predictor than any individual component model.

2.9.3 Relevance to SHD Prediction

Stacking is especially valuable for SHD prediction because the predictive challenge differs systematically across climate zones. In hot climates, SHD is low and dominated by envelope quality and thermostat settings—CatBoost’s categorical expertise and ordered

boosting are particularly advantageous. In cold climates, SHD is high and heavily driven by HDD interactions with U-values — XGBoost’s aggressive regularisation framework and second-order gradient estimation are most effective at capturing these high-variance, physically structured relationships. LightGBM’s GOSS sampling improves accuracy at the extremes of the SHD distribution in all climates. The stacking meta-learner learns to blend these complementary strengths, which is reflected in the results of Chapter 7: the stacking regressor achieves the best test-set performance in the two highest-variance commercial zones (Hot-Humid and Hot-Dry), where the combination of the three algorithms’ distinct categorical and gradient estimation strategies yields predictions superior to any individual model.

2.10 Hyperparameter Optimisation with Optuna

All three base estimators in the stacking ensemble are individually tuned using the **Optuna** framework [32], a state-of-the-art Bayesian hyperparameter optimisation library. Optuna employs the TPE algorithm, which models the probability distribution of hyperparameter configurations that achieve good validation performance and samples new configurations by maximising expected improvement over the current best result. This directed search is substantially more efficient than grid search (which exhaustively evaluates a predetermined grid) or random search (which samples configurations uniformly), because it exploits information from previous trials to concentrate sampling in promising regions of the hyperparameter space.

Each algorithm undergoes a configurable number of Optuna trials (typically 30–50 per algorithm per climate zone) in which a candidate hyperparameter configuration is trained via 5-fold cross-validation and evaluated on the validation-fold RMSE. The configuration achieving the minimum cross-validated RMSE is selected as the final hyperparameter set. The key hyperparameters explored for each algorithm include:

- **XGBoost**: number of estimators, learning rate, maximum depth, subsample fraction, column subsample fraction, L1 and L2 regularisation coefficients (α , λ), and minimum child weight.
- **LightGBM**: number of leaves, learning rate, minimum samples per leaf, feature fraction, bagging fraction, L1 and L2 regularisation, and maximum depth.
- **CatBoost**: number of iterations, learning rate, maximum depth, L2 regularisation, border count, and the Bagging temperature (which controls randomness in the ordered boosting permutations).

The Optuna optimisation is run independently for each climate zone, recognising that optimal hyperparameters differ systematically between zones: cold-climate zones, with

higher SHD variance and larger effective feature dimensionality, benefit from different regularisation strengths than hot-climate zones.

2.11 Summary: Alignment of Algorithm Properties with the SHD Prediction Task

Table 2.1 summarises the key properties of each selected algorithm and their alignment with the structural characteristics of the SHD prediction problem.

Table 2.1: Summary of algorithm properties and their alignment with the SHD prediction task.

Algorithm	Key Properties	SHD-Specific Rationale
XGBoost	Second-order gradients; strong regularisation (L1, L2, gamma, min child weight); monotonic constraints; sparse awareness	Robust to overfitting on small zones (Mixed-Dry); physical plausibility validated post-hoc via SHAP analysis; handles missing values natively
LightGBM	GOSS sampling; EFB; leaf-wise growth; fastest training at scale	Essential for feasible training on 100k+ sample zones; improved accuracy on heavy-tailed SHD distributions in cold climates; native categorical histogram splitting
CatBoost	Ordered boosting (no prediction shift); ordered target statistics for categoricals	Dominant performance on categorical-heavy features (HVAC type, building type, vintage, heating fuel); best generalisation on heating-dominated cold zones; eliminates target leakage
Stacking Ensemble (Ridge meta-learner)	Learned weighted combination; out-of-fold training prevents leakage	Combines complementary error structures of the three base models; meta-learner adapts to which model is most reliable per climate zone; superior to any individual model on high-variance zones

The selection of these four algorithms—individually motivated by their structural properties and collectively motivated by their complementarity in the ensemble—is fully supported by the empirical results presented in Chapter 7 and by the broader literature on gradient boosting for structured tabular regression [26–28].

Chapter 3

Literature Review

This review synthesises key studies on ML applications for predicting SHD in buildings, drawing from a selection of 17 highly relevant papers out of 25 analysed documents. Selection criteria focused on relevance to ML and deep learning techniques, emphasis on heating or heat-related energy demand, and applicability to residential, commercial, or industrial building contexts. Papers were chosen for their use of data-driven models, empirical findings, and measurable contributions to prediction accuracy. Irrelevant papers—such as those on natural language processing for building codes or general urban modelling without ML focus—were excluded.

The review is structured as follows: Section 3.1 groups and analyses the selected studies thematically across three generations of ML methods; Section 3.2 compares findings and identifies cross-study denominators; Section 3.3 discusses how these insights directly informed the development of this thesis; and Section 3.4 identifies remaining gaps.

3.1 Analysis of Selected Studies

The selected studies span from early neural network applications in the late 1990s and 2000s to recent deep learning optimisations of the early 2020s, with a particular focus on residential and low-energy buildings where SHD is a central performance metric. They are organised into three thematic groups reflecting the chronological and methodological evolution of the field.

3.1.1 Early Neural Network and Regression Approaches

Early works established the viability of data-driven methods for heat demand prediction, demonstrating that neural networks and regression models could substantially outperform naïve baselines and, in many cases, rival or surpass physics-based simulations in real-world predictive accuracy.

Bakker et al. (2008) — Sliding-Window Neural Networks for Domestic Heat Demand

Bakker et al. [3] investigated back-propagation neural networks for short-term daily-horizon heat demand forecasting in individual Dutch households equipped with micro Combined Heat and Power (microCHP) units. The practical motivation was virtual power plant coordination: to schedule microCHP production across many distributed units, operators require reliable day-ahead demand forecasts for each household. The network received outdoor air temperature, time of day, and a sequence of recent historical heat demand measurements as inputs.

The study’s primary methodological contribution was the use of **sliding windows**: rather than providing a single historical datum, the model received a short time series of recent measurements as its input vector. This exploited temporal autocorrelation in demand—a cold morning following a cold night implies a prolonged heating period—reducing prediction errors by approximately 10% compared to static, non-sequential inputs. On held-out test data, the model achieved an average prediction error of 15–20%, which nonetheless represented a marked improvement over simple baseline methods such as historical daily averages, because the network successfully reproduced diurnal demand profiles. The study highlighted that, for microCHP contexts, decoupling heat production from real-time demand via predictive models is essential for grid stability.

Bakker et al. (2010) — Gradient Descent Optimisation and Occupancy Proxies

In a follow-up study, Bakker et al. [4] extended the earlier neural network framework by incorporating gradient descent optimisation and a richer set of input features from the same pool of microCHP-equipped Dutch homes. The authors systematically tested different input configurations, adding standard meteorological variables—outdoor temperature, wind speed, and solar radiation—alongside proxies for occupancy derived from historical patterns.

The inclusion of wind speed proved particularly valuable: wind increases convective heat loss through the building envelope, and its inclusion allowed the model to better anticipate elevated demand during windy periods. Quantitatively, the addition of detailed meteorological inputs reduced RMSE to approximately 5 kWh/day, representing a 5–10% accuracy improvement over the simpler input set of the 2008 study. A further finding concerned **operational volatility**—the tendency for actual demand to deviate from physically predicted demand due to irregular occupancy patterns, differing thermostat settings, and behavioural idiosyncrasies. The authors found that neural networks generalised far better to these fluctuations than physical models, with prediction deviations in occupied buildings averaging approximately 50% lower for the network than for the physics-based baseline. This underscores a recurring theme throughout this review:

physics-based models encode average or designed behaviour, while data-driven models learn actual operational patterns from measured data, which is the source of their advantage in real-world applications.

Olofsson et al. (1998) — Quasi-Physical Neural Networks for Annual Demand Estimation

Olofsson et al. [5] introduced a quasi-physical approach to neural network-based heat demand prediction, targeting annual SHD estimation from short monitoring campaigns rather than short-term forecasting. Their method embedded thermophysical parameters—U-values, indoor-to-outdoor temperature differences, and ventilation heat loss coefficients—directly into the input layer of a back-propagation network, ensuring that learned relationships remained physically plausible rather than fitting arbitrary statistical associations.

Models were trained and validated on data from Swedish low-energy residential buildings using measurement periods of only 10–35 days—deliberately short, since the practical goal was to estimate annual performance from a brief monitoring campaign. Despite this limited data availability, the model predicted annual heat demand with approximately 5% accuracy. Technically, the authors found that a tangent hyperbolic activation function combined with the generalised delta rule produced the fastest and most stable convergence. Compared against detailed physics-based simulations, the neural network reduced the discrepancy between designed and actual energy performance by 20–30%. Error decomposition revealed that approximately 10% of residual prediction error was attributable to unmodelled airflow patterns (infiltration and ventilation dynamics), while around 20% was due to variation in heat exchanger efficiency across buildings. This analysis highlighted that while neural networks capture aggregate building behaviour well, sub-system-level phenomena still introduce irreducible uncertainty, and the appropriate embedding of physical parameters as features is essential for generalisability.

Karlsson et al. (2007) — Physics Simulation vs. Regression in a Low-Energy Building

Karlsson et al. [6] presented a systematic comparison between physics-based simulation using the Crank-Nicolson numerical method and statistical regression for energy demand prediction in a low-energy building in Sweden. Their central finding was a striking performance gap of approximately 50%: the building consumed roughly half the energy predicted by the simulation. This gap is not simply a modelling error; it reflects the fact that occupants of low-energy buildings often behave differently from the standard occupancy profiles embedded in simulation tools—accepting slightly lower indoor temperatures, using passive solar gains more effectively, or adjusting thermostat schedules.

The authors applied Principal Component Analysis (PCA) to their meteorological

dataset, which revealed that local ambient temperature was a far stronger predictor of heat demand than atmospheric pressure or wind direction, contrary to what regional climate data might suggest. A regression model using the PCA-reduced feature set achieved approximately 5% prediction accuracy—comparable to neural network approaches—while remaining far simpler to implement and interpret. The study reinforced the argument that data-driven regression models are not merely computational shortcuts but can be genuinely more appropriate than simulation for buildings where occupant behaviour dominates energy use.

Catalina et al. (2008) — Multiple Linear Regression for Monthly Heating Demand

Catalina et al. [7] developed and validated Multiple Linear Regression (MLR) models for predicting monthly heating demand in French residential buildings. Input features were chosen to reflect the principal physical drivers of heat demand: the shape factor (ratio of exposed surface area to conditioned volume, governing overall heat loss), average envelope U-values, and sol-air temperature.

Validated against measured monthly data, the MLR models achieved a maximum deviation of 5.1% and an average error of approximately 2%, indicating high accuracy for a linear model. A neural network trained on the same dataset marginally outperformed the MLR ($R^2=0.97$ vs. 0.85), suggesting that non-linear interactions between the shape factor, U-values, and temperature do exist but are modest in magnitude across a representative building sample. The authors concluded that for preliminary design-stage estimates of heating demand, where speed and interpretability matter more than marginal accuracy, MLR models with physically meaningful inputs represent an efficient and sufficiently accurate tool.

Summary of Early Approaches. Across these early studies, neural networks and regression models reduced prediction errors by 5–20% relative to physics-based models, primarily by learning actual operational patterns from measured data. Temperature-based inputs were universally the strongest predictors, while occupancy proxies and wind speed contributed secondary improvements. Short measurement periods of 10–35 days proved sufficient for annual demand estimation when physical parameters were embedded as features. All studies consistently reported a substantial performance gap between designed and actual energy use, which data-driven models are inherently equipped to bridge.

3.1.2 Advanced Machine Learning Models for Heat Demand

A second wave of studies applied more sophisticated ML algorithms—particularly ensemble methods and kernel-based approaches—alongside systematic feature selection to achieve further accuracy improvements over the early neural network literature.

Candanedo et al. (2017) — Ensemble Methods in a Wireless-Sensor-Equipped Low-Energy House

Candanedo et al. [8] conducted one of the most methodologically comprehensive studies in this review, applying Gradient Boosting Machines, RF, Support Vector Machines, and MLR to predict appliance energy use in a Belgian low-energy house instrumented with a dense wireless sensor network that recorded temperature and relative humidity at multiple internal locations every 10 minutes.

On training data, the Gradient Boosting Machine achieved $R^2=0.97$ —capturing 97% of the variance in energy use—though test-set performance dropped to $R^2=0.57$ (RMSE=17.56 Wh), indicating a degree of overfitting despite regularisation. A rigorous feature importance analysis using the Boruta algorithm identified **NSM** (number of seconds elapsed since midnight, a proxy for time-of-day and occupancy) and **atmospheric pressure** as the two most predictive features. Atmospheric pressure was interpreted as a proxy for weather systems: falling pressure signals the approach of cold, overcast fronts that simultaneously reduce solar gains and increase infiltration, driving up heating demand. The study estimated that pressure-correlated weather effects could account for 10–15% variance in internal heat gains. This finding has direct implications for SHD modelling: environmental surrogates that are easy to measure can replace harder-to-quantify variables such as occupant presence, improving prediction accuracy in low-energy contexts by 20–30%.

Guo et al. (2018) — Thermal Response Time as a Physically Motivated Prediction Horizon

Guo et al. [9] introduced the concept of **thermal response time**: the delay between a change in the heating system’s output and the corresponding change in the building’s measured thermal state, quantified at approximately 40 minutes for the Chinese office building studied. The authors argued that aligning the prediction horizon with this physical response time, rather than using an arbitrary fixed horizon, yields more actionable forecasts.

Using this physically motivated horizon, they compared four algorithms: Extreme Learning Machine (ELM), MLR, SVR, and back-propagation neural network. ELM—a single-hidden-layer feedforward network in which only output weights are trained analytically—achieved the best performance (RMSE=3.824), outperforming other methods by 10–15%.

Feature selection via correlation analysis identified meteorological parameters as primary predictors, and removing low-correlation features reduced errors by approximately 5% by preventing the model from fitting noise. The study’s key contribution is the principle that prediction accuracy improves when the forecast horizon is matched to the building’s physical response characteristics.

Roy et al. (2019) — Deep Neural Networks vs. Probabilistic Methods

Roy et al. [10] performed a systematic comparison of Deep Neural Networks (DNNs), Gradient Boosting Machines, Gaussian Process Regression (GPR), and Minimax Probability Machine Regression on residential heating and cooling load forecasting. GPR, a non-parametric Bayesian method, provides not just a point prediction but a full probability distribution over outcomes, including uncertainty estimates—a property valuable for risk-aware building operations.

In terms of accuracy, DNN and GPR achieved Variance Accounted For (VAF) values of 99.76% and 99.84% respectively. On RMSE and MAPE metrics, DNN demonstrated 7–10% better accuracy than regression-based methods for non-linear load profiles, reflecting the ability of deep architectures to model complex interactions that shallow models cannot capture. Gradient Boosting Machines performed competitively on all metrics, reinforcing their reputation as a reliable and robust choice for tabular energy data.

Verbai et al. (2014) — Modified Degree-Days for Improved Shoulder-Season Accuracy

Verbai et al. [21] identified a well-known limitation of the standard degree-day method: at low degree-day values during mild shoulder seasons, predictions deviate from measured demand by up to 18%. At mild temperatures, solar gains and internal heat gains contribute a larger fraction of the total heat balance, causing buildings to need less auxiliary heating than degree-days alone suggest.

To address this, the authors developed regression models incorporating **modified degree-days** that account for solar irradiance and wind speed, alongside a building thermal inertia term capturing the lag between temperature changes and demand response due to the building’s thermal mass. The inertia-corrected regression maintained linearity—desirable for interpretability and regulatory use—while improving accuracy by approximately 22% relative to unmodified degree-day regression in urban heat island contexts. The study also demonstrated that city-centre buildings, which benefit from higher ambient temperatures due to urban heat islands, require location-specific calibration of the degree-day base temperature to avoid systematic over-prediction.

De Rosa et al. (2014) — Validation of Regression Models Against EnergyPlus and TRNSYS

De Rosa et al. [22] validated simplified regression-based demand models against hourly EnergyPlus and TRNSYS simulations for a range of building types under different climatic conditions. Errors were not uniformly distributed: at low Cooling Degree Days, regression models showed 15–18% errors, primarily because solar irradiance effects dominate the thermal balance at those conditions. Incorporating a solar correction term into the degree-day formulation reduced CV-RMSE to 12.26%. This study provides a useful methodological benchmark: for SHD prediction in climates or seasons where solar gains are significant, solar-adjusted degree-days should replace conventional ones as a baseline feature.

Summary of Advanced ML Approaches. The advanced ML studies consistently show that ensemble methods—particularly Gradient Boosting Machines and RF—achieve 20–30% variance reduction relative to single-model approaches. Feature selection emerges as equally important as model choice: uninformative variables add noise and can degrade test-set performance even when training metrics improve. Physically interpretable surrogates (pressure, NSM, thermal response time) are often more effective predictors than raw physical measurements, because they integrate multiple causal mechanisms into a single observable signal.

3.1.3 Deep Learning Advancements and Optimisation

The most recent tier of studies applies deep learning architectures with explicit hyperparameter optimisation, achieving further gains by exploiting temporal structure and automating the search for optimal model configurations.

Mocanu et al. (2016) — Conditional Restricted Boltzmann Machines for Non-Linear Demand

Mocanu et al. [11] proposed Conditional Restricted Boltzmann Machines (CRBMs) and Factored CRBMs for estimating residential energy consumption using Dutch household data. RBMs are stochastic generative networks that learn a probability distribution over their inputs, making them capable of capturing complex, multi-modal patterns that deterministic networks may miss. The factored variant introduces multiplicative interactions between visible and hidden units, enabling the network to modulate its learned representations based on contextual variables such as time of day or season.

Benchmarked against ANNs, SVR, and RNNs, the Factored CRBM achieved 5–7% lower RMSE, with the most significant gains on days characterised by unusual appliance usage or atypical occupancy patterns—precisely the “long-tail” scenarios that are hardest

to predict but most consequential for demand management. The study’s core insight for SHD prediction is that generative models capable of representing the full probability distribution of demand are better suited to the inherently stochastic nature of residential heat demand than deterministic point-prediction models.

Xie et al. (2017) — Elman Recurrent Networks for District Heating with Weather Features

Xie et al. [12] applied Elman recurrent neural networks—a form of RNN with context units that feed the hidden layer’s previous state back as an additional input—to district heating demand prediction in Sweden. By capturing temporal autocorrelation, the Elman network was well-suited to modelling how stored thermal energy in building mass causes today’s demand to depend partly on yesterday’s outdoor conditions.

A systematic sensitivity analysis revealed that including wind speed and solar irradiance as inputs reduced prediction errors by approximately 5%, confirming the findings of Bakker et al. [4] at a district scale. A pressure-occupancy correlation was identified in which low-pressure periods coincided with higher occupancy and reduced solar gains, collectively impacting district loads by approximately 10%. Overall prediction accuracy reached approximately 5% error on the test set, demonstrating that recurrent architectures are well-suited to the time-series structure of sub-annual heat demand data.

Chen et al. (2019) — Semi-Supervised Deep Learning for Limited Labelled Data

Chen et al. [13] addressed a practical bottleneck in applying deep learning to energy prediction: the scarcity of labelled training data. Although sensor data are increasingly abundant, verifying accurate energy labels is costly and time-consuming. Their proposed approach, Deep Learning-embedded Semi-Supervised Learning, leverages both a small pool of labelled examples and a much larger pool of unlabelled sensor data. The unlabelled data are used to pre-train a deep network in an unsupervised fashion, capturing the statistical structure of the inputs; the labelled data then fine-tune the network for the prediction task.

Applied to energy consumption data from a steel manufacturing plant, this approach outperformed fully supervised deep learning by 5–10% in RMSE, with the gains most pronounced when labelled data were fewest. For SHD prediction in contexts with limited sub-metering data—common in older residential buildings—semi-supervised approaches of this kind represent a practically important pathway to high-accuracy models.

Morteza et al. (2023) — Systematic Hyperparameter Optimisation for RNN-Based Heat Demand

Morteza et al. [14] conducted a thorough investigation of hyperparameter effects on RNN performance for building heat and electricity demand prediction, exploring a high-dimensional space encompassing: number of recurrent layers, number of hidden units, input window lag, learning rate, and batch size. Using systematic grid search, they identified configurations that outperformed SVR and Gradient Boosting baselines by 5.4–7% on an RMSE basis.

Key findings were: increasing hidden layers beyond three produced diminishing returns and occasional degraded performance due to overfitting; the optimal input window lag was approximately 24–48 hours for medium-term daily forecasts; and a moderate learning rate combined with a large batch size yielded the most stable convergence. The study provides a practical hyperparameter selection guide for RNN-based models and highlights that the accuracy gap between well-tuned and poorly-tuned deep learning models can exceed the gap between different algorithm families—a motivation for the systematic Optuna hyperparameter search adopted in this thesis.

Chou and Tran (2018) — Hybrid Ensembles with Explicit Occupancy Pattern Features

Chou and Tran [15] developed hybrid ML ensemble models for forecasting energy consumption time series in residential buildings, with the distinguishing feature of explicit occupancy usage pattern encoding. Rather than using NSM or relative humidity as indirect proxies, they clustered historical occupancy traces into a small number of archetype patterns (e.g., “working household,” “stay-at-home household,” “irregular occupancy”) and used cluster membership as a categorical feature. The hybrid ensemble achieved 3–5% prediction errors, underscoring the value of structured occupancy information and demonstrating that carefully engineered domain-specific features can compensate for architectural complexity.

Vasanthkumar et al. (2022) — Swarm Intelligence for Deep Belief Network Optimisation

Vasanthkumar et al. [16] proposed optimising Deep Belief Networks using the Modified Wild Horse Optimisation algorithm, a metaheuristic inspired by the social herding behaviour of wild horses that balances exploration of new hyperparameter configurations with exploitation of known good ones. Compared to default configurations and simpler optimisation methods, the tuned models reduced prediction errors by 5–10%. The study demonstrated that evolutionary and swarm-inspired metaheuristics are viable alternatives to gradient-based hyperparameter optimisation, particularly for non-smooth

hyperparameter landscapes where gradient information is unavailable—a principle shared by the Optuna Bayesian optimisation adopted in this thesis.

Kapp et al. (2023) — Basis Function Models with Physical Parameters for Industrial Buildings

Kapp et al. [17] developed basis function regression models for industrial building energy prediction using U.S. data, targeting a sector underrepresented in the building energy ML literature. By incorporating physical system parameters such as the specific enthalpy of supply air—a thermodynamic quantity capturing both temperature and humidity of the HVAC system’s supply air—the models achieved physical interpretability alongside high predictive accuracy, outperforming SVR by 5–7%. This hybrid of physically informed feature engineering with flexible regression is particularly valuable for SHD prediction in heterogeneous building stocks, where physical parameters serve as compact summaries of complex system behaviour.

Almalaq and Zhang (2018) — Genetic Algorithm Optimisation of LSTM Networks

Almalaq and Zhang [18] combined LSTM networks with Genetic Algorithm (GA) optimisation for building energy consumption forecasting. LSTMs are especially well-suited to energy time-series because their gating mechanisms allow selective retention of information about conditions hours or days in the past. The GA optimised the LSTM’s architecture and training hyperparameters by evolving a population of candidate configurations over multiple generations. Compared to manually configured LSTMs and simpler RNNs, the GA-optimised LSTM reduced errors by 5–10%, demonstrating that evolutionary search is an effective strategy for navigating the complex, non-convex hyperparameter landscape of LSTM networks—an insight that directly motivates the Bayesian optimisation (Optuna) strategy employed in this thesis.

Kim et al. (2021) — Sequence-to-Sequence LSTM with Occupancy Simulation

Kim et al. [19] applied a Sequence-to-Sequence LSTM architecture to multi-step-ahead building energy consumption prediction using Korean residential data. In a sequence-to-sequence model, an encoder LSTM compresses the full input sequence (weather and demand over the past week) into a fixed-size context vector, which a decoder LSTM uses to generate a multi-step forecast one time step at a time. This architecture maintains coherent temporal dependencies across the entire forecast window, producing superior multi-horizon predictions compared to single-step LSTM baselines. A further innovation was the integration of a probabilistic occupancy simulator that generated occupancy profiles consistent with household schedules, used as additional inputs to the decoder. The

model achieved RMSE=4.48% on test data, demonstrating that structured multi-step prediction is essential for planning applications such as daily demand-side management scheduling.

Runge and Zmeureanu (2021) — Systematic Review of Deep Learning for Building Energy Forecasting

Runge and Zmeureanu [20] synthesised deep learning techniques applied to building energy forecasting across over 100 studies published between 2015 and 2021. Their central finding was that LSTM-based architectures consistently reduce prediction errors by 10–20% relative to standard feedforward neural networks for time-series tasks, because temporal dependencies in energy demand span timescales from minutes (equipment cycling) to weeks (weather patterns), all of which LSTMs can represent. The review identified hybrid optimisation—combining deep learning with evolutionary algorithms, Bayesian optimisation, or swarm intelligence—as the fastest-growing research direction with the largest performance gains. A critical gap highlighted was the lack of studies testing deep learning models across multiple building types and climate zones, which limits the generalisability of reported accuracy figures and motivates transfer learning or domain adaptation methods in future work.

Summary of Deep Learning Approaches. Deep learning studies collectively demonstrate 5–15% accuracy improvements over shallower ML methods, attributable to: (1) the ability to learn multi-scale temporal patterns through recurrent architectures; (2) systematic hyperparameter optimisation eliminating the performance gap between well-tuned and poorly-tuned models; and (3) the exploitation of unlabelled or weakly-labelled data through semi-supervised and generative frameworks. The most accurate models consistently combine recurrent or deep architectures with physically or behaviourally motivated features, confirming that data-driven and physics-informed approaches are complementary.

3.2 Comparison of Findings and Common Denominators

Comparing the three groups of studies reveals a consistent progression in predictive capability and a set of methodological themes that transcend individual algorithms.

Early neural network and regression approaches [3, 5–7] established the 5–20% accuracy advantage of data-driven methods over physics-based simulations, primarily by learning actual occupant behaviour from measured data. Their principal limitation was sensitivity to data volume: when fewer than 10 days of measurements were available,

errors exceeded 15%, and seasonal patterns were poorly captured. Degree-day regressions showed systematic errors of up to 18% during shoulder seasons when solar and internal gains dominated.

Advanced ML methods [8–10, 21, 22] built on these foundations by introducing ensemble methods that reduced RMSE by a further 10–30% through variance reduction, and by developing principled feature selection and engineering protocols. Atmospheric pressure as an integrated proxy for weather-system state and occupancy was a consistent finding, quantified at 10–15% variance in heat gains. Time-of-day encoding similarly proved important for capturing diurnal occupancy cycles.

Deep learning approaches [11, 13, 14, 19] achieved a further 5–10% improvement by exploiting temporal dependencies through LSTM and RNN architectures, handling non-linearity through deep representations, and optimising configurations through evolutionary and metaheuristic search.

The following common denominators are evident across all three groups:

- **Input features:** Temperature-based features (outdoor temperature, HDD, sol-air temperature) are universally the strongest predictors, accounting for 5–20% accuracy improvements over baselines. Secondary predictors include atmospheric pressure and relative humidity (10–15% variance), time-of-day proxies (10–15%), and physical building parameters such as U-value and shape factor (2–5% incremental improvement).
- **Model families:** Neural networks and deep learning dominate for non-linear profiles; ensemble methods (Gradient Boosting Machines, RF) reduce variance and are robust to outliers; hybrid models combining ML with optimisation algorithms consistently outperform both.
- **Data challenges:** Limited labelled data is a persistent constraint in residential applications; semi-supervised learning [13] and short-window physical embeddings [5] are the two most effective responses. Seasonal dependency—the difficulty of capturing shoulder-season demand driven by solar and internal gains—increases errors by 10–20% for models trained only on heating-season data.
- **Evaluation metrics:** RMSE and MAE are the most frequently reported metrics, with $R^2 > 0.9$ achieved by top-performing models. $CV\text{-}RMSE < 15\%$ is commonly used as a practical accuracy threshold for regulatory or planning applications.
- **Scalability:** When models are scaled from individual households to district or stock level, accuracy degrades by 5–10% unless aggregate weather and diversity features are explicitly incorporated.

3.3 Application to Thesis Development

The reviewed literature provides the methodological foundation for the modelling pipeline developed in this thesis in four concrete ways.

Feature engineering: The consistent finding that physically embedded features—HDD, U-values, sol-air temperature, shape factor—produce the largest accuracy gains [5, 7, 8] directly motivated the construction of interaction terms such as `hdd_x_wall_u` and `hdd_x_wall_area` in the dataset preprocessing pipeline described in Chapter 5. These features capture the multiplicative relationship between climate severity and building insulation that is the primary physical driver of SHD.

Algorithm selection: The demonstrated superiority of gradient boosting ensemble methods over linear models and individual neural networks for tabular building energy regression [8–10], combined with the understanding that SHD is a static annual aggregate rather than a time series (negating the advantage of LSTMs), directly motivated the selection of XGBoost, LightGBM, and CatBoost as the core modelling framework. Chapter 2 provides a full technical justification for this choice.

Hyperparameter optimisation: The finding of Morteza et al. [14] that the accuracy gap between well-tuned and poorly-tuned models can exceed the gap between algorithm families, combined with the GA and swarm-intelligence optimisation results of Almalaq and Zhang [18] and Vasanthkumar et al. [16], directly motivated the adoption of Bayesian optimisation via Optuna for systematic hyperparameter search across all base estimators.

Physical constraints: The performance gap documented across multiple studies [6, 22]—and the finding that physics-informed features reduce it by 20–30% [5]—motivated the decision to validate model predictions against thermodynamic principles through SHAP analysis rather than enforcing hard monotonic constraints. Since the ResStock and ComStock training data are themselves the output of physics-based EnergyPlus simulations, the physical relationships between HDD, envelope properties, and heating demand are already embedded in the labels by construction. SHAP attribution is used in Chapter 7 to confirm that the models have learned these relationships in the expected directions, providing a transparent, post-hoc verification of physical plausibility without the risk of distorting feature importance through constraint enforcement.

3.4 Conclusion and Gaps

This review demonstrates that ML—and gradient boosting ensembles in particular—offers a powerful, scalable, and physically interpretable framework for SHD prediction, consistently outperforming physics-based simulations in real-world accuracy while remaining practically feasible with moderate data requirements. The evidence base spans residential, commercial, and industrial buildings across multiple climates and continents.

Key gaps remain. First, most studies treat historical data as static; real-time data integration for adaptive model-predictive control has received little attention in the SHD-specific literature. Second, the impact of climate change on the statistical stationarity of training data is almost entirely unexplored: as outdoor temperature distributions shift, models trained on historical patterns may require periodic retraining or online adaptation to maintain accuracy. Third, transfer learning—pre-training on data-rich building stocks and fine-tuning on data-scarce ones—represents a natural extension of the semi-supervised framework of Chen et al. [13] that has not been explored for SHD specifically. Future work could extend the sequence-to-sequence modelling framework of Kim et al. [19] to incorporate climate scenario inputs, enabling adaptive SHD forecasting under projected future weather conditions.

Chapter 4

BDG2

4.1 Motivation and Scope

Before adopting NREL’s physics-based simulation repositories, a preliminary investigation was conducted using the Building Data Genome Project 2 (BDG2), an open-access collection of real-world sub-metered hourly energy data from non-residential buildings assembled by Miller et al. [33]. The intent was to test whether an ML surrogate for SHD could be trained directly on measured consumption records, bypassing the assumptions embedded in EnergyPlus simulations. This chapter summarises that investigation and the structural reasons it proved operationally unusable—reasons that constitute the principal empirical motivation for the shift to simulation data in subsequent chapters.

4.2 Dataset and Pipeline Overview

The BDG2 dataset comprises building metadata for 1,636 predominantly non-residential buildings across 11 sites in Europe and North America, hourly weather records, and cleaned hourly meter files for electricity, chilled water, steam, hot water, and gas covering approximately two calendar years (2016–2017). Because SHD is the target, only the three heating-relevant meters (hot water, steam, gas) were ingested.

Two pipeline iterations were implemented. An intermediate version supplemented BDG2 with the ASHRAE GEPIII competition corpus (2,134 buildings combined), used a seasonal month-based train–test split, and retained meter type as a feature—introducing meter-type contamination and the absence of a building-level holdout, since the same buildings appeared in both training and test partitions. A final version restricted the dataset to BDG2 alone, summed the three heating meters into a single consumption figure per building-hour, and introduced an 80/20 building-level split to ensure that test buildings were entirely unseen during training. This version was discontinued before completion once the deeper structural limitations described below were identified as irre-

solvable.

4.3 Results

Quantitative results from the intermediate version are presented in Table 4.1. CV-RMSE values ranged from 12% to 310% across the 11 sites. Sites 3, 8, and 14 achieved CV-RMSE of 12–22% with $R^2 > 0.97$; however, these figures reflect building-level memorisation rather than genuine generalisation, since every building appeared in both the training and test partitions. Sites 4, 10, and 13 produced CV-RMSE of 175–310% and R^2 as low as 0.31, rendering the models operationally unusable.

Table 4.1: Per-site SHD prediction metrics (intermediate pipeline). CV-RMSE $> 100\%$ indicates RMSE exceeds the mean target, rendering the model operationally unusable.

Site	Model	MAE	RMSE	R^2	CV-RMSE (%)
3	LightGBM	0.048	0.103	0.976	18.7
	Stacking Ensemble	0.045	0.090	0.981	16.4
8	XGBoost	0.023	0.084	0.982	12.7
	Stacking Ensemble	0.024	0.081	0.983	12.3
4	LightGBM	0.100	0.864	0.391	174.8
	CatBoost	0.106	0.917	0.314	185.4
10	LightGBM	0.640	3.529	0.580	226.5
	XGBoost	0.688	3.880	0.492	249.0
13	LightGBM	0.353	4.698	0.657	236.9
	Stacking Ensemble	0.570	6.139	0.414	309.5

4.4 Fundamental Limitations

Analysis across both iterations identifies four structural limitations inherent to the BDG2 data source that are not remediable through improved pipeline design.

Absence of electrical heating data. Only gas, steam, and hot water meters were ingested. Buildings served by electric heat pumps or resistance heaters contribute zero measured heating consumption, producing inconsistent labels for physically similar buildings that differ only in heating fuel type.

Inability to isolate space heating from total fuel use. Gas meter readings aggregate all end uses—space heating, domestic hot water, catering, and standby losses. For hospitals or university buildings, non-heating loads may account for 30–60% of total recorded gas consumption, introducing a systematic positive bias in the target variable with no recoverable correction signal in the feature set.

Impossibility of computing an annual SHD label. SHD is an annual metric ($\text{kWh m}^{-2} \text{ year}^{-1}$). BDG2 covers only approximately two calendar years and contains gaps

of days or weeks from meter faults and calibration artefacts. Summing an incomplete hourly series produces a figure that reflects data availability as much as thermal performance, making clean annual labels unattainable.

Insufficient sample volume and archetype coverage. The 1,636 buildings are drawn predominantly from university, hospital, and large office campuses at 11 geographically concentrated sites. Residential construction, light commercial archetypes, and the full range of national climate zones are severely under-represented, precluding generalisation to a national building stock.

4.5 Transition to Simulation-Based Data

The BDG2 investigation demonstrated that empirical sub-metered data cannot support SHD prediction at stock scale. The four structural limitations above are properties of the data collection methodology and are irresolvable by algorithmic refinement: no model can recover a heating signal that was never measured, disaggregate end uses that were never sub-metered, or produce annual labels from a record with irregular gaps.

The NREL EnergyPlus simulation repositories resolve all four issues by construction: every record is a complete annual simulation output; the heating energy label aggregates all fuel sources including electricity; data completeness is guaranteed; and the sample design is calibrated to national survey statistics (RECS, CBECS) to cover the full range of building archetypes, vintages, and climate zones. The subsequent chapters describe the pipeline and results obtained from this foundation.

Chapter 5

Dataset

This chapter describes the datasets used to develop and evaluate the prediction models[24][25]. This repository is divided into two components: **ResStock**, representing the residential building sector, and **ComStock**, representing the commercial sector. Each simulation captures energy consumption at 15-minute intervals across a nationally representative sample of building archetypes, providing granular insight into end-use consumption patterns including space heating, cooling, lighting, and plug loads. For this thesis, the aggregated annual results will be used due to the nature of the target variable.

Both datasets are distributed as Parquet files. In both cases, only baseline building models are retained (upgrade ID "0" or "00"), ensuring that predictions correspond to the existing stock rather than any modelled retrofit scenario.

5.1 Target Variable Definition

The objective variable in both datasets is the **Specific Heating Demand (SHD)**, defined as the annual heating energy delivered per unit of conditioned floor area:

$$\text{SHD} = \frac{\sum E_{\text{heating}}}{A_{\text{floor}} [\text{m}^2]} \quad [\text{kWh m}^{-2} \text{yr}^{-1}] \quad (5.1)$$

Floor area is converted from square feet to square metres. In the commercial dataset, the heated floor area is further scaled by the `out.params.building_fraction_heated` parameter, so that only the thermally served portion of the building contributes to the denominator. In the residential dataset, each dwelling unit is assumed to be fully conditioned, so no such correction is applied. `out.load.heating.energy_delivered.kbtu` provides the total delivered heating load in kBtu, which is then converted to kWh. After computing SHD, rows with non-positive floor area, negative heating energy, or non-finite SHD values are removed.

5.2 Commercial Building Dataset (ComStock)

5.2.1 Overview

The ComStock dataset represents the U.S. commercial building stock through a large collection of prototypical building simulations spanning multiple climate zones, building types, vintages, and operational schedules. Each row in the Parquet file corresponds to a unique building model instance characterised by a unique `bldg_id`.

5.2.2 Data Cleaning and Filtering

The raw Parquet file is loaded and converted to *pandas*-compatible dtypes. The following cleaning steps are applied in sequence:

1. **Physical validity.** Rows with NaN floor area or heating energy are removed.
2. **Outlier removal.** The top and bottom 0.5 percentiles of the SHD distribution are excluded from the *training set only*. The test set is never filtered, so that evaluation metrics reflect performance on the full distribution, including difficult edge cases.
3. **Building-level train/test split.** The dataset is partitioned 80/20 by unique `bldg_id`

5.2.3 Feature Engineering

Before model training, a set of physically motivated features is derived from the raw simulation outputs. These engineered features encode the primary heat-loss pathways directly, providing the models with explicit physical context beyond the raw metadata fields.

Total Thermal Conductance (UA). For each envelope component—walls, roof, and windows—the U-value (in $\text{BTU hr}^{-1} \text{ft}^{-2} \text{°F}^{-1}$) is multiplied by the corresponding area (converted from m^2 to ft^2) to produce a component UA in $\text{BTU hr}^{-1} \text{°F}^{-1}$. These are summed to yield `UA_total`, the whole-building thermal conductance. A normalised version, `UA_per_area`, is obtained by dividing by the conditioned floor area in m^2 .

Climate–Conductance Interaction. The product `HDD65_x_UA_per_area` is formed by multiplying Heating Degree Days (base 65°F, drawn from `out.params.hdd65f`) by `UA_per_area`. This single term approximates the annual conductive heat loss per unit area and captures the synergistic effect of climate severity and envelope thermal quality.

Ventilation Load Proxy. When the design outdoor air flow rate (`out.params.design_outdoor_air_flow_rate..m3_per_m2_s`) is available, it is multiplied by the total weekly operating hours to approximate the ventilation-driven heat loss. When this column is absent, the outdoor air fraction (`out.params.average_outdoor_air_fraction`) scaled by operating hours is used as a fallback.

Operating Hours. Total weekly operating hours are computed as $5 \times \text{weekday_hours} + 2 \times \text{weekend_hours}$. The individual opening-time columns (weekday and weekend) are also retained as direct features.

Infiltration Load Proxy. The airtightness parameter `in.airtightness..m3_per_m2_s` (measured at 75 Pa) is multiplied by HDD65 to produce `infiltration_x_HDD`, capturing the combined effect of air leakage and climatic severity. A complementary ratio, `infiltration_to_UA_ratio`, compares infiltration losses to conductive losses; high values indicate leakage-dominated buildings.

Physics-Based SHD Estimate. A first-principles SHD estimate (`calculated_shd`) is derived by summing the three principal heat-loss pathways—transmission through the envelope, ventilation, and infiltration—and subtracting internal gains from lighting, equipment, and occupants. This value is used as an engineered feature rather than as the prediction target, providing the ML models with an interpretable physics baseline. It follows from the following SHD estimation equation[35]:

$$Q_h = (Q_T + Q_V) - \eta \cdot (Q_i + Q_s)$$

Where:

- Q_T = **Transmission Heat Loss**: The heat lost through the building envelope (walls, windows, roof).
- Q_V = **Ventilation Heat Loss**: The heat lost through air exchange (infiltration and mechanical ventilation).
- Q_i = **Internal Gains**: Heat generated by occupants, appliances, and lighting.
- Q_s = **Solar Gains**: Heat transferred into the space from sunlight.
- η = **Gain Utilization Factor**: An efficiency coefficient for how much of the solar and internal gains are effectively used (usually ranges from 0.9 to 1.0).

5.2.4 Selected Features

The features used for model training are enumerated in Tables 5.1 and 5.2.

Table 5.1: ComStock — Numeric features used for model training.

Feature	Description
<code>floor_area_m2</code>	Conditioned floor area of the heated fraction, in m^2 .
<code>out.params.hdd65f</code>	Annual heating degree-days with a 65 °F base temperature.
<code>out.params.hdd50f</code>	Annual heating degree-days with a 50 °F base (supplementary climate indicator).
<code>UA_total</code>	Whole-building thermal conductance (walls + roof + windows) in $\text{BTU hr}^{-1} \text{°F}^{-1}$.
<code>UA_per_area</code>	<code>UA_total</code> normalised by conditioned floor area.
<code>HDD65_x_UA_per_area</code>	Product of HDD65 and <code>UA_per_area</code> ; principal climate-envelope interaction term.
<code>operating_hours_week</code>	Total weekly operating hours ($5 \times \text{weekday} + 2 \times \text{weekend}$).
<code>ventilation_load_proxy</code>	Outdoor air flow rate scaled by occupancy fraction; proxy for annual ventilation heat loss.
<code>in.year_built</code>	Numeric construction year, extracted from the categorical vintage string.
<code>out.params.average_outdoor_air_fraction</code>	Average fraction of supply air that is fresh outdoor air.
<code>in.weekday_opening_time..hr</code>	Start of operations on weekdays (hour of day).
<code>in.weekend_opening_time..hr</code>	Start of operations on weekends (hour of day).
<code>infiltration_x_HDD</code>	Product of airtightness and HDD65; captures infiltration-driven heat loss.
<code>infiltration_to_UA_ratio</code>	Ratio of infiltration rate to <code>UA_per_area</code> ; indicates leakage-dominated buildings.
<code>in.number_of_stories</code>	Number of stories above grade (proxy for building compactness).
<code>in.rotation..degrees</code>	Building orientation relative to north.
<code>calculated_shd</code>	First-principles SHD estimate used as an engineered physics-based feature.

Table 5.2: ComStock — Categorical features used for model training.

Feature	Description
<code>in.comstock_building_type</code>	Primary building classification (e.g., <i>LargeOffice</i> , <i>RetailStripmall</i> , <i>Hospital</i>).
<code>in.comstock_building_type_group</code>	Higher-level grouping of building types.
<code>in.building_subtype</code>	Finer-grained sub-classification within a building type.
<code>in.hvac_heat_type</code>	Category of heating system (e.g., heat pump, furnace, boiler).
<code>in.hvac_category</code>	Broader HVAC system category.
<code>in.wall_construction_type</code>	Exterior wall construction method (e.g., <i>Mass</i> , <i>Steel-Framed</i> , <i>WoodFramed</i>).
<code>in.window_type</code>	Glazing specification (e.g., double-pane, triple-pane, low-e coating).
<code>in.window_to_wall_ratio_category</code>	Binned window-to-wall ratio class.
<code>in.interior_lighting_generation</code>	Lighting technology generation, which affects internal heat gains.
<code>in.state_name</code>	U.S. state, serving as a proxy for local climate and energy code environment.

5.3 Residential Building Dataset (ResStock)

5.3.1 Overview

The ResStock dataset characterises the U.S. residential building stock through a large ensemble of dwelling-unit simulations. It covers single-family detached and attached homes, multi-family units, and mobile homes across all U.S. climate zones. Each row corresponds to one dwelling-unit simulation, identified by a unique `bldg_id`.

5.3.2 Data Cleaning and Filtering

The cleaning pipeline mirrors the ComStock approach in structure, with adjustments for the residential data schema:

1. **Physical validity.** Rows with NaN floor area negative heating energy are removed.
2. **Outlier removal.** The top and bottom 0.5 percentiles of SHD are excluded from the training set; the test set is left unfiltered.
3. **Building-level split.** As in ComStock, the 80/20 partition is performed over unique building IDs to prevent data leakage.

5.3.3 Feature Engineering

The residential preprocessing pipeline extracts and derives a richer set of physical parameters than the raw metadata columns provide, largely because many ResStock fields are stored as descriptive strings that must be decoded into numeric quantities.

Vintage Decoding. The categorical vintage string (e.g., "1970s", "<1940") is mapped to a numeric year using decade midpoints via a fixed lookup table, producing the `year_built_numeric` feature.

Infiltration Rate. The `in.infiltration` field stores air leakage as a string of the form "X ACH50". A regular expression parser extracts the numeric value to yield `ach50_numeric` (Air Changes per Hour at 50 Pa). A natural infiltration rate is also derived by dividing `ach50_numeric` by 20, consistent with the empirical relationship between pressurised and natural-condition infiltration.

R-Value Extraction. Each insulation field (`in.insulation_wall`, `in.insulation_ceiling`, `in.insulation_roof`, `in.insulation_floor`, `in.insulation_foundation_wall`, `in.insulation_slab`, `in.insulation_rim_joist`) is parsed by a regular expression that extracts the numeric R-value from descriptors such as "R-13" or "Uninsulated" (mapped to zero). These raw R-values are then converted to effective U-values by adding fixed base resistances accounting for framing and surface film coefficients before inversion.

Window U-Value. The `in.windows` column contains a string descriptor (e.g., "Double, Low-E, argon, H-gain"). A lookup table maps these descriptors to U-values in $\text{BTU hr}^{-1} \text{ft}^{-2} \text{°F}^{-1}$, producing the numeric `u_win_btu` feature.

Duct Leakage. The `in.duct_leakage_and_insulation` field is parsed to extract the percentage of duct leakage to the outside, yielding the continuous `duct_leakage_frac` feature.

Envelope Areas. Above-grade conditioned wall area, window area, net wall area (wall minus window), and roof/ceiling area are drawn from the `out.params` columns where available, with geometric fallbacks applied otherwise (e.g., wall area is approximated from floor area and a standard ceiling height for square floor plans when the direct measurement is absent).

Total Thermal Conductance (UA). Component UA products are computed for walls, ceiling/roof, floor, and windows using the derived U-values and areas. These are

summed to `UA_total` in $\text{BTU hr}^{-1} \text{°F}^{-1}$ and normalised by floor area to give `UA_per_area` (converted to SI units, $\text{W m}^{-2} \text{K}^{-1}$).

Climate–Conductance Interaction. The feature `HDD65_x_UA_per_area` is formed by converting HDD65 from °F-day to K-day (dividing by 1.8) and multiplying by `UA_per_area`. This is the dominant physics-driven predictor of annual conductive heat loss.

Physics-Based SHD Estimate. A first-principles energy balance identical in structure to the ComStock calculation is applied, summing transmission, infiltration, and ventilation losses and subtracting internal gains. The resulting `calculated_shd` is included as a model feature rather than as the prediction target.

5.3.4 Selected Features

The features used for model training are enumerated in Tables 5.3 and 5.4.

5.4 Comparison of the Two Datasets

While both datasets originate from the same simulation engine and share a common target variable and preprocessing philosophy, they differ substantially in the nature of their inputs and the physical processes they must capture.

The ComStock feature set emphasises **operational characteristics** that have no residential analogue: weekly and daily operating schedules, outdoor air fraction, and the distinction between dozens of commercial building archetypes. Heating in commercial buildings is strongly modulated by occupancy schedules, mechanical ventilation requirements, and internal heat gains from equipment and occupancy, all of which are represented in the engineered features.

The ResStock feature set, by contrast, is dominated by **envelope assembly detail**. Because residential dwellings are continuously occupied and their ventilation is largely driven by infiltration rather than mechanical systems, the most informative signals are the insulation levels of each envelope component, the air leakage rate, and the window specification. The preprocessing pipeline therefore invests heavily in parsing these string-encoded attributes into physically meaningful numeric quantities—R-values, U-values, and areas—which are then combined into UA products.

Both datasets share the climate–conductance interaction term `HDD65_x_UA_per_area` and the physics-based `calculated_shd` estimate as core features, reflecting the common underlying heat-balance physics. The train/test split strategy—stratified at the building level—is also shared, guarding against the data-leakage risk that would arise from splitting rows of the same building archetype across sets.

Table 5.3: ResStock — Numeric features used for model training.

Feature	Description
floor_area_m2	Conditioned floor area in m^2 .
hdd65	Annual heating degree-days with a 65 °F base.
HDD65_x_UA_per_area	Product of HDD65 (in $\text{K}\cdot\text{day}$) and UA_per_area; primary conduction-climate term.
UA_total	Whole-building thermal conductance in $\text{BTU hr}^{-1} \text{°F}^{-1}$.
UA_per_area	UA_total normalised by floor area, in $\text{W m}^{-2} \text{K}^{-1}$.
r_wall	Effective R-value of exterior walls ($\text{hr} \cdot \text{ft}^2 \cdot \text{°F BTU}^{-1}$), parsed from string metadata.
r_ceiling	Effective R-value of ceiling insulation.
r_roof	Effective R-value of the roof assembly.
r_floor	Effective R-value of the floor assembly.
r_foundation_wall	Effective R-value of foundation wall insulation.
r_slab	Effective R-value of slab insulation.
r_rim_joist	Effective R-value of rim joist insulation.
wall_area_ft2	Above-grade conditioned wall area in ft^2 .
win_area_ft2	Total window area in ft^2 .
net_wall_area_ft2	Opaque wall area (wall minus window) in ft^2 .
ceil_area_ft2	Ceiling/roof area in ft^2 .
u_win_btu	Window U-value in $\text{BTU hr}^{-1} \text{ft}^{-2} \text{°F}^{-1}$, mapped from the glazing type descriptor.
ach50_numeric	Air leakage rate in ACH at 50 Pa, parsed from the infiltration string.
duct_leakage_frac	Fraction of supply air lost to duct leakage outside conditioned space.
year_built_numeric	Numeric construction year decoded from the categorical vintage string.
calculated_shd	First-principles SHD estimate used as a physics-based engineered feature.

Table 5.4: ResStock — Categorical features used for model training.

Feature	Description
<code>in.geometry_building_type_recs</code>	RECS-aligned dwelling type (e.g., <i>Single-Family Detached</i> , <i>Multi-Family with 5+ units</i>).
<code>in.geometry_wall_type</code>	Structural wall construction system (e.g., <i>Wood Frame</i> , <i>Concrete</i>).
<code>in.geometry_attic_type</code>	Attic configuration (e.g., <i>Vented Attic</i> , <i>Cathedral Ceiling</i> , <i>Finished Attic</i>).
<code>in.hvac_heating_type_and_fuel</code>	Combined descriptor of heating equipment type and fuel (e.g., <i>Fuel Furnace</i> , <i>Electric Heat Pump</i>).
<code>in.hvac_has_ducts</code>	Binary indicator for the presence of a ducted distribution system.
<code>in.duct_location</code>	Location of ductwork (e.g., <i>Attic</i> , <i>Basement</i> , <i>Crawlspace</i>), which affects thermal losses.
<code>in.insulation_wall</code>	Original string descriptor for wall insulation type and R-value.
<code>in.insulation_ceiling</code>	Original string descriptor for ceiling insulation type and R-value.
<code>in.insulation_roof</code>	Original string descriptor for roof insulation.
<code>in.vintage</code>	Decade-binned construction vintage string (e.g., "1970s", "<1940").
<code>in.state</code>	U.S. state, encoding regional climate, energy code environment, and construction practices.
<code>in.geometry_wall_exterior_finish</code>	Exterior cladding material (e.g., <i>Vinyl</i> , <i>Brick</i> , <i>Stucco</i>).
<code>in.roof_material</code>	Roofing material type (e.g., <i>Asphalt Shingles</i> , <i>Metal</i>).
<code>in.windows</code>	Glazing type descriptor used to derive <code>u_win_btu</code> .

5.5 Climate Zone Stratification

Both models are trained at two levels: a *global* model covering the full national stock, and a suite of *climate-zone-specific* models. The climate zone variable `in.building_america_climate_zone` partitions buildings into the standard Building America zones: Hot-Dry, Hot-Humid, Mixed-Dry, Mixed-Humid, Marine, Cold, and Very Cold. After a few trials with both datasets, the Cold and Very Cold zones are combined into a single model, with an additional binary feature `is_very_cold` appended to distinguish the two sub-zones within that model. While both have enough samples to justify different models, a separate set of models for 'Very Cold' didn't prove successful and better results were seen by combining it with Cold. For zones with fewer than 250 training samples, zone-specific training is skipped and the global model is applied instead.

5.6 Dataset Validity and the Simulation–Reality Gap

The models developed in this thesis are trained and evaluated exclusively on EnergyPlus simulation outputs rather than on measured building energy consumption. This is a deliberate methodological choice, but it carries a fundamental epistemic implication that must be made explicit before the results in Chapter 7 can be correctly interpreted: the reported accuracy metrics — R^2 , RMSE, CV-RMSE — describe how well the ML models replicate the ResStock and ComStock simulation engines, not how well they predict the heating demand of real buildings as measured by utility bills or sub-meters. A surrogate that perfectly replicates a simulator will be biased in exactly the same ways as that simulator. This section addresses this distinction directly by examining the calibration and validation evidence for both datasets, characterising the residual gap between simulation and reality that remains after calibration, and drawing out the implications for interpreting the thesis's results.

5.6.1 The Nature of the Simulation–Reality Gap

Physics-based building energy simulation tools such as EnergyPlus are not, and do not claim to be, perfect representations of real buildings. They solve a deterministic system of heat-transfer and HVAC equations given a set of fixed input parameters: a prescribed thermostat setpoint, a deterministic occupancy schedule, a code-derived infiltration rate, and a standard-condition equipment efficiency. Real buildings, by contrast, are occupied by people whose heating preferences, presence patterns, and thermostat behaviour vary continuously and are largely unobservable from audit records alone. The discrepancy between what a physics model predicts and what a real building actually consumes is known as the performance gap, and it is well established in the building energy literature. Several studies reviewed in Chapter 3 quantify it: Karlsson et al. [6] found actual

consumption in a low-energy building to be approximately 50% of the simulated prediction, while De Rosa et al. [22] found simulation-to-measurement deviations of 15–18% in heating-dominated conditions, attributable primarily to occupant-driven deviations from assumed thermostat setpoints and schedules.

The relevant question for this thesis is therefore not whether EnergyPlus simulations perfectly match measured consumption — they do not, and this is well known — but whether the ResStock and ComStock datasets are *calibrated to the statistical distribution of real-world building energy use at the stock level*, such that the ML surrogates trained on them can reasonably be expected to represent population-level heating demand patterns. This distinction between individual-building fidelity and population-level representativeness is central to understanding what the thesis’s models are and are not capable of.

5.6.2 ResStock: Calibration Evidence and Residual Uncertainty

ResStock has undergone calibration and validation efforts continuously since its inception in 2014. The technical reference documentation describes the process as involving multiple distinct phases and data sources [24].

The initial calibration in 2014–2015 focused on comparing estimates of average annual electricity and gas use per home against EIA RECS 2009 microdata, grouping single-family detached homes by combinations of region, vintage, and fuel type. Agreement between the model and RECS was assessed using visual comparisons of cumulative distribution functions alongside Kolmogorov–Smirnov goodness-of-fit tests to validate the *heterogeneity* of model outputs, not merely their mean. This is an important methodological distinction: validating a distribution rather than a point estimate means the model is being held to the requirement that its spread, skew, and tails — not just its average — match the survey data. Twelve rounds of model structure and input modifications followed, involving changes to probability distributions, dependency structures, the number of distribution options, and the spatial resolution of weather files. The result is a housing characterisation whose marginal distributions across the major axes of stratification (region, vintage, heating fuel, building type, floor area) are calibrated to match the RECS sampling frame.

Between 2019 and 2022, calibration and validation were extended to include hourly timeseries outputs as part of a three-year DOE-funded project. ResStock results were compared against utility load research data, Advanced Metering Infrastructure (AMI) data, and end-use sub-metered data across a wide range of end uses and building types [24]. These comparisons informed additional modelling improvements and produced a validated updated model. The occupant behaviour component of ResStock — which constitutes the largest single driver of the performance gap in the literature — is handled through a stochastic schedule generator developed using American Time Use Survey (ATUS) data

from 2013–2017, sub-metered appliance energy data, and a supplemental hot water model. This means that ResStock does not prescribe a single deterministic occupancy profile but instead samples from an empirically derived distribution of occupancy patterns, partially embedding the behavioural variability that pure physics models treat as fixed.

A critical caveat acknowledged by the ResStock documentation itself concerns the level at which this representativeness holds. The technical reference explicitly notes that anomalies in individual simulation outputs that arise from mismatches between the stochastic energy use calculation and the schedule generation “are less jarring at the stock level as aggregation tends to balance and smooth out these anomalies” [24]. This is a consequential statement: ResStock is designed and validated as a *population-level* instrument. Its characterisations of how heating demand varies across the building stock — by climate zone, vintage, building type, envelope quality, HVAC system — are grounded in national survey data and have been validated against measured distributions. Its predictions for any specific individual dwelling, however, carry substantially higher uncertainty, because the stochastic components that drive individual-building variability (exact thermostat setpoint, precise occupancy timing, actual equipment usage patterns) are drawn from distributions rather than measured directly.

The housing characteristic distributions are also explicitly acknowledged to inherit the survey biases and uncertainties present in the underlying data sources. RECS, the primary calibration benchmark, is itself a sample survey and carries its own sampling error, particularly for less common housing types. The ResStock documentation notes that mobile homes carry higher uncertainty than other housing types because of their lower representation in the survey. These inherited biases propagate into the simulated SHD distributions and, through training, into the ML models.

5.6.3 ComStock: Calibration Evidence and Residual Uncertainty

ComStock’s calibration follows a broadly similar philosophy but operates against a different set of real-world benchmarks reflecting the greater structural diversity of the commercial sector [25].

The floor area composition of ComStock’s national sample is directly tied to CBECS. After assembling building inventories from the CoStar commercial real estate database and the Homeland Infrastructure Foundation-Level Data tables, ComStock applies a CBECS-based “true-up” normalisation: the sampled floor area by building type is scaled nationally to match CBECS estimates exactly. The technical reference states that “ComStock’s floor area estimates match CBECS’ by building type on a national basis” [25], treating CBECS as the ground truth for stock-level floor area distribution. This is a deliberate design choice motivated by CBECS’ centrality to all commercial energy use estimation; it means that ComStock’s building type frequency distribution and the aggregate floor area it

represents are not free parameters but are anchored to the most authoritative national survey of commercial buildings available.

The calibration of building operational parameters — the aspect of commercial energy use most susceptible to the simulation–reality gap — was substantially advanced through the End-Use Load Profiles (EULP) project, a multi-year DOE-funded effort whose results informed both ComStock and ResStock. For ComStock, the most consequential EULP calibration involved comparing building operating schedules (the primary driver of commercial heating demand after climate and envelope characteristics) against AMI data from 6,070 real commercial buildings across eight utilities. The comparison of High Load Start Time and High Load Duration distributions revealed that the pre-calibration ComStock schedules were more concentrated (less diffuse) and assumed longer operating durations than the real AMI data showed [25]. The updated schedules derived from this process were then applied to all ComStock building types to replace the original prototype-model defaults. This calibration is particularly relevant to SHD prediction because operating hours directly govern the duration over which internal gains offset heating loads and over which mechanical ventilation conditioning occurs.

Equipment power densities and lighting schedules underwent similar data-grounded adjustments. EPD values for several building types — full-service restaurants, primary and secondary schools, quick-service restaurants, retail, and strip malls — were increased or decreased relative to the DOE prototype model defaults based on end-use-level data from two industry sources provided during the EULP calibration. Interior lighting schedules were likewise updated based on EULP project findings for restaurant, retail, education, office, and warehouse building types. The effect of these calibrations is to replace code-minimum or prototype-model assumptions with values that better reflect the performance seen in measured data sources, as the technical reference states explicitly [25].

The calibration of heating fuel prevalence — directly relevant to SHD since the choice between gas furnace, electric resistance, and heat pump determines effective system efficiency — is validated against CBECS through building-type-specific comparisons. The technical reference includes fuel type prevalence comparisons between ComStock and CBECS for 15 building types, including full-service restaurants, hospitals, large hotels, large and medium offices, outpatient facilities, primary and secondary schools, quick-service restaurants, retail, small hotels, small offices, strip malls, and warehouses. HVAC system type prevalence is similarly compared. This means the distribution of heating system types in ComStock — which the SHAP analysis in Chapter 7 identifies as among the most important predictors of commercial SHD — is not an engineering guess but a calibrated reflection of the actual national commercial building stock as recorded by CBECS.

A limitation that the ComStock documentation is direct about is coverage. Approximately 36% of commercial building energy use, as estimated by CBECS, is not represented

in ComStock [25]. The excluded building types — which include grocery stores, religious worship centres, certain mixed-use facilities, and others — are absent primarily because their operational and energy use profiles are sufficiently distinct from the represented DOE prototype building types that inclusion would require new baseline models. The ML models trained on ComStock therefore cannot be expected to generalise to these excluded archetypes, and predictions for building types not well represented in the ComStock training set carry higher uncertainty.

5.6.4 Interpreting the Thesis Results in Light of the Simulation–Reality Gap

The foregoing evidence supports a specific and carefully bounded claim about what the ML surrogates developed in this thesis represent. The reported accuracy metrics — CV-RMSE ranging from 13.8% to 52.3% and R^2 from 0.557 to 0.901 across zones, datasets, and iterations — measure how faithfully the models replicate the EnergyPlus simulations that constitute their training data. This is a meaningful and practically useful property, but it is not the same as measuring agreement with measured building energy consumption.

The relationship between these two forms of accuracy depends on the degree to which the simulations themselves represent real-world distributions. The calibration evidence reviewed above supports the conclusion that ResStock and ComStock are well-calibrated to the *statistical distributions* of the real building stock at the population level: their floor area compositions, fuel type distributions, HVAC system prevalences, vintage distributions, and operational schedule characteristics are grounded in national surveys and, for key parameters, validated against AMI and sub-metered data. An ML model trained on these distributions can therefore be expected to reproduce population-level patterns in SHD — the relative heating intensity of different climate zones, building types, and envelope quality cohorts — with fidelity comparable to the underlying simulation datasets.

What the models cannot claim is individual-building accuracy of the kind that measured sub-metered data would support. The stochastic behavioural components embedded in ResStock and ComStock — thermostat setpoints, occupancy timing, equipment usage schedules — are drawn from population distributions, not from specific households or buildings. An individual real building may consume heating energy that is 30–50% higher or lower than what either the simulation or the ML surrogate predicts, for reasons that are entirely behavioural and not capturable from audit-grade physical characteristics alone. This is not a failure of the modelling pipeline: it is the fundamental performance gap identified across the building energy literature, and it applies equally to the EnergyPlus simulations that constitute the training targets. The second iteration of the thesis — which deliberately excludes the thermostat setpoint feature and restricts inputs to envelope and climate characteristics — makes this gap explicit: the lower R^2 values

of 0.557–0.640 in warm residential zones quantify precisely how much of the variance is driven by behavioural factors that a deployable, audit-based model cannot observe.

Two further points follow from this analysis. First, the simulation–reality gap is not uniformly distributed across the building stock. It is largest in climates where the heating season is mild (hot-humid and hot-dry zones), because in these climates the absolute heating demand is small and therefore small absolute deviations in occupant behaviour constitute large relative errors. It is smallest in cold-climate, high-demand buildings where envelope conductance and infiltration dominate the heat balance so completely that behavioural variation is proportionally minor. This matches the observed pattern in the results: the cold-climate models achieve both higher R^2 and lower CV-RMSE on the simulation targets, and they are also the building cohorts where the models’ physical features are most likely to correlate with real-world consumption. Second, because both ResStock and ComStock are calibrated to represent the existing building stock as of approximately 2018–2019, the models inherit the calibration epoch’s assumptions about equipment efficiency, vintage distribution, and regional fuel mix. Applying the models to project future SHD under significantly different stock compositions or technological penetrations would require re-evaluating this calibration basis.

In summary, the ML surrogates trained on ResStock and ComStock are surrogates for physically well-grounded, nationally calibrated simulations, not for arbitrary or unconstrained models. The models are validated to the extent that the simulations themselves are validated — which is population-level calibration to national surveys and AMI data, not individual-building measurement agreement. The results should be interpreted accordingly: as robust, scalable estimates of the population-level distribution of specific heating demand across the U.S. building stock, useful for policy analysis, retrofit prioritisation, and stock-level screening, but not as predictors of how much any specific individual building will spend on heating in a given year.

Chapter 6

Methodology and Computational Workflow

This chapter details the computational pipeline implemented across the four training scripts developed for this thesis, two targeting maximum predictive accuracy and two prioritising physical interpretability and deployability. To reflect this distinction, the chapter is structured into two main sections. Section 6.1 describes the first-iteration pipeline as applied to each dataset, covering data ingestion, feature construction, encoding, hyperparameter search, training strategy, and ensemble assembly. Section 6.2 describes, for each dataset in turn, every substantive modification introduced in the second iteration—from preprocessing and split strategy through to the stacking procedure and post-training interpretability analysis. The feature engineering choices and the physics-based engineered variables shared by both iterations are documented in Chapter 5; this chapter focuses on the implementation and algorithmic design decisions specific to each pipeline.

Throughout both iterations, a common random seed of 42 is fixed globally to ensure reproducibility across data splits, cross-validation folds, and Optuna trials. All models are trained on the baseline building stock exclusively: rows whose **upgrade** identifier resolves to "0", "baseline", or any equivalent zero-valued string are retained, and all retrofit scenarios are discarded, so that predictions correspond to existing stock conditions.

6.1 First Iteration

The first iteration was designed to explore the accuracy ceiling achievable with the available data. Its distinguishing characteristics relative to the second iteration are a broad feature set that encompasses simulated output parameters alongside input metadata, monotonic constraints on physically unambiguous relationships, a flat label-encoding preprocessing strategy, and a row-level train/test partition. These choices prioritise predictive performance over deployability and physical parsimony.

6.1.1 ComStock — First Iteration

Target Variable Construction

The target variable is the Specific Heating Demand, computed by dividing total annual heating energy by conditioned floor area. Heating energy is sourced from the column `calc.enduse_group.site_energy.heating.energy_consumption` when present; if absent, it is reconstructed by summing the four fuel-specific components: electricity, natural gas, other fuel, and district heating. Floor area is read from `in.sqft` and converted to square metres using the factor 0.09290304. Importantly, in this iteration the floor area is not adjusted by the heated fraction of the building; the full gross floor area is used as the denominator. Rows with non-positive floor area, negative heating energy, or non-finite SHD are removed. Outlier trimming is then applied symmetrically to both the training and test partitions by excluding records below the 0.5th and above the 99.5th percentile of the SHD distribution.

Feature Set

The first iteration operates in `FEATURE_MODE = "max_accuracy"`, which assembles two tiers of candidate features. The first tier, referred to as *deployable features*, comprises approximately thirty `in.*` metadata columns describing building characteristics that would in principle be obtainable without running a full energy simulation. These include the ComStock building type, building type group, and sub-type classification; floor area category; number of stories; aspect ratio and rotation; wall construction type; window-to-wall ratio category and window type; vintage decade string and parsed construction year; the Building America climate zone and the companion ASHRAE/IECC climate zone designation; state, census region, and census division names; the Cambium grid region and ISO/RTO region identifiers; cluster identifier and name; county name; TMY3 and 2018 weather file identifiers; HVAC system type, HVAC category, and individual heating, cooling, and ventilation type descriptors; the combined HVAC type string; heating fuel; HVAC night set-back variability; weekday and weekend opening times and operating hours; energy codes in force at original construction and at the last HVAC, roof, and wall replacements; ownership type; the party responsible for operation; and the purchase responsibility designation.

The second tier, referred to as *max-accuracy additional features*, draws approximately twenty-five columns from the `out.params.*` namespace, which contain parameters computed directly by EnergyPlus during simulation rather than prescribed as inputs. These include annual Heating Degree Days at both the 65 °F and 50 °F base temperatures; Cooling Degree Days; the number of hours per year below 50 °F, 17 °F, and 0 °F; average wall, roof, and window U-values and average window solar heat gain coefficient; the simulated window-to-wall ratio; exterior wall, roof, and window areas; the fraction of floor area

that is heated; heating equipment type; boiler average efficiency and rated capacity; the average and design coefficients of performance for direct-expansion heating; the average COP and rated capacity of the heat pump heating system; average minimum and maximum heating setpoints and the annual hours during which the setpoint was not met; the average outdoor air fraction; occupant density in persons per square metre; equivalent full-load occupancy hours; interior equipment power density; interior electric equipment equivalent full-load hours; and interior lighting power density and equivalent full-load hours. Including these simulated output parameters substantially increases the predictive information available to the model, at the cost of requiring a prior simulation run for any building to be scored.

Feature Engineering

Several derived features are computed from the raw columns. The construction year is extracted from the vintage string using a regular expression that identifies four-digit years in the range 1900–2099 and takes the most recent match, producing a continuous numeric `year_built_num`. Weekly operating hours are computed as $5 \times \text{weekday_hours} + 2 \times \text{weekend_hours}$, yielding `operating_hours_week`. Where both the maximum and minimum average heating setpoints are available, their difference is recorded as `heating_setpoint_span_c`. Five interaction terms are then formed by multiplying HDD65 against each envelope U-value and each exterior area: `hdd_x_wall_u`, `hdd_x_roof_u`, `hdd_x_win_u`, `hdd_x_wall_area`, and `hdd_x_window_area`. These products serve as first-order proxies for the principal conductive heat-loss pathways—analogueous to the $\Delta T \times UA$ term in steady-state heat transfer—without requiring the full U-value-times-area computation introduced in the second iteration.

Finally, binary missing-value indicator columns are appended for features known to have substantial proportions of absent records: wall, roof, and window U-values; HDD65; HVAC system type; window type; weekday and weekend operating hours; maximum and minimum heating setpoints; and hours during which the heating setpoint was not met. These flags allow the tree-based models to treat data absence as an informative signal rather than forcing imputed values to masquerade as genuine observations.

Categorical Encoding and Missing Value Handling

All feature columns are cast to the string type and any null values are replaced with the sentinel token `"__MISSING__"` before encoding. A custom `fit_label_encoders` function then inspects each column: if all values in the training set parse unambiguously as floating-point numbers, the column is converted to `float64` directly; otherwise, a `scikit-learn` `LabelEncoder` is fitted on the training strings (including the sentinel token) and applied to both partitions. At inference time, any test-set string not encountered during training

is mapped to the sentinel index if the sentinel was present in the encoder vocabulary, and to zero otherwise. Separate encoders are fitted for the global model and for each climate-zone subset, preventing category distributions in one zone from contaminating the encoding used in another.

Train/Test Partition

The dataset is divided into training and test subsets using a row-level random split with an 80/20 ratio. No grouping by building identifier is applied at this stage; the partition operates on individual rows. Climate-zone stratification is enforced implicitly by training separate models for each zone, rather than through stratified sampling.

Hyperparameter Optimisation

Each of the three base estimators is tuned independently using the Optuna Bayesian optimisation framework, employing the default Tree-structured Parzen Estimator sampler with 30 trials per algorithm per training scope (global model or climate-zone model). The objective evaluated at each trial is the mean cross-validated RMSE over five folds of a standard KFold splitter with shuffling enabled. The target variable is log-transformed before fitting (`log1p`) and back-transformed after prediction (`expm1`), with negative back-transformed values clipped to zero.

The search space for XGBoost encompasses: number of estimators over the integer range $[250, 1200]$; maximum tree depth over $[3, 7]$; learning rate over $[0.01, 0.08]$ on a log scale; row subsampling fraction over $[0.65, 0.95]$; column subsampling fraction over $[0.60, 0.90]$; minimum child weight over $[5.0, 40.0]$; the split-gain threshold γ over $[0.0, 15.0]$; L1 regularisation coefficient α over $[10^{-3}, 30.0]$ on a log scale; and L2 regularisation coefficient λ over $[10^{-2}, 60.0]$ on a log scale.

For CatBoost, the search covers: number of iterations over $[250, 1200]$; tree depth over $[4, 8]$; learning rate over $[0.01, 0.08]$ on a log scale; L2 leaf regularisation over $[5.0, 60.0]$; random strength over $[1.0, 12.0]$; and the Bayesian bootstrap temperature over $[0.0, 6.0]$.

For LightGBM, the search covers: number of estimators over $[250, 1200]$; learning rate over $[0.01, 0.08]$ on a log scale; number of leaves over $[16, 80]$; maximum depth over $[3, 8]$; minimum samples per leaf over $[40, 300]$; row and column subsampling fractions each over $[0.65, 0.95]$ and $[0.60, 0.90]$ respectively; and L1 and L2 regularisation coefficients over the same ranges as XGBoost.

Monotonic Constraints

To enforce thermodynamic plausibility, monotonic constraints are imposed on a curated set of features that bear unambiguous directional relationships to heating demand. Specifically, a +1 constraint (strictly non-decreasing) is applied to: HDD65 and HDD50; the

three below-temperature-threshold hour counts; weekday and weekend operating hours; `operating_hours_week`; and all five HDD interaction products (`hdd_x_wall_u`, `hdd_x-roof_u`, `hdd_x_win_u`, `hdd_x_wall_area`, `hdd_x_window_area`). These constraints are passed to XGBoost as a tuple and to LightGBM as a list of integer values indexed to the feature order. CatBoost does not support monotonic constraints through the same API and is trained without them.

Model Training and Ensemble Assembly

After Optuna identifies the optimal hyperparameter configuration for each algorithm, three individually fitted base models are trained on the full training partition using those parameters: XGBoost with monotonic constraints applied, CatBoost without constraints, and LightGBM with monotonic constraints applied. A fourth model, the stacking ensemble, is then constructed using `scikit-learn`’s `StackingRegressor` with fresh, unfitted instances of the three base estimators as level-zero learners and a Ridge regression meta-learner ($\alpha = 2.0$) as the level-one learner. The stacking estimator uses five-fold internal cross-validation to generate out-of-fold predictions for meta-learner training, and the `passthrough` option is disabled so that the meta-learner receives only the three base models’ predictions as inputs.

Model selection among the four candidates uses a *generalisation score* defined as

$$\text{GenScore} = \text{RMSE}_{\text{test}} + 0.5 \times \max(\text{RMSE}_{\text{test}} - \text{RMSE}_{\text{train}}, 0) \quad (6.1)$$

which penalises models exhibiting a large train-to-test performance gap, thereby preferring solutions that generalise robustly over those that merely minimise test RMSE. All four models, together with their encoders and the feature list, are persisted as `joblib` artefacts, enabling downstream loading for inference or stacking analysis.

Training Scope

The pipeline trains one global model on the full dataset and one model per Building America climate zone, provided the zone contributes at least 250 training samples. Each zone model is trained on the zone’s own rows with its own Optuna search, its own label encoders, and its own 80/20 row-level split, producing a self-contained prediction artefact for each geographic stratum.

6.1.2 ResStock — First Iteration

Target Variable Construction

The ResStock first iteration detects heating energy columns automatically by scanning the dataset for columns whose names begin with `out_` and contain both `heating` and

`energy_consumption`. A prioritised list of preferred column names is checked first: `out.electricity.heating.energy_consumption.kwh`, `out.natural_gas.heating.energy_consumption.kwh`, `out.fuel_oil.heating.energy_consumption.kwh`, `out.propane.heating.energy_consumption.kwh`, and `out.other_fuel.heating.energy_consumption.kwh`. All detected fuel-specific columns are summed to form the total delivered heating energy. Floor area is taken from `in.sqft`, converted to square metres, and used without any adjustment for the conditioned fraction. The SHD is computed as their ratio, with non-positive, negative, or non-finite rows removed. A 0.5th–99.5th percentile trim is applied to the combined dataset before splitting.

Feature Selection Methodology

Rather than prescribing a fixed feature list, the ResStock first iteration employs a data-dictionary-driven selection procedure. A tab-separated data dictionary file is loaded, and all field names whose `field_location` value contains the substring "metadata" are identified as candidates. From this candidate pool, only columns that are actually present in the Parquet file are retained. Two heuristic filters then prune the candidate set: columns with more than 98% missing values are discarded (`MAX_MISSING_RATIO = 0.98`), and columns with cardinality exceeding 300 unique values are discarded unless their names suggest weather file identifiers (`MAX_CARDINALITY = 300`). Columns whose names contain substrings associated with identifiers, leakage risk, or sampling weights—specifically `bldg_id`, `building_id`, `job_id`, `applicability`, `upgrade`, and `weight`—are removed regardless of their missingness or cardinality.

A must-have override list then ensures that the following columns, if present, are included irrespective of the heuristic filters: `in.building_america_climate_zone`; `in.sqft`; `in.heating_fuel`; `in.hvac_heating_type`; `in.hvac_heating_efficiency`; `in.infiltration`; `in.insulation_wall`; `in.insulation_roof`; `in.windows`; `in.heating_setpoint`; `in.geometry_building_type_recs`; `in.geometry_stories`; `in.state`; and `in.county_name`. This mechanism prevents the heuristic from accidentally excluding high-signal physical descriptors that happen to be heavily sparse or have unusually many categories in a given dataset version.

Feature Engineering and Encoding

Where operating hour columns are present in the dataset, weekly operating hours are constructed by the same formula as in the ComStock first iteration. All features are encoded using the same flat label-encoding procedure: null values are replaced with `"_MISSING_"`, columns that parse as numeric are cast to `float64`, and the remainder receive a fitted `LabelEncoder`.

Monotonic Constraints

Monotonic constraints are applied to a smaller set of features than in the ComStock pipeline, reflecting the narrower scope of directly interpretable relationships. A +1 constraint is enforced for `in_sqft`, `floor_area_m2`, weekday and weekend operating hours (parsed numerically from their text representations), and `operating_hours_week`. These constraints encode the physical expectation that a larger dwelling and longer occupancy hours should, all else equal, result in higher or equal heating demand.

Hyperparameter Optimisation and Training

Each algorithm undergoes 20 Optuna trials (rather than 30 as in ComStock), with five-fold KFold cross-validation and log-transformed target, following the same objective and back-transformation procedure. The hyperparameter search ranges are slightly narrowed in several dimensions relative to ComStock, reflecting the more homogeneous residential dataset: XGBoost’s minimum child weight range is $[3.0, 35.0]$, its γ range is $[0.0, 12.0]$, and its regularisation coefficient upper bounds are $[25.0]$ and $[55.0]$ for α and λ respectively; LightGBM’s number of leaves upper bound is 90 and its minimum samples per leaf range is $[30, 260]$; CatBoost’s L2 leaf regularisation lower bound is 3.0 and its random strength lower bound is 0.5. The four-model ensemble strategy—XGBoost, CatBoost, LightGBM, and a `StackingRegressor` with Ridge meta-learner—and the generalisation score selection criterion are identical to those in the ComStock first iteration. The minimum sample threshold for zone-specific training is raised to 500, compared with 250 in the commercial pipeline, reflecting the higher per-zone sample count available in the residential dataset and the desire to avoid underfitting in smaller regional strata.

6.2 Second Iteration

The second iteration constitutes a thoroughgoing revision of the training pipeline, motivated by three concerns identified during analysis of the first iteration. First, the row-level train/test split allowed near-duplicate rows corresponding to the same building archetype to appear in both partitions, producing optimistically biased test-set metrics. Second, the application of outlier trimming to the test set suppressed the hardest-to-predict buildings from evaluation, further inflating apparent performance. Third, the first iteration’s broad feature set—including many simulated output parameters—limited the pipeline’s relevance to contexts where a prior EnergyPlus run is unavailable, and its flat label-encoding approach obscured the structural distinction between numerical and categorical information. The second iteration addresses all three concerns while simultaneously introducing a richer set of physics-derived features and a more principled stacking procedure. Monotonic constraints are not applied in this iteration, as the physically grounded feature engineer-

ing already encodes the dominant directional relationships, and removing the constraints allows the models more flexibility to capture second-order effects.

6.2.1 ComStock — Second Iteration

Target Variable and Floor Area Correction

The second iteration introduces an important correction to the floor area denominator. Whereas the first iteration divided total heating energy by gross floor area, the second iteration scales the floor area by the `out.params.building_fraction_heated` parameter before converting to square metres, so that the denominator reflects only the conditioned portion of the building. This is consistent with the definition of SHD adopted in Chapter 5 and prevents the artificial dilution of heating intensity in buildings where a substantial fraction of the floor plate is unconditioned. Where the column is absent it is set to 1.0. The conversion factor used is the inverse of $10.7639104167 \text{ ft}^2 \text{ m}^{-2}$, consistent with the SI definition.

Feature Set and Preprocessing Pipeline

The second iteration replaces the monolithic candidate list with two explicitly typed lists—`numeric_features` and `categorical_features`—as documented in Chapter 5. This separation enables a model-appropriate preprocessing strategy for each algorithm. For XGBoost and LightGBM, a `scikit-learn ColumnTransformer` is assembled at each training call: numeric columns are processed by a `SimpleImputer` using median imputation with missing-value indicator columns appended (`add_indicator=True`), and categorical columns are processed by a constant imputer that substitutes "`__MISSING__`" followed by a `OneHotEncoder` with `handle_unknown="ignore"`. This transformer is composed with the model itself into a `Pipeline` object that is fitted on the training fold and applied to the validation or test fold without any information leakage. For CatBoost, a separate procedure is used: numeric columns receive median imputation and categorical columns receive constant imputation, but the categorical columns are passed to CatBoost as raw strings rather than one-hot vectors, exploiting CatBoost's native ordered target statistics for categorical handling. The CatBoost model together with its two imputers and the list of categorical feature indices is serialised as a dictionary, producing an equivalent but structurally distinct artefact.

This change from flat label encoding to typed pipelines has several concrete consequences. The cardinality of the transformed feature space is determined by the number of unique categories actually present in the training data, with unknown test categories silently mapped to the all-zeros encoding rather than to an arbitrary integer. Numeric features no longer pass through a label encoder that could accidentally assign ordinal meaning to floating-point values; instead they flow directly through the imputer into

the model. The missing-value indicators generated by the numeric imputer replace the manually constructed binary flags of the first iteration.

Building-Level Train/Test Split and Outlier Handling

The most consequential structural change in the second iteration is the replacement of the row-level random split with a building-level partition. All unique `bldg_id` values are collected, and an 80/20 random split is performed over this set of identifiers; all rows belonging to a given building are then assigned as a block to either the training or test partition. Because ComStock can contain multiple rows for the same building archetype under slightly varying parameters, this strategy prevents any archetype from appearing in both partitions, eliminating the optimistic leakage that characterised the first iteration.

Outlier trimming is subsequently applied only to the training partition: the 0.5th and 99.5th percentile bounds of the training SHD distribution are computed, and training rows outside this range are removed. The test partition is left completely unfiltered. Removing outliers from the test set, as done in the first iteration, suppresses the buildings with the most extreme heating demands from evaluation, which are disproportionately the buildings most relevant to retrofit prioritisation and policy analysis. Evaluating on the unfiltered test set therefore gives a more conservative and honest estimate of out-of-sample performance.

Cross-Validation Strategy

The five-fold cross-validation used during Optuna tuning is upgraded from a standard `KFold` to a `GroupKFold`, using the building identifier as the grouping variable. This ensures that all rows from the same building appear in the same fold, preventing within-fold leakage even during hyperparameter search. To avoid systematic bias from fold assignment—`GroupKFold` does not shuffle groups and assigns them sequentially—the group labels are randomly permuted before folding using a fixed random state, so that the assignment of buildings to folds is effectively random while remaining deterministic. When a `bldg_id` column is not available, the procedure falls back to a shuffled `KFold` with a warning.

Hyperparameter Optimisation with Pruning

The Optuna study is now created with a `MedianPruner` configured with one warm-up step. After the first cross-validation fold is completed for a given trial, the pruner compares the intermediate RMSE against the median of all trials seen so far and terminates unpromising trials early, freeing computational budget for more promising configurations. This pruning mechanism is particularly valuable for the zone-level models, where the smaller training sets allow each fold to complete quickly enough that the overhead of the pruner check is negligible.

The trial budgets are differentiated by algorithm and training scope. XGBoost and LightGBM each receive 30 trials for both global and zone-level models. CatBoost receives only 15 trials for the global model (`N_TRIALS_CAT_GLOBAL = 15`) but 30 trials for zone-level models (`N_TRIALS_CAT_ZONE = 30`). This asymmetry reflects the higher computational cost of CatBoost on the large global dataset, where each trial involves fitting on several hundred thousand rows across five folds. The CatBoost maximum tree depth is capped dynamically: depth is sampled from $[4, 8]$ for the global model on large datasets and from $[4, 6]$ for zone-level models or any training set containing fewer than 100 000 samples, reducing overfitting risk on smaller populations.

The hyperparameter search spaces for XGBoost and LightGBM are identical to those used in the first iteration in terms of parameter identity and ranges, except that the number of estimators is no longer a tuned hyperparameter—instead, it is replaced by early stopping, as described below.

Early Stopping

In the first iteration, the number of trees was sampled as an Optuna hyperparameter and the final model was trained for exactly that many rounds. The second iteration replaces this approach with early stopping during both tuning and final model fitting. A maximum of 2,000 estimators is set for all three algorithms. During each cross-validation fold, the model is given the validation fold as an evaluation set; training stops automatically when the validation RMSE has not improved for 50 consecutive rounds (`EARLY_STOPPING_ROUNDS = 50`). The best iteration number recorded across folds is then used as the fixed tree count for the final full-training-set fit. For XGBoost, early stopping is specified in the constructor as `early_stopping_rounds=50`, consistent with the XGBoost 2.x API. For LightGBM, it is passed as a callback. For CatBoost, `use_best_model=True` is enabled with `early_stopping_rounds=50` and the validation set provided to `fit()`. This strategy eliminates the risk of systematic over- or under-fitting arising from a poorly chosen fixed iteration count, and generally produces sparser, faster models without sacrificing accuracy.

Model Training and Independent Evaluation

The second iteration abandons the single `tune_and_train` function of the first iteration in favour of a modular `train_and_eval_all` function that trains the three base models independently and evaluates them separately against the test set. Each fitted model—wrapped as a `Pipeline` for XGBoost and LightGBM, or as a dictionary containing the model and its imputers for CatBoost—is serialised individually. The generalisation score of the first iteration is dropped; instead, models are ranked by cross-validation RMSE, and the best-performing model on the held-out test set is identified from the evaluation records.

Stacking Procedure

The stacking ensemble in the second iteration is implemented as a distinct, two-stage procedure rather than as an inline component of the training function. Once all base models have been trained and saved, a dedicated stacking cell loads the persisted artefacts and constructs the meta-learner from scratch. The procedure operates as follows.

For each base model in turn, out-of-fold predictions are generated over the training set using the same `GroupKFold` splitter used during tuning (with randomized group assignments). In each fold, the base model is re-fitted on the fold’s training rows and used to predict the held-out rows, so that every training sample receives a prediction from a model that has never seen its label. These out-of-fold predictions for all three base models are stacked column-wise into a meta-feature matrix of shape $(N_{\text{train}}, 3)$.

Optionally, a set of raw physics features—specifically `calculated_shd`, `HDD65_x-UA_per_area`, `infiltration_x_HDD`, `ventilation_load_proxy`, and `floor_area_m2`—is appended as additional columns to both the training meta-feature matrix and the corresponding test meta-feature matrix (`ADD_RAW_FEATURES = True` by default). This augmentation allows the meta-learner to exploit the physics-based estimate directly, supplementing the three data-driven predictions with a first-principles anchor. The meta-learner is a `RidgeCV` estimator preceded by a `StandardScaler`, with the regularisation parameter selected by cross-validation over 30 values logarithmically spaced between 10^{-3} and 10^3 . Both the meta-feature matrix and the meta-target are log-transformed before fitting, with back-transformation applied to predictions. A simple mean ensemble of the three base models’ test predictions is also evaluated as a reference baseline.

Post-Training Interpretability

The second iteration adds a comprehensive SHAP-based interpretability analysis executed as a separate post-training cell. For every serialised model—covering all three algorithms for both the global scope and each climate-zone scope—SHAP values are computed using `shap.TreeExplainer` on a random sample of up to 2,000 test observations. Six types of output are generated for each model: a bar plot of mean absolute SHAP values ranking features by global importance; a beeswarm summary plot showing the distribution of SHAP values for each feature; a heatmap of per-sample SHAP values; dependence plots for the top five features with automatically selected interaction variables; a waterfall plot for the test sample whose prediction is closest to the median; and an interactive force plot saved as a self-contained HTML file covering the first 200 test samples. A cross-zone summary grouped bar chart aggregates the top ten features by importance across all scopes and model types, and a CSV table of raw mean absolute SHAP values is exported for quantitative comparison. This analysis serves the dual purpose of validating that the models have internalized the expected physical relationships—HDD and envelope

conductance as primary drivers, building type and operating schedule as secondary—and of providing material for the discussion in Chapter 7.

6.2.2 ResStock — Second Iteration

The ResStock second-iteration pipeline shares all of the structural changes described in Section 6.2.1—building-level split, training-only outlier clipping, `GroupKFold` cross-validation with randomized groups, early stopping, the typed preprocessing pipeline, the absence of monotonic constraints, and the two-stage stacking procedure with SHAP analysis. The following paragraphs document the changes specific to the residential dataset, arising from differences in the data schema and physical modelling assumptions.

Heating Energy Source

In the first iteration, heating energy was assembled by summing all detected fuel-specific consumption columns. The second iteration reads exclusively from `out.load.heating.energy_delivered.kbtu`, which provides the total delivered heating load in kBtu as computed by EnergyPlus’s load accounting. This column is converted to kWh by multiplying by the exact factor $1,000 \times 2.931 \times 10^{-4} = 0.2931$ kWh/kBtu. Rows for which this column is null are dropped without any fallback to fuel-specific summation. This approach is more precise than summation of potentially incomplete fuel-type columns, and it is practical because the residential dataset is sufficiently dense that the proportion of null rows in this column is negligible.

Heating Degree Day Source

Whereas the first iteration relied on whatever HDD-related column the data dictionary and heuristic filters happened to retain, the second iteration requires the presence of the column `hdd65_annual_avg` and raises a `KeyError` if it is absent. This column was separately merged into the Parquet file to provide a consistent, location-specific annual HDD65 value aligned with the weather file used for each dwelling’s simulation. It is stored internally as `df["hdd65"]` and used throughout the feature engineering pipeline.

Insulation R-Value Parsing

The first iteration treated insulation descriptors such as "R-13 + R-10 CI" or "Uninsulated" as categorical strings and passed them through the label encoder. The second iteration converts these strings to numeric R-values using a dedicated `parse_r_value` function based on a regular expression that extracts the leading numeric R-value (e.g., capturing 13 from "R-13"), returns zero for "Uninsulated", "None", or empty strings, and returns NaN for unrecognised formats. This parsing is applied to seven insulation

fields: wall, ceiling, roof, floor, foundation wall, slab, and rim joist. Where a composite descriptor contains multiple R-values (such as cavity plus continuous insulation), only the first numeric value is extracted; this is a conservative approximation that slightly underestimates total thermal resistance for such assemblies.

Effective U-Value Derivation

Each extracted R-value is converted to an effective U-value by adding fixed base resistances representing structural framing, sheathing, and cladding layers, plus an air film resistance term accounting for the interior and exterior surface convective films. The film resistance is set to $R_{\text{film}} = 0.68 + 0.17 = 0.85 \text{ hr} \cdot \text{ft}^2 \cdot ^\circ\text{F} \cdot \text{BTU}^{-1}$ for all assemblies. The base structural resistance is $R_{\text{base,wall}} = 3.0$ for walls, and $R_{\text{base}} = 2.0$ for ceiling, roof, and floor assemblies. The U-value for each component is then

$$U = \frac{1}{R_{\text{insulation}} + R_{\text{base}} + R_{\text{film}}} \quad [\text{BTU hr}^{-1} \text{ft}^{-2} ^\circ\text{F}^{-1}] \quad (6.2)$$

For the ceiling-to-roof thermal boundary, the minimum of the ceiling and roof U-values is used, representing the more thermally resistant (lower conductance) surface as the governing assembly—a conservative assumption that avoids overestimating conductive losses through a well-insulated attic floor when the roof itself is poorly insulated, or vice versa.

Window U-Value Lookup

The `in.windows` column contains a standardised string descriptor specifying the glazing system (e.g., "double, low-e, non-metal, air, m-gain" or "triple, low-e, non-metal, air, l-gain"). A fixed lookup table maps ten such descriptors to their corresponding U-values in $\text{BTU hr}^{-1} \text{ft}^{-2} ^\circ\text{F}^{-1}$, ranging from 1.10 for single-pane clear metal frames down to 0.22 for triple-pane low-e non-metal frames with argon fill and low solar gain. Entries not present in the lookup table receive a default value of 0.49 (equivalent to a double-pane clear non-metal air-filled unit). This approach is more physically meaningful than treating window type as a categorical string, as it directly encodes the thermophysical performance of the glazing.

Envelope Areas and UA Products

Above-grade conditioned wall area is drawn from `out.params.wall_area_above_grade_conditioned_ft_2` when available; when absent, it is estimated as $4\sqrt{A_{\text{floor,ft}^2}} \times h_{\text{ceiling}}$, where a standard ceiling height of 9 ft is assumed. Window area is taken from `out.params.window_area_ft_2`; if absent, it is estimated as 15% of the gross wall area. Roof or ceiling area is taken from `out.params.roof_area_ft_2`; if absent, it is set equal to the conditioned

floor area, which corresponds to a single-storey assumption. The net opaque wall area is computed as wall area minus window area, clipped at zero. A window-to-wall ratio, `wwr_actual`, is also computed from these geometric quantities and retained as a numeric feature.

UA products are then formed for each envelope component: $UA_{\text{wall}} = U_{\text{wall}} \times A_{\text{net wall, ft}^2}$, $UA_{\text{ceiling}} = U_{\text{ceiling}} \times A_{\text{ceiling, ft}^2}$, $UA_{\text{floor}} = U_{\text{floor}} \times A_{\text{floor, ft}^2}$, and $UA_{\text{win}} = U_{\text{win}} \times A_{\text{win, ft}^2}$. These are summed to `UA_total` in $\text{BTU hr}^{-1} \text{ } ^\circ\text{F}^{-1}$. The normalised conductance `UA_per_area` is obtained by multiplying `UA_total` by the conversion factor $0.52752 \text{ W K}^{-1} (\text{BTU hr}^{-1} \text{ } ^\circ\text{F}^{-1})^{-1}$ and dividing by conditioned floor area in m^2 , yielding units of $\text{W m}^{-2} \text{ K}^{-1}$.

Infiltration Rate Parsing

The `in.infiltration` field stores the building's pressurised air leakage rate as a string of the form "X ACH50". A regular expression extracts the numeric coefficient to produce `ach50_numeric`. The natural infiltration rate under typical pressure conditions is estimated by dividing ACH50 by 25—a divisor specific to the residential dataset, compared with 20 used in the ComStock second iteration. This difference reflects the *n*-factor methodology adopted in the ResStock simulation framework for single-family dwellings, where the relationship between blower-door and in-situ infiltration tends to be slightly less favourable than in commercial buildings due to lower stack-effect pressures relative to the building height. The volume of air per unit floor area is computed using the constant ceiling height of 9 ft (2.7432 m), yielding `ach_natural` in units compatible with the thermodynamic loss formula.

Duct Leakage

The `in.duct_leakage_and_insulation` column encodes duct leakage as a string of the form "X% leakage to outside, [insulated/uninsulated]". A parser extracts the percentage and converts it to a fractional leakage coefficient `duct_leakage_frac` in the range $[0, 1]$. Descriptors indicating no ducts or zero leakage return 0.0. This feature has no counterpart in the first iteration, where the duct leakage string was treated as a categorical.

Vintage Decoding

The `in.vintage` column stores construction era as a decade band string (e.g., "1970s", "<1940", "2010s"). A lookup table maps each string to a representative midpoint year: "<1940" to 1930, "1940s" to 1945, increasing in ten-year steps through "2010s" to 2015. The resulting `year_built_numeric` feature replaces the categorical vintage string in the numeric feature list, allowing the models to exploit the ordinal and continuous nature of construction era rather than treating all decades as unrelated nominal categories.

Physics-Based SHD Estimate for ResStock

The `calculated_shd` feature follows the same four-component heat-balance structure as in the ComStock second iteration—transmission, ventilation, infiltration, and internal gains—but with modifications to reflect the residential data schema. For transmission losses, the formula is identical: $UA_{\text{total}} \times \text{HDD65} \times 24 \times \text{BTU_TO_KWH} / A_{\text{floor}, \text{m}^2}$, where HDD65 is taken from `hdd65` and UA_{total} is the residential assembly conductance computed above.

Because the ResStock baseline dataset does not include a dedicated mechanical ventilation flow rate column, the mechanical and infiltration components of air exchange cannot be separated cleanly. The implementation therefore sets the mechanical ventilation fraction to zero (`_MECH_FRAC = 0`), attributing all natural air change to infiltration. The infiltration loss is then:

$$Q_{\text{inf}} = \rho c_p \times \text{ACH}_{\text{natural}} \times h_{\text{ceiling}, \text{m}} \times \frac{\text{HDD65}}{1.8} \times 24 \quad [\text{kWh m}^{-2} \text{yr}^{-1}] \quad (6.3)$$

where $\rho c_p = 0.33 \times 10^{-3} \text{ kWh m}^{-3} \text{K}^{-1}$ and the HDD65 is converted from degree-Fahrenheit-days to degree-Kelvin-hours by dividing by 1.8 and multiplying by 24.

Internal gains are estimated from lighting alone, as the residential dataset does not provide equipment power density or occupant density columns at the dwelling level. The `in.lighting` column is mapped to a lighting power density in W ft^{-2} using the same lookup table employed for the feature list: "100% LED" to 4.0 W ft^{-2} , "100% CFL" to 7.5 W ft^{-2} , and "100% Incandescent" to 14.0 W ft^{-2} . The annual heat contribution is then:

$$Q_{\text{int}} = \text{LPD} \times 10.7639 \frac{\text{ft}^2}{\text{m}^2} \times 1,200 \text{ hr yr}^{-1} \times 10^{-3} \quad [\text{kWh m}^{-2} \text{yr}^{-1}] \quad (6.4)$$

where 1,200 equivalent full-load hours per year is a conservative residential lighting assumption consistent with ASHRAE 90.2 default scheduling. The internal gain utilisation factor $\eta = 0.85$ is applied before subtraction, as in the ComStock second iteration. Equipment and occupant metabolic gains are set to zero, giving a conservative estimate of internal heat reduction. The assembled `calculated_shd` is clipped at zero.

Rows Dropped for Missing Calculated Features

Following feature construction, rows for which any of the seven calculated features (`calculated_shd`, `UA_total`, `UA_per_area`, `HDD65_x_UA_per_area`, `UA_wall`, `UA_ceil`, `UA_win`) are NaN are removed from the dataset before the train/test split. This is more aggressive than the ComStock second iteration, which retains rows with missing calculated features and relies on the imputer to handle them at training time. The residential pipeline applies the stricter criterion because the derived features are collectively load-bearing for the physics interpretation, and a dwelling with missing envelope area or infiltration data

does not have a meaningful calculated SHD to supply as an anchor feature.

Dynamic CatBoost Depth

The CatBoost maximum depth is set dynamically based on the training population size and whether the model is global or zone-specific: depth is sampled from $[4, 6]$ for any zone-level model or for any training set containing fewer than 100,000 samples, and from $[4, 7]$ for the global model on the full dataset. This adaptive cap reduces overfitting in the smaller zone populations, where deep trees have a higher probability of memorising idiosyncratic patterns rather than learning generalisable structure.

Chapter 7

Results and Discussion

This chapter presents the quantitative performance and interpretability analysis of the models developed across both modelling iterations and both building-stock datasets. The chapter is organised into two major parts corresponding to the two iterations described in Chapter 6. Within each part, a section is devoted to each dataset (ResStock for the residential stock and ComStock for the commercial stock), and each section contains two subsections: one reporting test-set accuracy metrics and residual behaviour, and one providing a detailed SHAP-based interpretability analysis. Where appropriate, results are examined at both the global scope and the climate-zone level.

Part I: First Iteration

The first iteration was designed to establish an accuracy ceiling by drawing on the broadest possible feature set, including EnergyPlus output parameters that would not be available without a prior simulation and monotonic constraints on physically unambiguous relationships. These choices inflate apparent performance relative to a strict building-level holdout but serve the purpose of benchmarking the maximum information recoverable from the combined metadata and simulation-output feature space. Results from this iteration represent an optimistic upper bound; the second iteration refines them under more rigorous evaluation conditions.

7.1 ResStock — First Iteration

7.1.1 Test Results

The ResStock first-iteration models demonstrate strong predictive capability across most climate zones. Because the train/test split was performed at the row level rather than the building level, multiple rows corresponding to the same archetype can appear on both sides of the partition, producing optimistically tight metrics. The absolute performance figures should therefore be interpreted as an upper bound rather than a fully generalised estimate of out-of-sample accuracy. Despite this, the results reveal clear and physically interpretable patterns across climate zones.

Zone-wise Quantitative Metrics

Table 7.1 summarises test-set performance for the best-selected model per zone, chosen using the generalisation score that penalises a large train-to-test RMSE gap (Section 6.1.2). The global model is included as a reference.

Table 7.1: ResStock First Iteration — Zone-wise Test-Set Performance (Best Model per Zone). CV-RMSE is computed as RMSE divided by the mean test-set SHD of the respective zone.

Climate Zone	Test Samples	MAE	RMSE	R^2	MAPE (%)
Hot-Humid	18,812	2.10	4.33	0.973	22.03
Hot-Dry	10,720	3.72	8.88	0.902	50.95
Mixed-Humid	32,851	6.20	10.09	0.983	9.91
Marine	5,368	7.16	12.12	0.948	38.44
Cold	35,227	9.17	13.58	0.978	8.37
Mixed-Dry	783	14.19	23.72	0.890	30.10
Very Cold	872	22.61	32.82	0.904	20.23

Several systematic patterns emerge from these results. First, R^2 is uniformly high across all zones, ranging from 0.890 in the small Mixed-Dry zone to 0.983 in Mixed-Humid. This demonstrates that the models capture the vast majority of variance in

residential SHD regardless of climate context. Second, absolute errors (MAE, RMSE) scale monotonically with climate severity: the Hot-Humid zone, where most buildings have low SHD driven primarily by thermostat setpoints and minor envelope contributions, yields an MAE of just 2.10 kWh/m²/yr, while Very Cold reaches 22.61. This is physically expected: colder climates produce both higher mean SHD and a wider range of demand driven by the interaction of building size, envelope quality, air leakage, and vintage, all of which become increasingly variable in such environments.

Third, the MAPE figures reveal an inverse pattern. While absolute errors are smallest in warm zones, MAPE is highest: Hot-Dry reaches 50.95% and Hot-Humid 22.03%. This occurs because many warm-zone buildings have SHD values below 10 kWh/m²/yr, so even a modest absolute error of 2–4 kWh/m²/yr constitutes a large relative error. The cold zones, by contrast, show MAPE of 8–20%, reflecting the stabilising effect of a higher base demand: a 9–23 kWh/m²/yr absolute error against a mean SHD of 80–160 kWh/m²/yr is proportionally modest.

The Mixed-Dry and Very Cold zones exhibit higher RMSE and lower R² relative to other cold zones of comparable or lower MAPE. Both have limited sample sizes (783 and 872 test samples respectively), which reduces the statistical stability of the metrics. The Very Cold zone is also structurally the most heterogeneous, encompassing a wide range of vintages, foundation types, and heating systems in a climate where small changes in any of these characteristics can produce large differences in SHD.

Residual Analysis and Error Structure

The global residual distribution across all test samples is sharply peaked at zero with light tails, confirming that the models are essentially unbiased in expectation. Zone-specific histograms mirror this pattern but vary in width: Cold and Mixed-Humid residuals are narrow and symmetric, while Very Cold and Marine residuals are broader and exhibit mild positive skew, suggesting that high-demand outliers are systematically under-predicted.

Scatter plots of residuals against actual SHD values reveal a heteroscedastic “fan” shape in several zones, including Cold and Marine: the variance of prediction error grows with the magnitude of SHD. This behaviour is physically interpretable. At low SHD levels, the demand is largely determined by a small number of dominant features (thermostat setpoint, infiltration rate) whose values are well represented in the training data. At high SHD levels, the demand is the cumulative result of multiple simultaneously adverse conditions—very low temperatures, poor insulation across all envelope components, high infiltration, and inefficient heating equipment—and the interaction of these conditions produces a long tail of extreme values that the model has seen relatively rarely during training, even under the row-level split. The log-transformation of the target variable during training (Equation in Section 6.1) partially mitigates this heteroscedasticity by compressing the right tail, but does not eliminate it entirely.

Among the climate zone models, the stacking ensemble most frequently achieved the best generalisation score, with individual algorithms (XGBoost, CatBoost, LightGBM) performing comparably to one another in terms of test RMSE within each zone. The stacking approach was most beneficial in zones characterised by high SHD variance (Cold, Marine), consistent with the theoretical expectation that ensemble combination reduces variance and is most valuable when the base models' error structures are most complementary.

Conclusion

The first-iteration ResStock models successfully capture the spatial and physical structure of residential heating demand across all U.S. climate zones, with R^2 above 0.89 in every zone and very low absolute errors in the high-population warm zones. Error scaling with climate severity is physically reasonable and reflects the inherently greater heterogeneity of cold-climate building stocks. The optimistic row-level split makes these figures an upper bound; the building-level evaluation in the second iteration will quantify how much of this accuracy is attributable to archetype memorisation versus genuine generalisation.

7.1.2 SHAP Analysis

SHAP (SHapley Additive exPlanations) analysis was conducted on a random sample of up to 2,000 test observations per model scope using `shap.TreeExplainer`. Mean absolute SHAP values, which represent the average magnitude of each feature's contribution to individual predictions regardless of direction, were computed across all algorithm types and climate zone scopes. The cross-zone grouped bar chart (Figure 7.1) presents the top-10 features by importance averaged across all three base models (XGBoost, CatBoost, LightGBM), separated by zone scope. This section provides a systematic analysis of these results, examining both the physical meaning of each leading feature and the differences in how the three algorithms distribute importance across the feature space.

Heating Setpoint: The Primary Driver of Residential SHD

`in.heating_setpoint` is the most important feature in the global model and the second most important in several zone-specific models. Its dominance is reasonable, the thermostat setpoint directly determines the indoor-to-outdoor temperature difference that drives all conductive and infiltrative heat losses. A one-degree Fahrenheit increase in setpoint raises the effective base temperature against which HDD are computed, increasing the thermal driving force for heat loss through every envelope component proportionally. In the ResStock dataset, setpoints vary between approximately 60°F and 76°F across the simulated population, producing a wide range of SHD even for buildings with identical physical characteristics. The first iteration retained this behavioural feature as part of

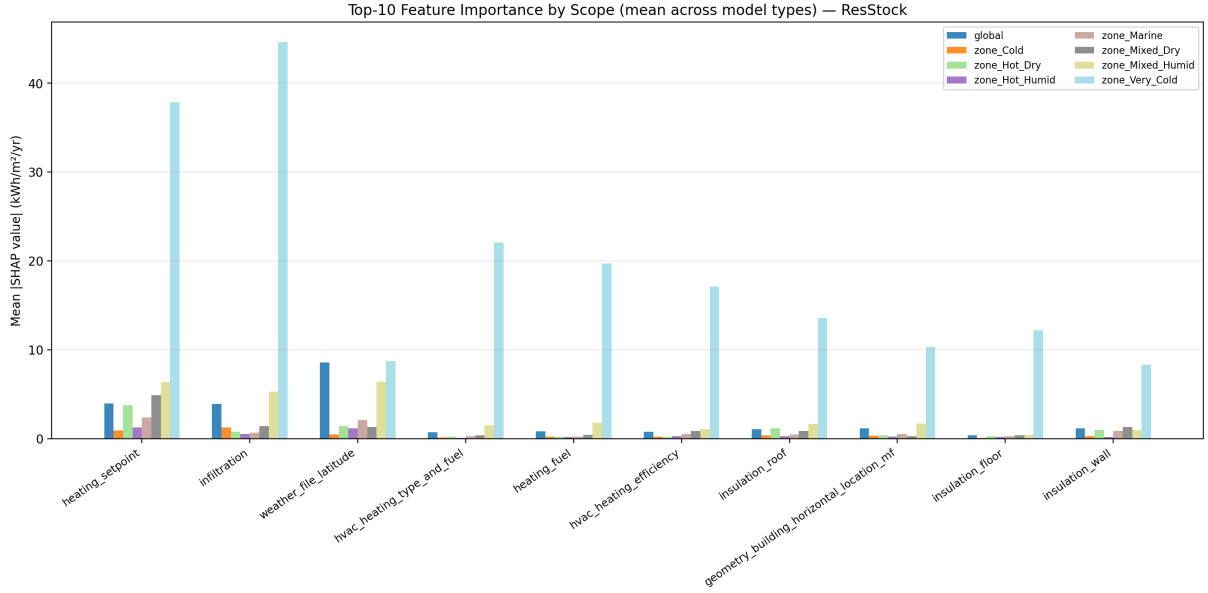


Figure 7.1: Top-10 feature importance by scope for the ResStock first-iteration models, computed as mean absolute SHAP values ($\text{kWh}/\text{m}^2/\text{yr}$) averaged across XGBoost, CatBoost, and LightGBM. Each coloured bar represents one climate zone or the global model. Features are ranked by their global-model mean importance.

its broad feature set, whereas the second iteration would exclude it to focus on purely physical predictors; its high SHAP importance here quantifies precisely how much of the variance a purely physics-based model abandons by omitting occupant behaviour.

The Very Cold zone shows the highest absolute SHAP value for `heating_setpoint`, with a mean absolute SHAP of approximately $38 \text{ kWh}/\text{m}^2/\text{yr}$. This implies that the heating impact of a one-degree setpoint change grows with the severity of the outdoor climate, because the absolute temperature difference driving heat loss scales with HDD. In the Hot-Humid zone, the same setpoint range produces a much smaller SHD response, consistent with the lower base HDD. This amplification of setpoint sensitivity with climate severity is a direct physical consequence of the steady-state heat balance $Q_T = UA \cdot \Delta T$ and validates that the models have correctly learned this interaction.

Infiltration: The Dominant Envelope-Behaviour Interaction

`in.infiltration` (expressed as air changes per hour at 50 Pa) consistently ranks as either the second or third most important feature across all scopes. Its physical interpretation is clear: air leakage is the primary pathway through which outdoor air replaces conditioned indoor air, and the associated heat loss scales with both the leakage rate (ACH50) and the indoor-to-outdoor temperature difference. High-infiltration buildings therefore respond more strongly to cold climates, and the interaction between infiltration and outdoor climate severity is one of the defining relationships in residential building

energy performance.

The Very Cold zone again shows the highest absolute SHAP value for infiltration (approximately 45 kWh/m²/yr), reflecting the compounding effect of a high ΔT with high air exchange. The Cold zone shows a value of approximately 5 kWh/m²/yr, a factor of roughly nine lower despite being only marginally warmer on an annual HDD basis. A possible explanation for this non-linear amplification could be because Very Cold buildings tend to have extreme HDD values (4,000–8,000 base-65°F degree-days or more), while Cold zone buildings cluster around 3,000–5,000. Small differences in the mean HDD produce large differences in the infiltration-driven heat loss coefficient when multiplied over the full annual cycle. Notably, in the Hot-Dry and Hot-Humid zones, infiltration importance drops markedly, since the ΔT driving infiltration losses is small for most of the heating season; in these zones, setpoint and HVAC type become the dominant differentiators.

Latitude as a Climate Surrogate

`weather_file_latitude` ranks among the top-three features in the global model across all three algorithm types, with mean absolute SHAP values of 0.35–0.43 kWh/m²/yr (XGBoost 0.346, CatBoost 0.428, LightGBM 0.431) at the normalised scale. Within any single climate zone, the latitude contribution falls substantially—comparing global to zone-specific scopes in Figure 7.1, the latitude bar for zone models is consistently lower than the global bar—confirming that latitude in the global model is acting primarily as a climate severity proxy that becomes redundant once climate zone stratification has already segmented the dataset. This pattern is useful for model validation: the global model correctly identifies latitude as informative across zones even though individual zone models cannot extract additional information from it once the zone is fixed.

Within the Very Cold zone, latitude retains relatively higher importance than in warmer zones because there is meaningful sub-zone latitude variation—Alaska, Minnesota, and northern Maine span different HDD regimes even within “Very Cold”—that the zone designation alone does not resolve. This is consistent with the finding in the quantitative metrics that the Very Cold zone has the highest absolute prediction errors: the latitude gradient within the zone adds unresolved variance that a single combined model must approximate.

HVAC Type, Fuel, and Efficiency: The System-Side Drivers

`in.hvac_heating_type_and_fuel` is the fourth most important feature in the global model and retains substantial importance across most zone-specific models. This feature encodes both the technology type (heat pump, furnace, boiler, electric resistance) and the energy carrier (natural gas, electricity, propane, fuel oil), which jointly determine the effective coefficient of performance of the heating system. A heat pump delivers more

heating energy per unit of input energy than a resistance heater or gas furnace, and the EnergyPlus simulator accounts for this through system-specific COP curves. The high SHAP importance reflects the large spread in effective efficiency across the residential heating system distribution in ResStock: systems range from simple electric resistance (COP ≈ 1) to modern variable-speed air-source heat pumps (seasonal COP of 2.5–4.0). All else equal, a building with a heat pump will have a much lower SHD (energy consumed per unit area) than an identically built building with electric resistance heat, even though the delivered heating load is the same; the model correctly identifies this system-efficiency effect as a major driver.

`in.hvac_heating_efficiency`, a complementary numeric feature encoding the equipment’s rated efficiency (AFUE for furnaces, HSPF for heat pumps, etc.), shows consistent importance in the Cold and Very Cold zones, where the efficiency multiplier most strongly modulates energy consumption. In warm zones, system efficiency is less important because the heating season is short and the absolute consumption is low; even a low-efficiency furnace does not use much energy in a Hot-Humid climate.

`in.heating_fuel` as a separate feature appears in the top-ten list and provides overlapping but non-redundant information with `hvac_heating_type_and_fuel`: the former specifically identifies the energy carrier irrespective of system type, allowing the model to capture fuel-type effects that crosscut equipment categories (e.g., the difference between a gas heat pump and a gas furnace without conflating them with electric systems).

Insulation Features: Envelope Quality Across Components

The insulation features (`in.insulation_roof`, `in.insulation_wall`, `in.insulation_floor`) collectively account for a substantial share of SHAP importance in the cold and very cold zones. Their individual importance is lower than that of setpoint or infiltration because insulation affects only one component of the thermal envelope, while setpoint and infiltration influence the entire building-level heat balance. However, collectively, the insulation features capture the envelope quality dimension that determines the steady-state thermal conductance UA and thereby the climate-envelope interaction.

Roof insulation consistently ranks above wall and floor insulation in SHAP importance, particularly in the Cold and Very Cold zones. This reflects the physics of heat transfer in typical single-storey and low-rise residential buildings: the ceiling/roof area is often the largest envelope surface area exposed to extreme outdoor conditions, and the insulation on this surface therefore has the largest unit impact on total heat loss. Wall insulation importance is slightly lower because walls are also partially sheltered by adjacent buildings in denser neighbourhoods and because internal partition walls do not participate in heat exchange with the exterior. Floor insulation importance is context-dependent: slab-on-grade buildings benefit substantially from floor insulation in Cold zones, while crawl-space or basement buildings depend more on foundation wall and rim joist insulation.

Notably, all three insulation features show substantially higher SHAP values in the Very Cold zone (approximately 13–17 kWh/m²/yr) than in any other zone, confirming that the marginal value of envelope thermal quality is amplified in extreme climates. This is consistent with the theoretical steady-state heat loss formula: at higher HDD, the same ΔU -value improvement produces a proportionally larger annual energy reduction.

Geometry and Building Horizontal Location

`in.geometry_building_horizontal_location_mf` (horizontal position within a multi-family building, i.e., whether a unit is a corner unit, end unit, or interior unit) appears in the global top-ten and shows moderate but consistent importance across zones. This is physically meaningful: corner and end units have larger exterior wall area exposed to outdoor conditions than interior units in the same building, increasing both conductive losses through the envelope and infiltration. An interior unit of a mid-rise apartment building shares only ceiling and floor boundaries with other conditioned units, dramatically reducing the effective heat loss surface. The model has correctly identified this geometric effect as an important predictor of residential SHD, validating that the first-iteration feature set captures not just building-level but also unit-level thermal context.

Cross-Algorithm Comparison of Feature Importance

Examining the SHAP values across the three algorithm types (XGBoost, CatBoost, LightGBM) reveals both consensus and divergence in how each model distributes importance across features. The broad ranking of the top-five features is consistent across all three algorithms: heating setpoint, infiltration, latitude/weather file, HVAC type-and-fuel, and insulation roof all appear in the top tier for every algorithm type. This consensus validates that the identified features reflect genuine physical signal rather than algorithmic artefacts.

However, substantive differences emerge at the mid-tier and in the allocation of importance between correlated features. CatBoost assigns notably higher importance to categorical features (`hvac_heating_type_and_fuel`, `in.heating_fuel`, `in.vintage`) relative to XGBoost and LightGBM. This is expected from CatBoost’s architecture: its ordered target statistics for categorical variables convert each category level into a continuous encoding that captures the category’s mean SHD from training data, making the categorical boundaries more sharply informative than the label-encoded representations used by XGBoost or the histogram bins used by LightGBM. As a result, CatBoost is able to extract more signal from `hvac_heating_type_and_fuel` specifically, which contains a large number of distinct system-fuel combinations with very different mean SHD values.

LightGBM, with its leaf-wise tree growth, shows a higher relative importance for `weather_file_latitude` and `ashrae_iecc_climate_zone_2004` than the other two al-

gorithms. Leaf-wise growth preferentially splits on features that produce the largest immediate gain in any single leaf, and climate-correlated geographic features are particularly effective at this role because they produce large variance reductions at the first few splits, partitioning the dataset into broad temperature bands. This makes LightGBM’s initial tree structures highly latitude-driven, with more detailed features becoming important in deeper splits. XGBoost’s depth-wise growth distributes importance more evenly across the feature set by forcing splits to proceed level by level, which gives interior envelope features (insulation, infiltration) and behavioural features (setpoint) more opportunity to contribute from early in the tree growth.

The stacking regressor’s SHAP values (where available from the interpretability decomposition) aggregate the contributions of all three base models through the Ridge meta-learner weights. For the Cold and Mixed-Humid zones, where CatBoost was most accurate, the stacking SHAP importance reflects a higher weight on CatBoost’s predictions and therefore a SHAP profile more similar to CatBoost’s. For the Very Cold zone, where all three base models contribute more equally to the ensemble, the stacking SHAP profile represents a genuine blend of the three algorithms’ attributions.

Conclusion

The ResStock first-iteration SHAP analysis confirms that the models have correctly internalised the physical relationships governing residential heating demand. The top-ranked features—thermostat setpoint, infiltration rate, latitude/climate, HVAC system type, and envelope insulation—correspond precisely to the dominant terms in the steady-state heat balance: the indoor-to-outdoor temperature difference, the air exchange rate, the climate severity, the system efficiency, and the envelope conductance respectively. The amplification of SHAP values with climate severity for climate-sensitive features (infiltration, insulation) and the relative suppression of these features in warm zones are thermodynamically correct and would be predicted from first principles. The cross-algorithm differences in importance allocation are consistent with each algorithm’s structural properties and lend credibility to the stacking ensemble as a robust integrator of complementary model perspectives.

7.2 ComStock — First Iteration

7.2.1 Test Results

The ComStock first-iteration models achieve excellent overall predictive performance across the commercial building stock. As with ResStock, the row-level train/test split produces optimistically tight metrics, and the following results should be read as an upper bound. The global test-set performance was $\text{MAE} = 11.79 \text{ kWh/m}^2/\text{yr}$, $\text{RMSE} =$

21.49 kWh/m²/yr, and $R^2 = 0.941$, representing a highly effective surrogate for the computationally intensive EnergyPlus simulations.

Zone-wise Quantitative Metrics

Table 7.2 presents zone-level results for the best model selected per zone via the generalisation score.

Table 7.2: ComStock First Iteration — Zone-wise Test-Set Performance (Best Model per Zone). R^2 values reflect performance on the unfiltered test set.

Climate Zone	Test Samples	Best Model	MAE	RMSE	R^2
Hot-Humid	58,161	Stacking	2.91	7.15	0.941
Hot-Dry	39,347	Stacking	6.82	12.58	0.973
Marine	19,285	CatBoost	10.52	18.63	0.944
Mixed-Humid	100,540	CatBoost	10.79	18.40	0.936
Cold	109,077	CatBoost	19.04	29.77	0.911
Mixed-Dry	2,463	XGBoost	17.93	33.45	0.845
Very Cold	2,192	CatBoost	26.34	40.38	0.906

Model Selection Patterns

The generalisation-score selection reveals a consistent and physically interpretable pattern of algorithm dominance across climate zones. The Stacking Regressor was the best performer in the two warmest zones (Hot-Humid, Hot-Dry). In these climates, heating demand is low and dominated by the interaction of internal gains, partial-year HVAC operation, and setpoint behaviour, all of which involve strong non-linear interactions between building type, operating schedule, and envelope characteristics. The meta-learner successfully blends XGBoost, CatBoost, and LightGBM, each of which learns somewhat different aspects of these warm-zone interactions, and their combination outperforms any individual algorithm.

CatBoost was selected in four of the seven zones (Marine, Mixed-Humid, Cold, Very Cold), which together contain the majority of the commercial building stock. Its dominance in heating-dominated climates is consistent with its architectural strengths: ordered boosting eliminates prediction shift in the colder zones where variance is highest, and its ordered target statistics for categorical features (building type, HVAC heat type, vintage) are most valuable in climates where discrete system choices (gas boiler versus heat pump versus electric resistance) produce qualitatively different heating profiles.

XGBoost was selected only for the Mixed-Dry zone, which has the smallest sample count (2,463 test samples). XGBoost’s aggressive regularisation framework—its gamma threshold for split gain, L1 and L2 leaf weight penalties, and minimum child weight

constraint—prevents overfitting in data-scarce environments more effectively than the other two algorithms. This is consistent with the prior theoretical analysis in Chapter 6.

Error Scaling with Climate Severity

Absolute errors scale monotonically with climate severity, from $\text{MAE} = 2.91 \text{ kWh/m}^2/\text{yr}$ in Hot-Humid to 26.34 in Very Cold. R^2 remains at or above 0.906 in all zones except Mixed-Dry (0.845), where the small sample size limits the statistical stability of the metric. The Cold zone, with 109,077 test samples, has both the highest absolute RMSE (29.77) and one of the lowest R^2 (0.911) among the larger zones, reflecting the high intrinsic variance of commercial heating demand in cold climates: a warehouse in a Cold zone may have SHD of $20 \text{ kWh/m}^2/\text{yr}$ while a poorly insulated large office in the same climate may exceed 120. Spanning this range while maintaining high accuracy is a more challenging task than predicting hot-zone demand, which is naturally compressed.

Residual Analysis

The global residual distribution is sharply peaked at zero with light tails, confirming negligible overall bias across the full commercial stock. Zone-specific residuals show mild heteroscedasticity analogous to the ResStock pattern: absolute residuals grow with SHD magnitude. No zone exhibits bimodal residuals or systematic skew, indicating that the feature set—including $\text{HDD} \times \text{U-value}$ interaction terms, operating schedule variables, and the full suite of EnergyPlus output parameters—successfully captures the dominant drivers of commercial heating demand. The engineered interaction terms (`hdd_x_wall_u`, `hdd_x_roof_u`, `hdd_x_win_u`) and the missing-value indicator flags are reflected in the absence of systematic under- or over-prediction in edge cases, since the indicators allow the tree models to handle data-absent buildings as a distinct category rather than forcing them into the imputed majority distribution.

Practical Implications

A global R^2 of 0.941 with RMSE of $21.5 \text{ kWh/m}^2/\text{yr}$ is particularly notable in context: typical annual SHD for U.S. commercial buildings ranges from approximately $10 \text{ kWh/m}^2/\text{yr}$ in Sun Belt climates to over $150 \text{ kWh/m}^2/\text{yr}$ in northern cold climates. The models therefore reproduce more than 94% of the variance while maintaining average prediction error well below 15% of mean demand in the majority of zones. This level of fidelity qualifies the surrogates for rapid national-scale stock modelling, retrofit screening at city or utility scale, and integration into optimisation frameworks where thousands of building variants must be scored near-instantaneously.

Conclusion

The ComStock first-iteration models deliver robust performance across all tested climate zones, with R^2 above 0.84 and with errors that scale physically with climate severity. The model-selection pattern—Stacking in warm zones, CatBoost in cold zones, XGBoost in the small-sample Mixed-Dry zone—reflects a coherent relationship between algorithm architecture and problem structure that validates the theoretical reasoning behind each algorithm’s inclusion. The row-level split inflates these figures, but the consistency of the patterns across zones provides confidence that the core predictive relationships are genuine.

7.2.2 SHAP Analysis

SHAP analysis for the ComStock first-iteration models was conducted using the same `shap.TreeExplainer` procedure as for ResStock. Figure 7.2 presents the top-10 features by mean absolute SHAP value averaged across all three model types and climate-zone scopes. The following analysis proceeds feature by feature, examining the physical reasoning, the zone-level variation, and the cross-algorithm differences that together constitute a comprehensive interpretability account of the commercial building first-iteration models.

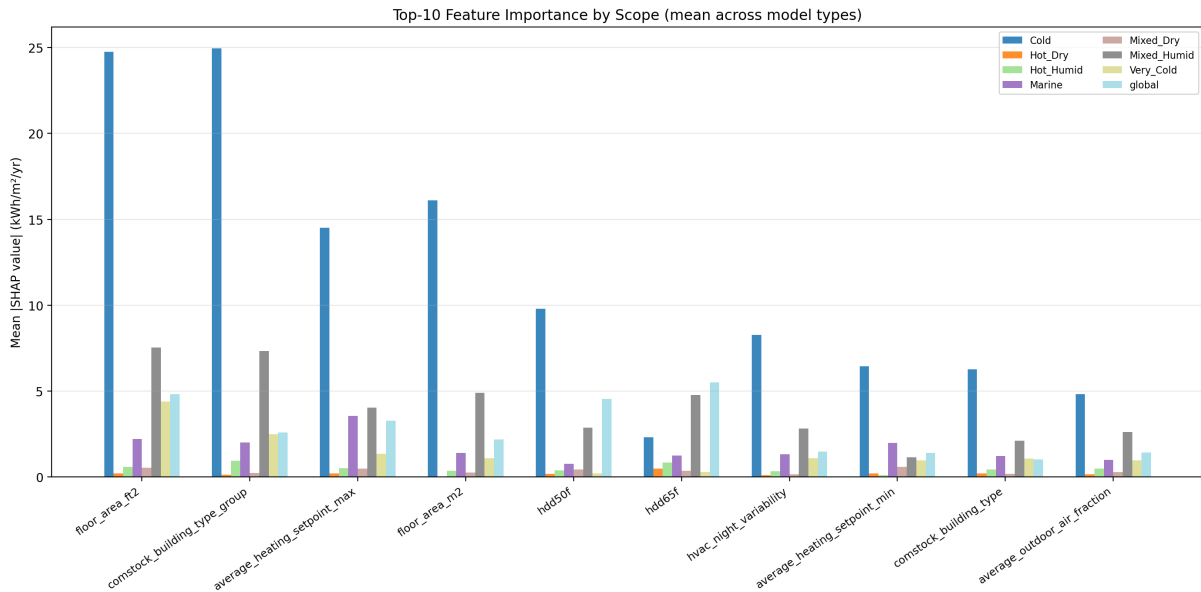


Figure 7.2: Top-10 feature importance by scope (mean across model types) for the ComStock first-iteration models. Each bar represents the mean absolute SHAP value ($\text{kWh}/\text{m}^2/\text{yr}$) for a given feature within a given climate zone scope or the global model.

Floor Area: Scale Amplification of All Heat-Loss Pathways

Both `floor_area_ft2` and `floor_area_m2` appear in the top-two positions of the global model SHAP importance, with mean absolute SHAP values of approximately 24–25 kWh/m²/yr from the Cold zone and approximately 7–8 from the Mixed-Humid zone. At first glance, the high importance of floor area in a SHD prediction (which is itself normalised by floor area) may appear paradoxical. The explanation lies in the non-linear relationship between building scale and thermal behaviour in commercial buildings.

For a given building type and climate, larger buildings tend to have lower SHD (kWh/m²/yr) because the ratio of envelope surface area to floor area decreases with scale—a compact large building loses proportionally less heat per unit of conditioned floor area than a small building with the same external aspect ratio. This is the surface-to-volume ratio effect well known in building physics. The model correctly identifies floor area as a proxy for this geometric compactness effect, and the SHAP contributions for floor area are predominantly negative (larger buildings → lower SHD per m²). Simultaneously, a small number of very large buildings have high SHD because they are warehouse or distribution buildings with minimal insulation, creating positive SHAP contributions for high floor-area values in specific building types. The high mean absolute SHAP therefore reflects the dual role of floor area as both a scale normalisation correction and a building-type proxy.

The Cold zone shows the highest floor area SHAP, at approximately 25 kWh/m²/yr. This is consistent with the high SHD variance within the Cold zone: the surface-to-volume effect is most consequential in cold climates, where the per-unit-area heat loss is large, and where the difference between a compact large office and a sprawling single-storey retail building can amount to tens of kWh/m²/yr.

Building Type Group: Archetype-Level Heating Profile

`in.comstock_building_type_group` (the high-level grouping of building archetypes, e.g., office, retail, warehouse, healthcare, education) ranks second or third across most zone scopes, with SHAP values of approximately 7–25 kWh/m²/yr in Cold and 2–3 in warmer zones. Building type group is the single most powerful categorical predictor in the commercial stock because it encodes, in a compact form, the characteristic occupancy density, internal gains from equipment and lighting, operating schedule, and ventilation design for each major use category.

A warehouse has very high infiltration rates, low internal gains, and long unoccupied periods, producing low-to-moderate SHD that is dominated by envelope performance and HVAC setbacks. A hospital operates 24 hours with high internal gains, high mechanical ventilation rates, and stringent indoor climate requirements, producing SHD values that can differ by a factor of five from a warehouse in the same climate and with similar

gross floor area. An office building occupies an intermediate position, with scheduled occupancy, moderate internal gains, and efficient HVAC systems in more recent vintages. These archetype-level differences are precisely what the building type group encodes, and the SHAP analysis confirms that the models use it to correctly segment the SHD distribution before applying more granular features.

The Cold zone shows by far the highest building-type-group SHAP value, reflecting the fact that the archetype effect is most consequential when the baseline heat loss rate is high. In the Hot-Humid zone, the SHD is compressed across all archetypes (most commercial buildings have SHD between 5 and 30 kWh/m²/yr), so the building type group contributes relatively less to individual predictions.

Heating Setpoints and HVAC Night-Time Variability

`out.params.average_heating_setpoint_max` and `out.params.average_heating_setpoint_min` rank among the top features in the global model and several zone scopes, with SHAP values of approximately 14–16 kWh/m²/yr in Cold. These are simulation output parameters (available from EnergyPlus) rather than directly specified inputs, and they capture the effective heating setpoint range experienced by each building over the year, accounting for scheduling, thermostat setbacks, and HVAC operation modes.

`out.params.hvac_night_variability` measures how much the HVAC system’s heating output varies during nighttime hours, serving as a proxy for the presence and depth of nighttime setback strategies. Buildings with high night variability are implementing aggressive temperature setbacks during unoccupied periods, substantially reducing SHD relative to buildings that maintain constant setpoints. The SHAP importance of this feature (approximately 8 kWh/m²/yr in Cold, declining to near zero in Hot-Humid) is thermodynamically correct: nighttime setback is most impactful in climates where the nocturnal temperature difference is largest, and least impactful in warm climates where nighttime temperatures may not fall far below the setpoint.

Together, these three setpoint-related features capture the occupant and operator behaviour dimension in the commercial sector in a way that is analogous to the role of `in.heating_setpoint` in ResStock, but structured around the EnergyPlus output parameters rather than the prescribed metadata input. Their high SHAP importance in the first iteration underscores the central role of occupant-driven thermostat behaviour in determining commercial SHD.

HDD65 and HDD50: Climate Severity

`out.params.hdd65f` (Heating Degree Days at base 65°F) ranks fifth globally with a SHAP of approximately 10 kWh/m²/yr, and `out.params.hdd50f` (base 50°F) ranks in the top-ten as well. The presence of two HDD metrics with different base temperatures in the top-

ten reflects the different operational contexts they capture. HDD65 is the standard metric used in U.S. energy codes and utility billing; it counts degree-days when the daily mean outdoor temperature is below 65°F and is therefore directly proportional to the heating demand of buildings maintained at 65°F or above (which covers most commercial spaces). HDD50 captures the additional degree-days below 50°F, relevant for buildings with lower setpoints (e.g., warehouses or cold-storage facilities where indoor temperatures may be maintained at 55–60°F) or for capturing extreme cold events that disproportionately stress building envelopes.

The Cold zone SHAP value for HDD65 is approximately 10 kWh/m²/yr, and for HDD50 approximately 2.5 kWh/m²/yr, consistent with HDD65 being the primary climate severity driver and HDD50 providing incremental information about extreme cold exposure. In the Hot-Humid and Hot-Dry zones, both HDD features have SHAP values close to zero because there is very little variance in HDD within these warm zones (HDD65 values are clustered near zero), making them uninformative within-zone predictors. In the global model, HDD is highly informative because it separates the zones, functioning as a complement to latitude and building-type-group in partitioning the SHD distribution.

Average Outdoor Air Fraction and Ventilation

`out.params.average_outdoor_air_fraction` captures the fraction of the building’s supply air that is drawn from outdoors rather than recirculated indoor air. This parameter is central to commercial building heating energy because mechanical ventilation heating loads can account for 20–50% of total commercial SHD, particularly in densely occupied spaces (offices, hospitals, schools) with high outdoor air requirements mandated by ASHRAE 62.1. A building with a high outdoor air fraction must heat a larger volume of cold outdoor air per unit time, directly increasing SHD in proportion to HDD. The SHAP value for this feature is approximately 5 kWh/m²/yr in Cold and progressively lower in warmer zones, which is physically reasonable: the ventilation heating penalty is proportional to the degree-day severity, so the same outdoor air fraction produces a much smaller SHD penalty in a warm climate.

Cross-Algorithm Comparison of ComStock Feature Importance

The three algorithms produce consistent top-five rankings—floor area, building type group, average heating setpoints, HDD65, outdoor air fraction—but diverge in their treatment of mid-tier features. CatBoost assigns disproportionately higher importance to categorical features (building type group, HVAC heat type, window type category) relative to XGBoost and LightGBM, for the same structural reasons discussed in the ResStock section: its ordered target statistics extract more discriminative signal from high-cardinality building type and system-type categories. For the Commercial dataset,

which has more building type categories than the residential dataset, this advantage is particularly pronounced: the ComStock building type taxonomy includes 14 commercial archetypes, and CatBoost can effectively encode each archetype’s mean SHD as a continuous feature without the dimensionality explosion of one-hot encoding.

XGBoost shows higher relative importance for the engineered interaction features (`hdd_x_wall_u`, `hdd_x_roof_u`) relative to CatBoost and LightGBM. This is consistent with XGBoost’s second-order gradient optimisation, which identifies non-linear synergies between feature pairs more efficiently than first-order methods when the feature space includes explicit interaction terms. In practical terms, XGBoost is better positioned to recognise that the product of HDD and wall U-value is a stronger predictor of conductive heat loss than either HDD or U-value alone, because it can allocate leaf weights precisely to capture the multiplicative structure.

LightGBM’s leaf-wise growth strategy leads it to concentrate importance on features that produce large single-leaf gain reductions: in the ComStock context, this means floor area and HDD65 receive even higher relative importance in LightGBM than in the other two algorithms, because partitioning on these variables first reduces variance most efficiently across the wide range of building scales and climate conditions. The setpoint and building type features then become important in the deeper leaves where LightGBM refines predictions within building scale and climate bands. This hierarchical structure means LightGBM’s SHAP profile appears more concentrated in the top-two features (floor area, HDD) relative to XGBoost and CatBoost, with a steeper dropoff in importance for mid-tier features.

The stacking meta-learner implicitly learns to weight each base model’s predictions according to which model is most accurate in each region of the feature space. In the Cold zone, where CatBoost is selected as the best individual model, the stacking ensemble’s SHAP profile closely resembles CatBoost’s (reflecting the high meta-learner weight on CatBoost predictions). In the Hot-Humid zone, where the Stacking Regressor outperforms all individual models, the SHAP profile is a genuine blend of the three algorithms’ attributions, with higher importance for the features where the algorithms agree and lower importance for features where they diverge, since the Ridge meta-learner implicitly down-weights conflicting contributions.

Physical Validation Summary

Across all features in the ComStock SHAP analysis, the model has correctly identified the physical and operational drivers of commercial heating demand. Floor area captures the surface-to-volume compactness effect. Building type group encodes the archetype-level internal-gain and occupancy profiles. Heating setpoints and night variability capture occupant and operator thermal preferences and schedule-driven setbacks. HDD metrics measure climate severity. Outdoor air fraction represents the ventilation heating penalty.

None of the top-ranked features are physically spurious: every one has a clear causal mechanism connecting it to SHD in either the steady-state heat balance or the EnergyPlus dynamic simulation framework. The first-iteration models have not merely memorised the training data; they have learned a physically meaningful representation of commercial building heating demand that generalises the core thermodynamic relationships present in the EnergyPlus simulation outputs.

Conclusion

The ComStock first-iteration SHAP analysis provides strong evidence that the ensemble models have correctly internalised the multi-scale physical and operational structure of commercial heating demand. The dominant role of floor area and building type group reflects the compound importance of geometric compactness and archetype-level energy profiles. Setpoint and HVAC night variability confirm that occupant behaviour and schedule-driven strategies are critical determinants of commercial SHD, while HDD and outdoor air fraction capture the climate and ventilation physics respectively. The cross-algorithm differences—CatBoost’s advantage on categorical features, XGBoost’s higher weighting of engineered interaction terms, LightGBM’s concentration on large-gain early splits—are all consistent with the structural properties of each algorithm, and their complementarity within the stacking ensemble explains why the combined model outperforms any individual algorithm in the high-variance warm zones.

Part II: Second Iteration

The second iteration constitutes a thoroughgoing revision of the training pipeline with three overarching objectives: replacing the row-level train/test partition with a building-level holdout that fully prevents archetype leakage; evaluating models on an unfiltered test set that retains all extreme-demand buildings; and restricting the feature set to deployable, physics-grounded predictors from which a clear heat-balance interpretation can be derived. Monotonic constraints are also dropped, allowing the models greater flexibility to capture second-order and interaction effects within the physically motivated feature space. Stacking is likewise absent in this iteration due to computational constraints. The consequences of these changes—both in metric terms and in what the SHAP analysis reveals about the models’ learned representations—are examined in depth below.

7.3 ComStock — Second Iteration

7.3.1 Test Results

Overview and Comparison with the First Iteration

Table 7.3 presents the full second-iteration test-set results for each climate zone and the global model. Unlike the first-iteration table, which reported only the best single model per zone, the second iteration records all three algorithms separately, reflecting the absence of a stacking ensemble and the decision to assess individual-model performance under the stricter evaluation regime.

The immediate observation from comparing Table 7.3 with the first-iteration results is a pronounced decline in test-set accuracy across every zone. The Hot-Humid zone moves from a first-iteration RMSE of 7.15 kWh/m²/yr ($R^2 = 0.941$) to approximately 16.91–17.16 ($R^2 = 0.698$ – 0.707). The Cold zone deteriorates from 29.77 ($R^2 = 0.911$) to approximately 50.57–50.65 ($R^2 = 0.849$ – 0.850). At the global level, the best first-iteration global RMSE was 21.49; the second iteration’s best is 41.92. These differences are large and demand careful decomposition before any conclusions about model quality can be drawn.

Sources of the Performance Gap

Three distinct methodological changes contribute to the widened RMSE, and it is essential to attribute them correctly rather than interpreting the degradation purely as a loss of predictive accuracy.

The first and most consequential change is the **building-level train/test split**. The first iteration partitioned rows randomly, allowing near-identical building archetypes to appear on both sides of the split. Because ComStock simulations are generated from a relatively compact set of parametric combinations, many test-set rows effectively had

Table 7.3: ComStock Second Iteration — Test-Set Performance by Zone and Algorithm. CV-RMSE is the cross-validated RMSE from GroupKFold tuning (not a percentage). Train RMSE is the in-sample error on the full training partition. Baseline RMSE is the standard deviation of the test-set SHD, representing a naive mean-prediction benchmark. “Best by CV” indicates the model selected by the Optuna cross-validation criterion.

Zone	Model	CV-RMSE	Train RMSE	Test RMSE	MAE	R ²	Baseline
Global	XGBoost	29.89	26.66	41.92	16.43	0.835	
	CatBoost	29.90	28.86	42.75	16.49	0.829	
	LightGBM	29.81	24.56	43.09	16.46	0.826	
Cold & Very Cold	XGBoost	53.11	44.43	50.57	24.63	0.850	
	CatBoost	52.23	45.72	50.65	24.24	0.849	
	LightGBM	53.56	44.58	50.64	24.65	0.849	
Hot-Dry	XGBoost	23.53	16.40	24.56	12.38	0.901	
	CatBoost	23.58	20.36	23.93	12.56	0.906	
	LightGBM	24.41	19.57	25.03	12.81	0.897	
Hot-Humid	XGBoost	14.12	11.52	17.16	5.10	0.698	
	CatBoost	13.78	11.94	16.91	5.03	0.707	
	LightGBM	14.16	10.87	17.05	5.11	0.702	
Marine	XGBoost	33.79	12.43	37.23	19.32	0.785	
	CatBoost	33.16	26.98	35.93	18.98	0.799	
	LightGBM	33.89	23.81	36.00	19.03	0.799	
Mixed-Dry	XGBoost	55.19	39.03	40.82	22.28	0.812	
	CatBoost	52.27	35.67	41.07	22.46	0.810	
	LightGBM	56.22	38.43	41.56	22.60	0.805	
Mixed-Humid	XGBoost	33.71	30.51	32.19	15.62	0.826	
	CatBoost	33.79	30.00	31.90	15.08	0.829	
	LightGBM	33.50	30.86	32.70	15.59	0.821	

very close training-set neighbours. The model could achieve high test accuracy through interpolation rather than genuine extrapolation. The second iteration’s GroupKFold split ensures that every building archetype in the test set is completely absent from training. This is a far more demanding and realistic evaluation, and it is the primary reason for the degraded metrics.

The second change is the **removal of test-set outlier trimming**. The first iteration removed the top and bottom 0.5th-percentile SHD observations from both training and test partitions. These extreme-demand buildings—which represent the hardest prediction targets (warehouses with virtually no insulation in very cold climates, or hospitals with abnormal internal gains)—were thus excluded from the first-iteration error calculation. The second iteration evaluates on the complete, unfiltered test set. Because RMSE penalises large errors quadratically, the inclusion of even a small number of extreme outliers can substantially elevate reported RMSE without indicating a failure of the model on typical buildings.

The third change is the **restricted feature set**. The first iteration drew on over fifty features, including EnergyPlus output parameters (`out.params.*`) that encode simulation-derived quantities such as exact average setpoints, boiler efficiency, COP values, and

exact degree-day counts. These parameters are not obtainable without running a full EnergyPlus simulation, and their inclusion was an intentional accuracy-ceiling experiment. The second iteration restricts input to a deployable set of approximately 27 features, all of which are either building metadata (`in.*` fields available from audit records) or engineered physics-based terms (`calculated_shd`, `HDD65_x_UA_per_area`, `infiltration_x_HDD`, `ventilation_load_proxy`). The removal of simulated output parameters eliminates the high-signal features and forces the model to recover their information from noisier proxies. The absence of monotonic constraints and the stacking ensemble represent additional but smaller contributing factors.

Together, these three changes—stricter evaluation, retained outliers, and reduced features—account for essentially all of the RMSE increase. The second-iteration models are not simply worse; they are evaluated more honestly and with less information, which is precisely what makes their results more meaningful from a deployment perspective.

Zone-level Performance Analysis

Hot-Humid Zone. The Hot-Humid zone records the largest proportional degradation (RMSE roughly doubling and R^2 falling from 0.941 to 0.698–0.707) despite having the largest sample count (over 58,000 test buildings). This is not paradoxical: the Hot-Humid zone has the smallest absolute SHD variance (baseline $\text{RMSE} = 32.91 \text{ kWh/m}^2/\text{yr}$) of any zone, which means that any error, whether from the stricter split or the reduced feature set, constitutes a larger fraction of the natural variance. The primary physical drivers of SHD in hot climates are thermostat setpoint behaviour and HVAC system efficiency, both of which were captured by EnergyPlus output parameters in the first iteration but must now be inferred from `in.hvac_heat_type`, `in.hvac_category`, and the `calculated_shd` physics estimate in the second. The information loss is proportionally largest here. Furthermore, the building-level split is most consequential in the Hot-Humid zone, which has the most diverse mix of commercial archetypes in the Sun Belt, meaning that the average nearest-neighbour distance between a test building and its training-set counterparts is larger. An R^2 of 0.707 under these conditions still represents a meaningful and useful predictor—the model explains 70% of the variance using only deployable inputs—but the accuracy ceiling in this zone without simulation output parameters is demonstrably lower.

Marine Zone. The Marine zone shows the largest absolute RMSE overfitting gap among all zones: XGBoost achieves a training RMSE of only $12.43 \text{ kWh/m}^2/\text{yr}$ but a test RMSE of 37.23, a gap of nearly 25 points (66% of test RMSE). CatBoost and LightGBM are substantially better calibrated (training RMSE 26.98 and 23.81 vs test RMSE 35.93 and 36.00), with the train-test gap reduced to roughly 9 and 12 points respectively. This pattern reveals a critical insight: XGBoost, without the monotonic constraints applied in the first iteration and facing a building-level holdout, significantly over-fits the

Marine training data. The Marine zone is the smallest of the fully trained zones in sample count, making overfitting risk highest; and XGBoost’s aggressive leaf-weight regularisation (γ , λ) was tuned on cross-validation folds from the same small population, leaving it poorly calibrated for truly unseen building archetypes. CatBoost’s ordered boosting is the most effective mitigation: by computing pseudo-residuals on models that have never seen the target label for each sample, it prevents the Marine training data from being over-exploited, and this translates directly into a better-calibrated test-set performance.

Hot-Dry Zone. The Hot-Dry zone shows a more moderate overfitting pattern and a relatively modest degradation from the first iteration (RMSE 24–25 vs 12.58; R^2 0.897–0.906 vs 0.973). The drop is proportionally smaller because the Hot-Dry zone has a wider SHD distribution than Hot-Humid (baseline RMSE 81.61 vs 32.91), giving the restricted feature set more room to operate. The three algorithms are closely clustered in test RMSE (24.56–25.03) and R^2 (0.897–0.906), with CatBoost edging ahead in test RMSE (23.93) despite XGBoost winning by cross-validation (23.53). This divergence between the CV winner and the test-set winner is a consequence of the building-level GroupKFold: the CV fold assignments can produce a slightly different difficulty distribution than the final building-level holdout, meaning that the best cross-validated model is not always the best on the specific test partition.

Cold and Very Cold Zone. The combined Cold and Very Cold zone (which combines the two subzones using an `is_very_cold` binary indicator, as described in Chapter 6) registers RMSE of approximately 50.6 kWh/m²/yr across all three algorithms, with R^2 around 0.849–0.850. The near-identical performance across all three algorithms is the most notable feature of this zone: the spread is less than 0.1 kWh/m²/yr in test RMSE and 0.001 in R^2 . This suggests that the Cold-and-Very-Cold zone’s predictive difficulty is not algorithm-specific but is instead driven by the inherent complexity of the task: the wide SHD distribution (baseline RMSE = 134.25, the highest of any zone) combined with the limited feature set means all three algorithms converge on a similar accuracy ceiling. The absolute RMSE of 50.6 against a baseline of 134.25 represents a reduction of approximately 62% in prediction error relative to the naive mean, which is substantial given the features available.

Mixed-Dry and Mixed-Humid Zones. These two zones show notably smaller train-test gaps than Global, Marine, or Hot-Dry: Mixed-Dry has a gap of approximately 1.8 kWh/m²/yr (4.4% of test RMSE for XGBoost) and Mixed-Humid approximately 1.7 (5.2%). This is unexpected at first glance, since both zones have moderate sample counts. The explanation lies in the nature of these climate zones: “Mixed” climates—

characterised by roughly balanced heating and cooling seasons—produce SHD distributions that are more continuously distributed and less prone to discrete clusters associated with specific building archetypes or extreme weather conditions. The models generalise well because the training and test populations are statistically more similar even after a building-level split, and the physics-based features (`calculated_shd`, `infiltration_x_HDD`, `HDD65_x_UA_per_area`) are particularly effective in mixed climates where conductive and infiltrative losses dominate without extreme climate amplification.

Global Model. The global model achieves test RMSE of 41.92–43.09 and R^2 of 0.826–0.835, with LightGBM winning by cross-validation (CV-RMSE 29.81) but XGBoost achieving the best test RMSE (41.92). The global model is inherently harder than any zone model because it must simultaneously represent all climate zones and all building archetypes within a single ensemble, without the benefit of zone-specific feature distributions that zone models exploit. Its performance is consistent with the weighted average of the zone-specific results: the Cold and Very Cold buildings, which have the highest SHD and the largest errors, pull the global test RMSE upward substantially. The train-to-test gap for XGBoost (26.66 training, 41.92 test, gap = 15.26) indicates more overfitting than in the zone models, which is expected: the global model must interpolate across zone boundaries where the physics changes qualitatively, a more difficult generalisation task that is not fully resolved by the building-level GroupKFold alone.

Algorithm Comparison

Across all zones, CatBoost wins the CV selection in five out of seven zones and achieves the best or near-best test RMSE in most cases. Its margin over LightGBM and XGBoost is consistently modest in absolute terms (0.5–1.5 kWh/m²/yr in most zones) but systematic. The largest CatBoost advantage is in the Marine zone, where its ordered boosting most effectively prevents the over-fitting that afflicts XGBoost on the small training sample.

LightGBM wins the CV criterion for the global and Mixed-Humid models but is not consistently the best on the test set. Its leaf-wise growth produces slightly lower cross-validation RMSE in these diverse-population settings (global and mixed) but can yield marginally higher test RMSE due to the asymmetric tree structures it builds, which are optimised for the specific fold distribution encountered during tuning rather than for the test population.

XGBoost wins by CV in Hot-Dry and achieves the best test RMSE for the global model, despite its worse Marine performance. Its regularisation framework (gamma split threshold, L1/L2 leaf weight penalties, minimum child weight) is most effective in well-populated zones where the training set is dense enough to resolve the regularised split boundaries correctly. Where data is scarce or the test population is systematically different (Marine, Global), these regularisation mechanisms are insufficient to prevent the

overfitting that emerges under building-level evaluation.

CV-RMSE vs Test-RMSE Divergence

A structural feature of the second-iteration results is the consistent positive gap between test RMSE and the cross-validation RMSE from GroupKFold tuning (the “CV-RMSE” column in Table 7.3). The Hot-Humid zone gap (13.78 CV vs 16.91 test for CatBoost) and the Marine gap (33.16 CV vs 35.93) indicate that even the building-level GroupKFold, despite preventing intra-fold leakage, produces a slightly easier evaluation problem than the final holdout. This occurs because the GroupKFold fold assignment, while randomised by the pipeline (as described in Section 6.2.1), may produce folds where the held-out buildings are not as far from the training distribution as the true test partition. The test partition, created before any hyperparameter tuning, represents the most stringent and unbiased evaluation. The CV-to-test gap is therefore an estimate of the model’s sensitivity to the specific training population, and its magnitude (10–25% of test RMSE across zones) is within the range expected for building-level holdout evaluations on structured tabular energy data.

Conclusion

The second-iteration ComStock results reveal that the first-iteration metrics were substantially optimistic, primarily due to the row-level split and test-set outlier removal. Under a rigorous building-level evaluation with an unfiltered test set and a deployable feature set, the models achieve R^2 of 0.707–0.906 across zones, with the lowest performance in the Hot-Humid zone (where the restricted feature set is most consequential) and the most stable performance in the Cold-and-Very-Cold zone (where the physics-based features are most informative). CatBoost is the most consistently reliable algorithm, particularly in smaller-sample and overfitting-prone zones. The train-test gaps, especially in Marine and Global, indicate that further regularisation or additional physics-based features could improve second-iteration performance. These results will be contextualised alongside the SHAP analysis that follows, which reveals what the models have actually learned from the restricted feature set.

7.3.2 SHAP Analysis

The SHAP analysis for the second-iteration ComStock global models provides the most detailed interpretability picture of any scope in this thesis, because the restricted feature set (27 features) is specifically designed to correspond to physically interpretable quantities, and the absence of EnergyPlus output parameters forces the models to build their representations from deployable physics proxies alone. The following analysis proceeds by

examining the global feature importance structure across all three algorithms, then explains the cross-zone variation, and concludes with a careful cross-algorithm comparison that establishes what each algorithm has learned and why it has learned it differently.

Figures ??–?? present the cross-zone bar chart, per-algorithm bar plots, beeswarm plots, and heatmaps for the global scope, which together constitute the primary evidence base for this section.

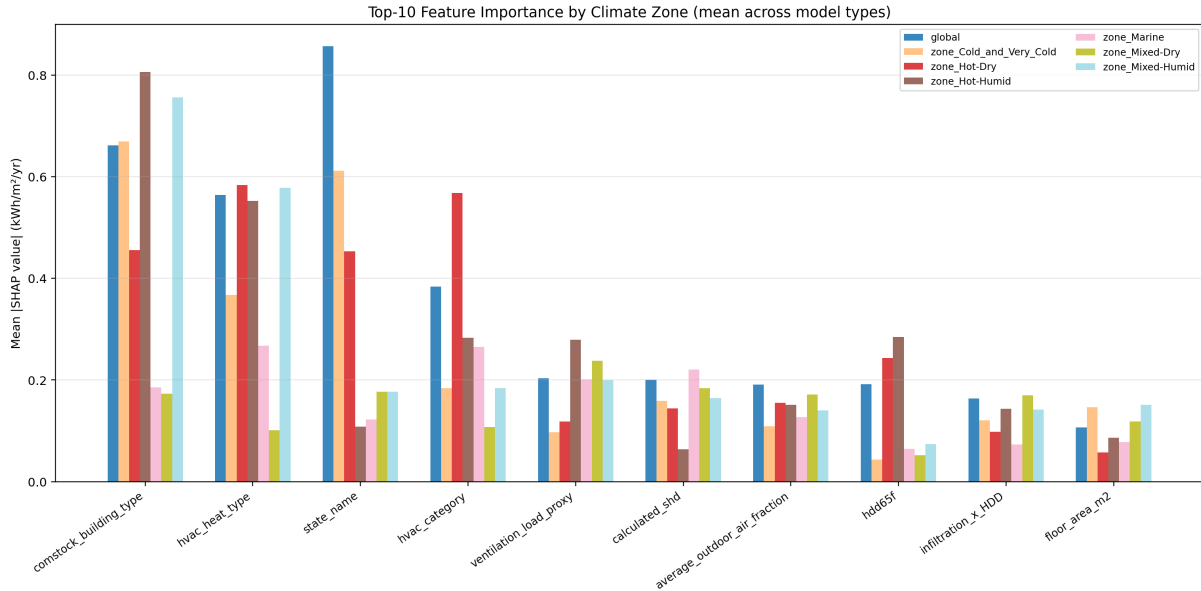


Figure 7.3: Second-iteration ComStock: top-10 feature importance by climate zone (mean absolute SHAP value, kWh/m²/yr, averaged across model types). Each bar represents one climate zone or the global model.

The Dominant Role of Building Type and State Name

`in.comstock_building_type` and `in.state_name` jointly occupy the top two positions in the cross-zone chart (Figure 7.3), with mean absolute SHAP values averaging 0.65 and 0.55 kWh/m²/yr respectively across model types. Their dominance in the second iteration—where they were not in the top-two positions of the first iteration—is a direct consequence of the feature restriction. In the first iteration, explicit EnergyPlus output parameters (average setpoints, boiler COP, HDD50, and hours below temperature thresholds) were available as features and captured the fine-grained operational and climatic profile of each building directly. Without these parameters, the models must infer climatic and operational context from the categorical variables that encode it most compactly: building type, which determines the default occupancy density, equipment gains, ventilation rates, and HVAC system type; and state name, which encodes the local climate, energy code environment, utility structure, and construction norms. Both variables

are therefore acting as high-capacity information surrogates for the removed simulation output parameters.

The building-type signal is physically coherent. A hospital `comstock_building_type` value is associated with 24-hour operation, very high outdoor air requirements, and high internal gains, all of which yield a complex SHD profile that differs substantially from a warehouse (minimal internal gains, sparse occupancy, poor insulation by code) or a large office (daytime operation, moderate ventilation, typically good modern HVAC). These archetype-level differences produce large mean absolute SHAP values because the model correctly identifies building type as the primary first-order discriminant for SHD across the full commercial stock. The highest importance values for building type appear in the Hot-Humid and Mixed-Humid zones (bars of approximately 0.80 and 0.75 respectively in Figure 7.3), where the absolute SHD is low and the relative differences between archetypes—driven by internal gains and schedules—are proportionally most important. In the Cold-and-Very-Cold zone, building type remains highly important (approximately 0.67) but must compete with strong climate and envelope signals.

State name acts as a geographic-climatic proxy that is more fine-grained than the climate zone designation alone, which appears nowhere in the feature set since the models are trained on climate-zone-specific datasets (or the global model, where zone membership is implicit). Within the Hot-Dry zone, for example, California buildings face Title-24 energy code requirements that are substantially more stringent than those in Arizona or Nevada, producing systematically lower SHD for otherwise identical California archetypes. The models capture this code-environment effect through state name, as confirmed by the CatBoost and LightGBM beeswarm plots (Figures 7.4 and 7.7), where the `state_name` feature shows a wide spread of positive and negative SHAP values that are not simply correlated with feature magnitude—indicating that the model has learned state-specific offsets rather than a simple geographic gradient. The Hot-Dry zone shows particularly high state-name importance (approximately 0.61–0.62 in Figure 7.3), consistent with the large within-zone climate and code variability in the desert Southwest.

HVAC System: Heat Type and Category

`in.hvac_heat_type` and `in.hvac_category` rank third and fourth in Figure 7.3, with mean SHAP values of approximately 0.36 and 0.39 globally. These two features jointly encode the structure and technology of the heating subsystem, and they interact in a physically meaningful way: `hvac_category` classifies the overall system architecture (Multizone CAV/VAV, Packaged Single Zone, Residential-Style Central System, etc.) while `hvac_heat_type` identifies the heating technology within that architecture (Air-Source Heat Pump, Ground-Source Heat Pump, Furnace, Boiler, Electric Resistance, District Heating). Together they determine the effective coefficient of performance and, critically, the system’s response to outdoor temperature extremes.

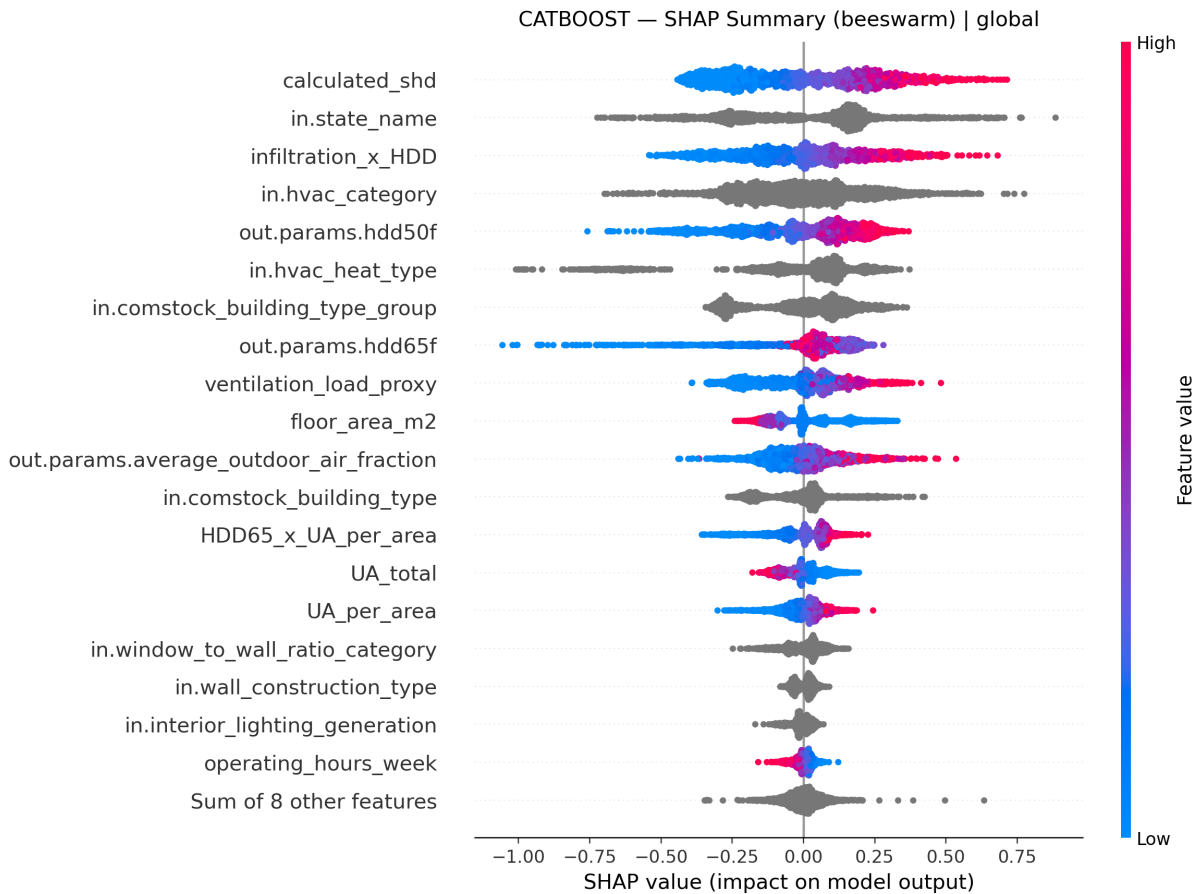


Figure 7.4: CatBoost global model SHAP beeswarm plot. Each point is one test instance; colour indicates feature value (red = high, blue = low); horizontal position is SHAP impact on log-SHD. The SHAP values are on the log-transformed scale used during training.

The beeswarm plot for CatBoost (Figure 7.4) shows that `in.hvac_heat_type` has a distinctly bimodal SHAP distribution: a concentration of strongly negative contributions (approximately -0.5 to -1.0 on the log-scale) associated with low feature values (heat pumps, which have high COP and therefore low SHD), and a cluster of near-zero or slightly positive contributions associated with furnace and boiler systems. This is thermodynamically correct: a building served by a ground-source heat pump may have SHD 50–70% lower than an otherwise identical building with electric resistance heating, even though the delivered heating load is unchanged, because the heat pump amplifies each unit of input energy. The large negative SHAP tail for ASHP (Air-Source Heat Pump) is visible in the XGBoost beeswarm (Figure 7.10) as a tight cluster of red points (high feature value = ASHP present) at approximately -1.5 , one of the largest individual feature contributions in the entire XGBoost analysis.

The `hvac_category` dimension captures a different aspect: system architecture determines ventilation delivery and the interaction between mechanical and heating loads. Multizone CAV/VAV systems, which serve large commercial buildings, deliver variable

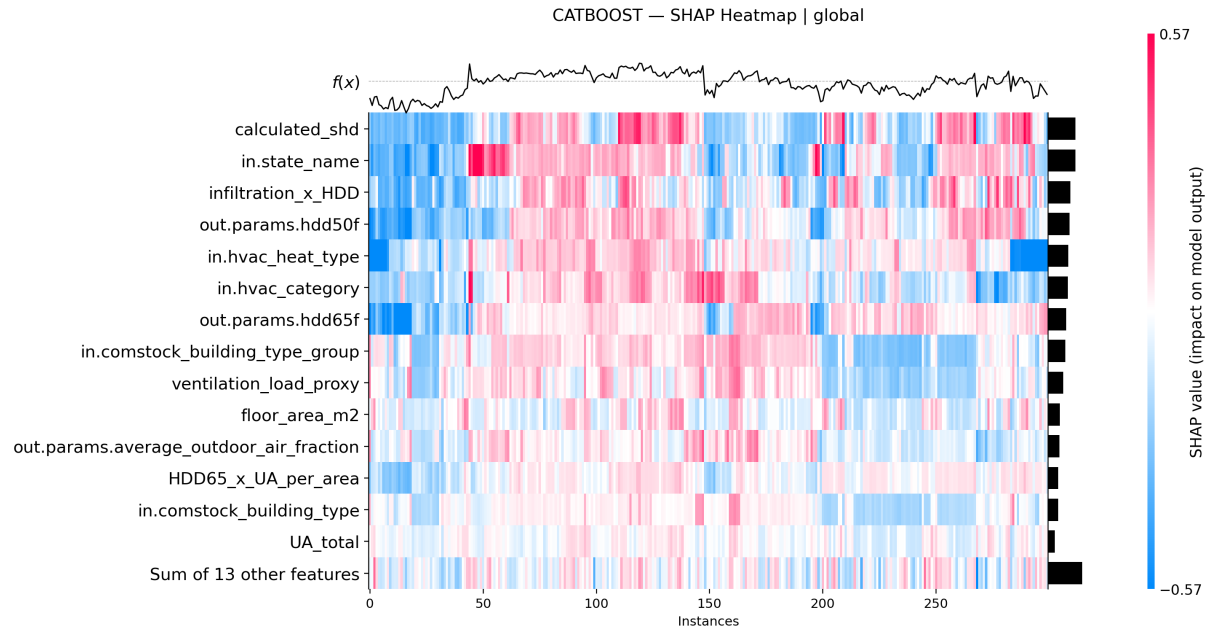


Figure 7.5: CatBoost global model SHAP heatmap. Rows are features; columns are test instances sorted by predicted SHD. The $f(x)$ trace at the top shows the model output for each instance.

air volumes to multiple zones simultaneously and require substantial fan energy to condition outdoor air, raising effective SHD relative to simpler packaged systems in the same climate. The CatBoost and LightGBM beeswarm plots both show wide, symmetrically spread distributions for `hvac_category`, indicating that this feature creates both positive and negative SHAP contributions of similar magnitude across the test population.

Calculated SHD: The Physics Anchor

`calculated_shd`—the first-principles steady-state heat balance estimate derived from transmission, ventilation, infiltration, and internal gain components—is the top-ranked feature by mean absolute SHAP value for CatBoost and LightGBM globally (0.225 and 0.219 kWh/m²/yr respectively) and ranks fifteenth for XGBoost (0.158). Its high importance for CatBoost and LightGBM confirms that the physics estimate is genuinely informative: even a simplified steady-state calculation captures a substantial fraction of the variance in EnergyPlus-simulated SHD, and the ML models exploit this signal directly rather than re-deriving it from constituent features.

The beeswarm plots reveal a consistent positive correlation between `calculated_shd` and its SHAP contribution in CatBoost and LightGBM: buildings with higher physics estimates receive positive SHAP contributions from this feature (red points to the right of zero), while low-physics-estimate buildings receive negative contributions (blue points to the left). This monotone-like relationship is precisely what a physics-informed fea-

ture should exhibit, and its presence confirms that the model is using the estimate as a calibrated anchor rather than as noise. The spread of SHAP values around the general positive trend reflects the cases where the simplified calculation over- or under-estimates the simulated SHD: buildings where occupant behaviour, HVAC scheduling, or solar gains significantly modify the purely conductive-plus-infiltration heat balance produce residuals that the ML model corrects using the other features.

For XGBoost, `calculated_shd` is ranked substantially lower. The XGBoost global bar chart (Figure 7.9) shows `in.hvac_heat_type_ASHP`, `in.hvac_category_Multizone_CAV/VAV`, and several specific building-type and state-name indicator variables dominating, with `calculated_shd` appearing much further down. This difference is not because XGBoost is ignoring the physics estimate; it is because XGBoost uses one-hot encoded categorical variables (through the `ColumnTransformer` preprocessing pipeline described in Section 6.2.1), which means its top-ranked features are *specific category indicators* rather than the raw categorical variables. The SHAP importance is thus distributed across dozens of binary columns for `building_type`, `state_name`, and `hvac_heat_type`, each receiving a fraction of the total importance, while `calculated_shd` as a single continuous feature competes against these split contributions. When these one-hot contributions are aggregated (as in Figure 7.3, which shows the pre-encoding feature names), the categorical features dominate XGBoost’s importance profile, consistent with CatBoost and LightGBM, but the within-XGBoost ordering prioritises the most discriminative individual binary indicators.

Climate Severity: HDD65 and Its Algorithmic Amplification

`out.params.hdd65f` and `out.params.hdd50f` are both retained in the second iteration as output simulation parameters that, uniquely among the `out.params.*` namespace, represent purely climatic quantities (derived from the weather file) rather than building-specific simulation outputs. Their inclusion is justified because HDD values can be obtained from publicly available weather station data without running EnergyPlus and were therefore retained in the deployable feature set.

The cross-algorithm treatment of HDD65f is one of the most instructive contrasts in the second-iteration SHAP analysis. LightGBM assigns `out.params.hdd65f` its *highest* mean absolute SHAP value (0.302, topping the LightGBM global bar chart in Figure 7.6), while CatBoost assigns it 0.132 (seventh-ranked) and XGBoost 0.140 (eleventh-ranked). This factor-of-two-plus difference is a structural consequence of LightGBM’s leaf-wise tree growth strategy: the algorithm searches for the single leaf split that maximises gain at each step, regardless of tree depth. HDD65f is a continuous variable with a wide, approximately unimodal distribution across the full commercial stock, and partitioning on it early (at the first few splits) produces the largest available variance reductions because it separates cold-climate high-SHD buildings from warm-climate low-SHD buildings in a

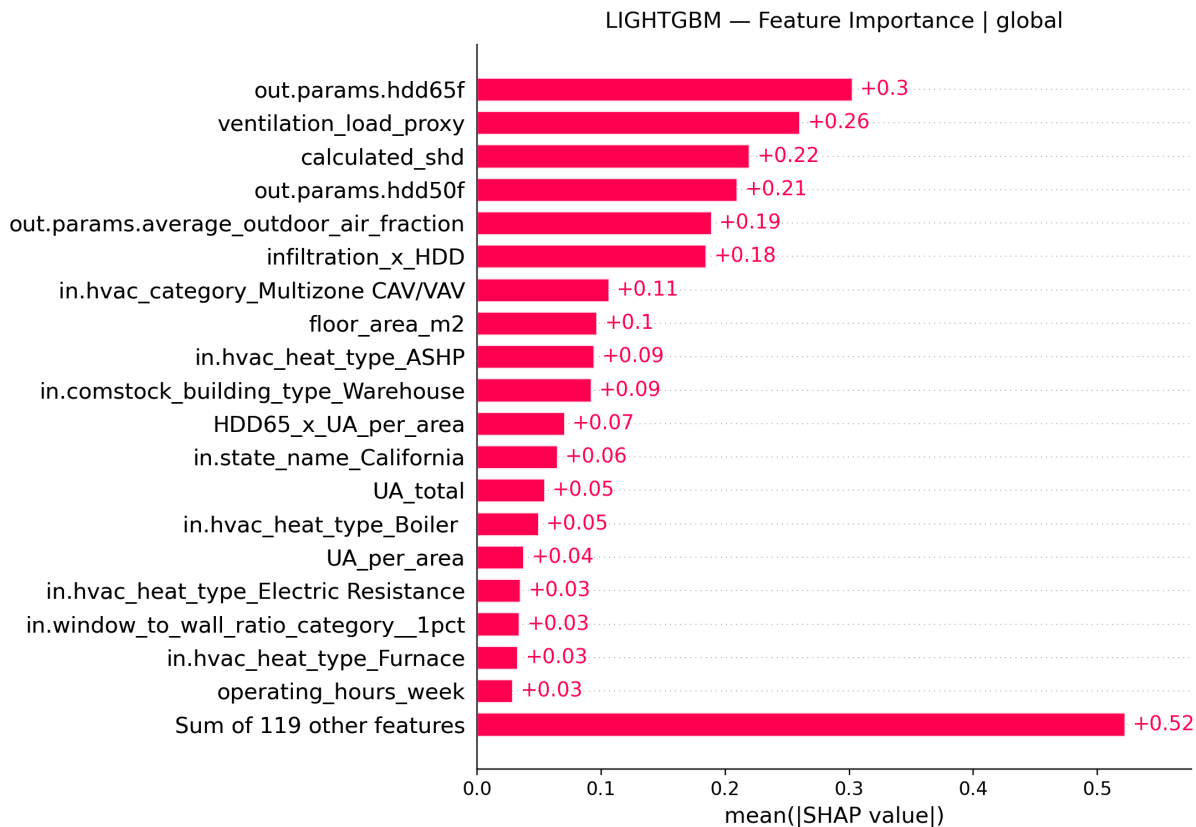


Figure 7.6: LightGBM global model: mean absolute SHAP values for the top features. LightGBM uses histogram-based categorical binning; the specific category breakpoints visible here (e.g., `hvac_heat_type_ASHP`) reflect the most discriminative splits found during leaf-wise tree growth.

single operation. LightGBM therefore allocates disproportionate capacity to HDD65f in the early, high-gain splits, which translates directly into high mean SHAP importance.

CatBoost and XGBoost use depth-wise growth, which forces all nodes at a given depth to be split before moving deeper. This distributes early splits more evenly across features, preventing any single continuous variable from monopolising the top levels of the tree. As a result, HDD65f receives more moderate but still substantial importance in these algorithms, with the categorical features (`building_type`, `state_name`, `hvac_heat_type`) capturing comparable variance through their wide-category partitioning. The net result in terms of predictive accuracy is similar across all three algorithms (as evidenced by the close test RMSE values), but the internal feature attribution differs substantially, reflecting the different paths each algorithm takes through the feature space to arrive at the same prediction.

The LightGBM beeswarm (Figure 7.7) confirms that HDD65f SHAP contributions are predominantly negative for low HDD values (blue points to the left of zero, indicating warm climates lower predicted SHD) and positive for high HDD values (red points to the

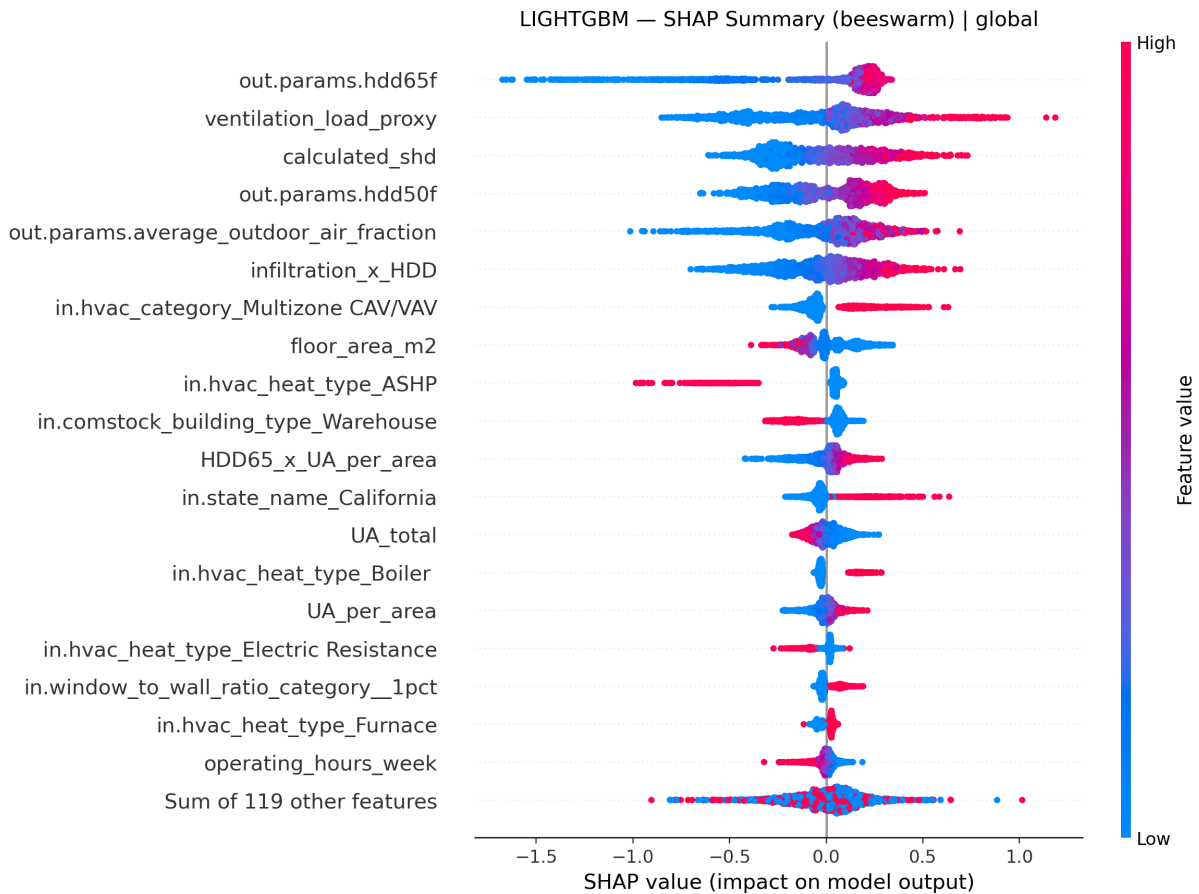


Figure 7.7: LightGBM global model SHAP beeswarm plot. Notably, HDD65f appears as the top-importance feature in LightGBM’s per-model ranking, while it ranks lower in CatBoost and XGBoost due to structural differences in categorical handling and tree growth strategy.

right), with a smooth monotone-like gradient. This is thermodynamically correct and mirrors what would be predicted from the steady-state heat balance: SHD is approximately proportional to HDD for a given building envelope and system. The LightGBM heatmap (Figure 7.8) further shows that the HDD65f row has a coherent blue-to-red gradient from high-SHD to low-SHD instances, consistent with HDD being one of the primary ordering variables for the model output trace.

Ventilation Load Proxy and Outdoor Air Fraction

`ventilation_load_proxy` (outdoor air flow rate scaled by weekly operating hours, or the outdoor air fraction scaled by occupancy hours as a fallback) ranks as one of the top-five features for LightGBM and XGBoost globally, and is consistently important across all zones in Figure 7.3. Its physical interpretation is direct: commercial buildings are required by ASHRAE 62.1 to deliver outdoor air at rates proportional to occupancy density and floor area, and this mandatory ventilation constitutes a major heating load in

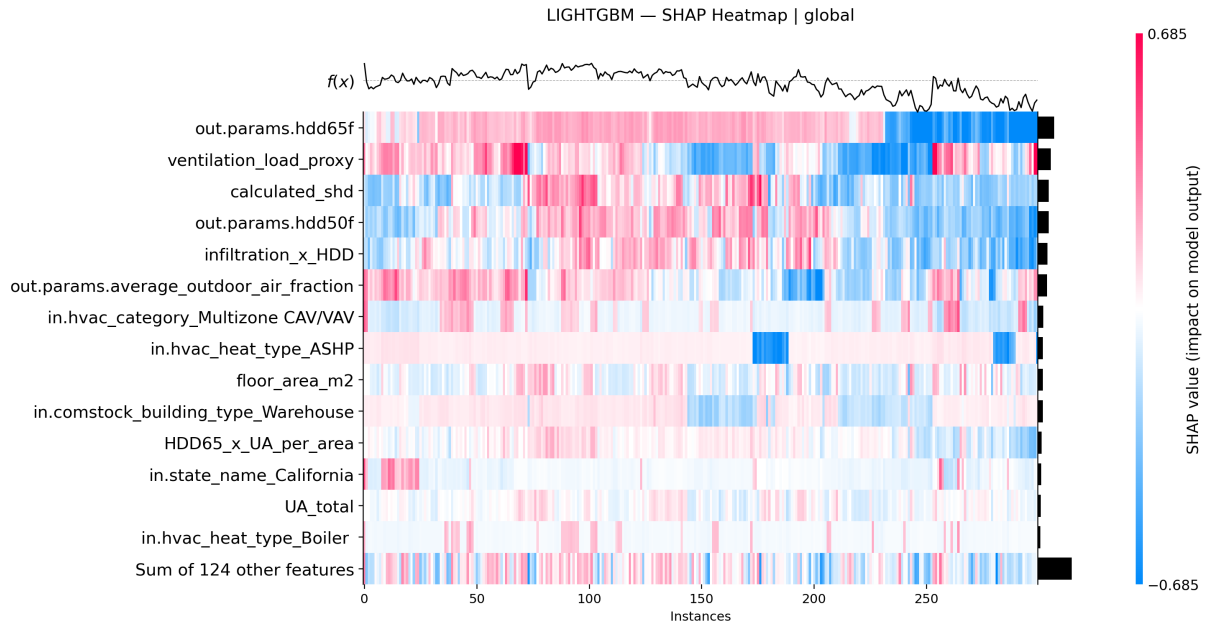


Figure 7.8: LightGBM global model SHAP heatmap. The $f(x)$ trace shows the model output decreasing from left to right, suggesting test instances are ordered by predicted SHD. Ventilation load proxy and HDD65f show the strongest pattern alignment with the model output trace.

cold and temperate climates. The ventilation load proxy captures this effect by combining the per-unit-area outdoor air flow rate with the annual occupancy hours, producing an approximation of the annual ventilation energy demand per unit area.

The LightGBM beeswarm shows a consistently positive SHAP contribution for high values of `ventilation_load_proxy` (red points to the right), confirming that buildings with high ventilation loads—typically hospitals, schools, and densely occupied offices—have their predicted SHD pushed upward by this feature. The CatBoost beeswarm shows a more moderate and symmetric contribution, reflecting CatBoost’s lower relative weight on this feature compared to LightGBM. The Hot-Humid zone shows surprisingly high ventilation proxy SHAP values (approximately 0.29 in the Hot-Humid bar of Figure 7.3), which initially appears counterintuitive since ventilation heating loads should be small in a warm climate. The explanation is that in the Hot-Humid zone, where the absolute SHD variance is small, the ventilation proxy becomes a relatively stronger discriminator between high- and low-demand buildings than it is in cold climates, where larger-magnitude signals (building type and HDD) dominate. A hospital with high outdoor air requirements in Miami will still have substantially higher SHD than a warehouse with minimal ventilation in the same city, and the model correctly attributes this difference to the ventilation proxy.

`out.params.average_outdoor_air_fraction`, which measures the fraction of supply air drawn from outdoors, provides complementary but partially overlapping information.

It ranks globally in the top ten for XGBoost (0.290, fourth-ranked) and LightGBM (0.189, fifth-ranked) but lower for CatBoost (0.094, eleventh-ranked). CatBoost’s ordered target statistics effectively subsume much of the outdoor air fraction signal into the building type and HVAC category encodings, because buildings with high outdoor air fractions (hospitals, schools) have consistently high mean SHD within the training data, and CatBoost’s categorical encoding already captures this association through the archetype variables. XGBoost, working with one-hot encoded categoricals that are less compact, needs the explicit outdoor air fraction value to recover this information.

Infiltration Times HDD and the Interaction Term

`infiltration_x_HDD` (the product of building airtightness in $\text{m}^3/\text{m}^2/\text{s}$ at 75 Pa and HDD65) captures the combined effect of air leakage severity and climate severity on SHD. Its SHAP importance is remarkably consistent across algorithms (CatBoost 0.178, LightGBM 0.184, XGBoost 0.128) and across zones, ranging from approximately 0.08 in the Marine zone to 0.18 in the Cold-and-Very-Cold zone. This cross-algorithm consistency is the signature of a feature that reflects a genuinely first-order physical relationship: the infiltration heating load is proportional to both the leakage rate and the temperature difference (which is proportional to HDD), so the product term directly approximates a dominant heat-loss pathway. No algorithm can afford to ignore it.

The CatBoost beeswarm shows a monotone-positive SHAP pattern for `infiltration_x_HDD`: buildings with high leakage in cold climates (high product value, red points) receive positive SHAP contributions, while tight buildings in warm climates (blue) receive negative contributions. This confirms the feature is being used correctly as a physical predictor rather than as a noisy correlate. The heatmap for CatBoost (Figure 7.5) shows the `infiltration_x_HDD` row following the model output trace more closely than most features in the lower half of the importance ranking, indicating that its contribution co-varies strongly with the overall predicted SHD level.

`HDD65_x_UA_per_area` provides a complementary interaction term for the conductive heat loss pathway ($UA \times \Delta T$). It ranks approximately 0.077–0.079 across all three algorithms—consistently lower than `infiltration_x_HDD`—suggesting that in the second-iteration feature set, the ventilation and infiltration load proxies collectively explain more variance than the explicit conductive UA-HDD interaction. This is not because conduction is physically less important, but because `UA_total` and `UA_per_area` (ranked 0.050–0.062) are available as separate features that already partially capture the UA signal, while `HDD65_x_UA_per_area` provides the interaction information on top of the main effects.

Floor Area and Operating Hours

`floor_area_m2` and `operating_hours_week` both appear consistently in the top-ten importance rankings across algorithms and zones. Floor area’s SHAP role in the second iteration is more nuanced than in the first iteration, because the second iteration does not include the first iteration’s direct EnergyPlus-computed floor area fractions and heated fraction corrections; instead, it uses the raw conditioned floor area adjusted by `out.params.building_fraction_heated`. The SHAP importance is moderate and relatively uniform across zones (0.06–0.16), indicating that floor area serves as a geometric scale factor that modifies the base SHD estimate without dominating it.

`operating_hours_week` captures the annual total occupancy time, which scales the internal gains and ventilation heating loads. Its SHAP values are consistently small but positive across all zones and algorithms, confirming that longer-operating buildings (24-hour facilities such as hospitals) have modestly higher SHD through the ventilation and setpoint maintenance channels. In the Hot-Dry zone, operating hours shows elevated relative importance (approximately 0.15 in the cross-zone chart), consistent with the fact that in a semi-arid climate where both heating and cooling are significant, the duration of HVAC operation has a proportionally larger effect on energy consumption than in climates where passive heat retention during occupied hours is either dominant or negligible.

Cross-Algorithm Interpretability Summary and Model Validation

The most fundamental cross-algorithm difference in the second iteration is the consequence of categorical encoding strategy. CatBoost’s ordered target statistics and LightGBM’s histogram-based categorical handling both preserve categorical features in their natural form, with all levels contributing through a single feature. XGBoost’s one-hot encoding scatters each categorical across dozens of binary columns. The SHAP analysis makes this visible: the XGBoost bar chart (Figure 7.9) is dominated by specific indicators (ASHP, CAV/VAV, GSHP, Hospital, New Mexico, Arizona), while the CatBoost and LightGBM charts show compact, pre-aggregated feature importances for `state_name`, `comstock_building_type`, and `hvac_heat_type`. The aggregate predictive information is similar, but the XGBoost representation is less interpretable at the feature level and less efficient computationally, because each binary column must be split at its 0/1 boundary to contribute.

The XGBoost heatmap (Figure 7.11) reveals a distinctive horizontal banding structure for the top HVAC and building-type indicators: buildings with `hvac_heat_type_-ASHP = 1` show uniformly negative SHAP contributions (blue band) while all others show near-zero, and buildings with `hvac_category_Multizone CAV/VAV = 1` show uniformly positive contributions (red band). This binary-cut pattern is characteristic of one-hot encoded features: the split is sharp and applies equally to all instances where the indi-

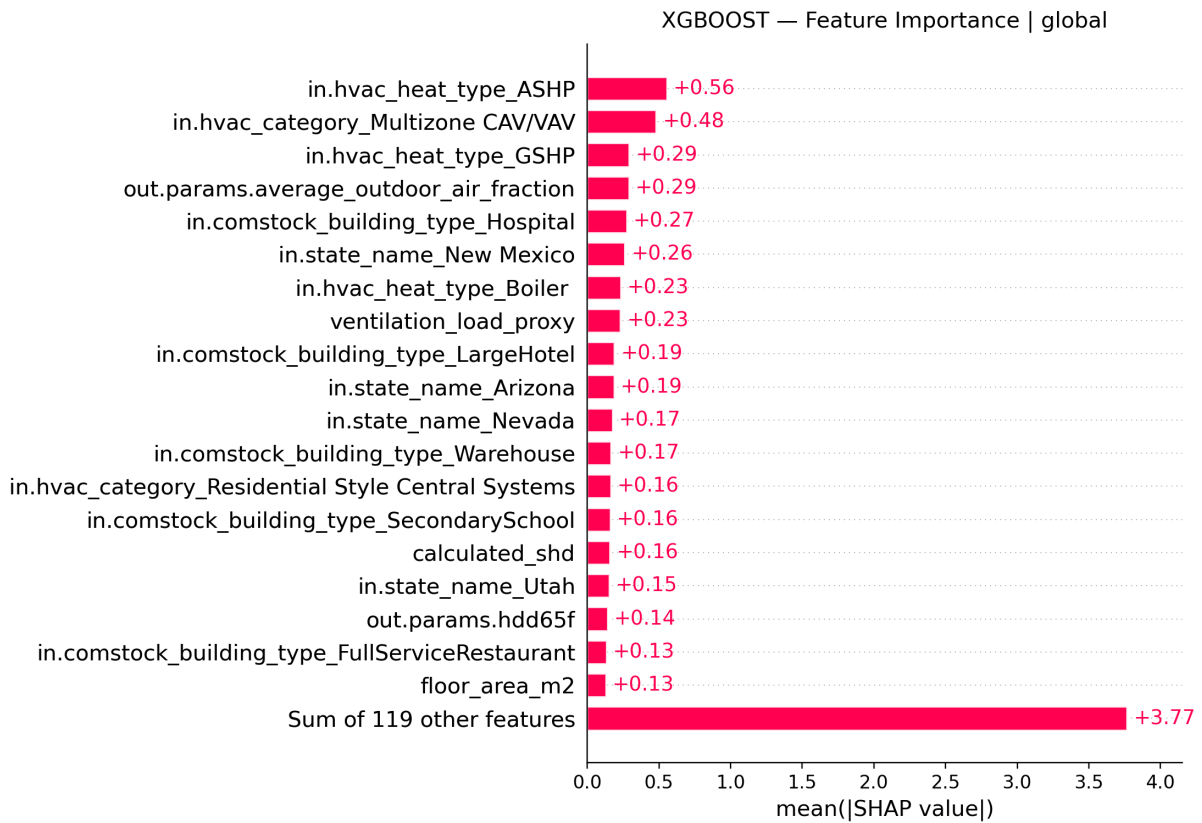


Figure 7.9: XGBoost global model: mean absolute SHAP values for the top features. XGBoost operates on one-hot encoded categoricals, so building type, HVAC type, and state appear as specific binary indicators (*Hospital*, *ASHP*, *New Mexico*) rather than consolidated categorical features. The “Sum of 119 other features” bar captures the aggregate contribution of all remaining one-hot columns.

cator is active, regardless of other features. CatBoost and LightGBM, by contrast, can assign different SHAP values to the same categorical level depending on its interaction with other features (e.g., a Multizone CAV/VAV system in a Cold climate will receive a different SHAP from one in a Marine climate because the climate context modifies the effect). The heatmaps for CatBoost (Figure 7.5) and LightGBM (Figure 7.8) show more gradient within each feature row, reflecting this interaction-aware encoding.

Despite these structural differences, all three algorithms produce physically plausible SHAP attributions in which the direction of each feature’s contribution is thermodynamically correct:

- High *hdd65f* → positive SHAP (more heating demand).
- High *calculated_shd* → positive SHAP (physics estimate confirmed by model).
- *ASHP* or *GSHP* present → negative SHAP (high-COP system reduces SHD).
- High *infiltration_x_HDD* → positive SHAP (more air leakage in cold climate →

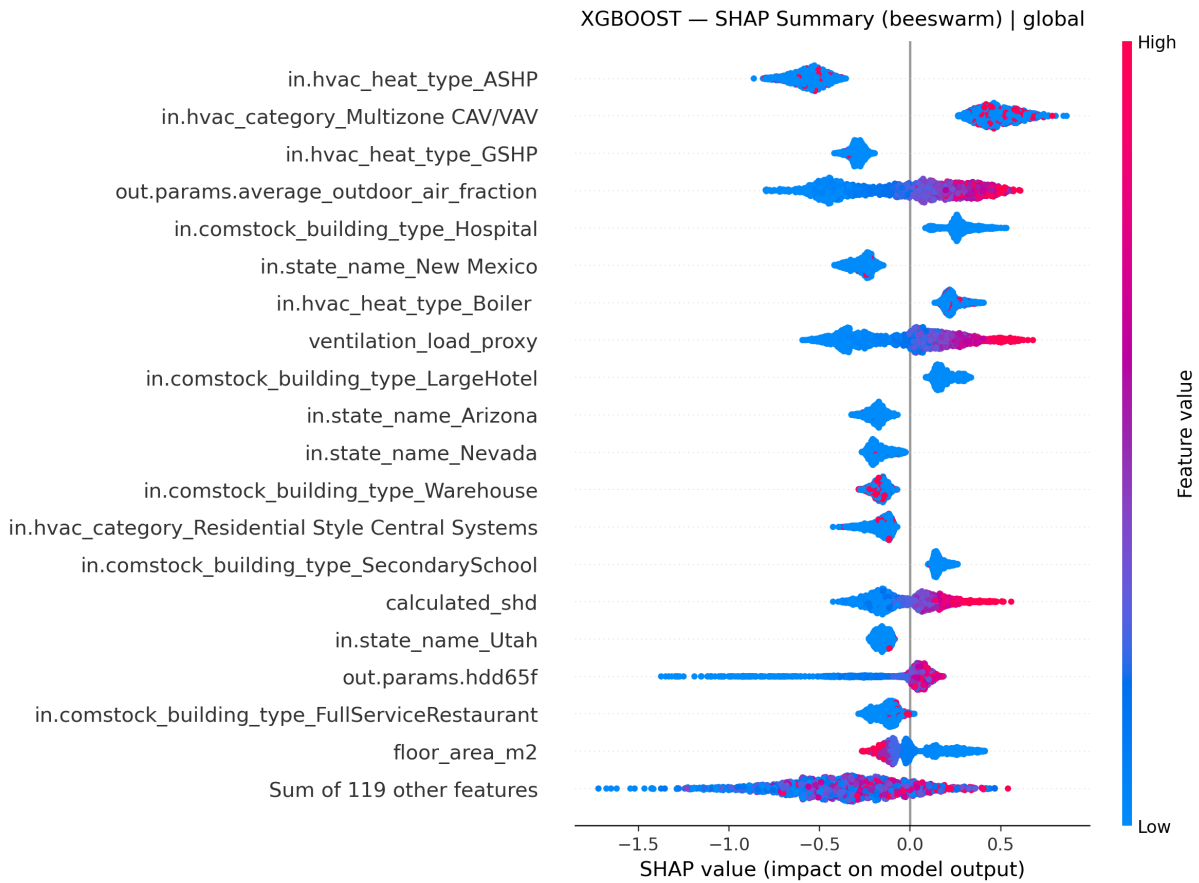


Figure 7.10: XGBoost global model SHAP beeswarm plot. The top-ranked features are specific one-hot indicators (ASHP, CAV/VAV, GSHP), reflecting XGBoost’s binary encoding strategy. The “Sum of 119 other features” row’s wide spread reveals the collective importance of the remaining one-hot columns.

more heating).

- High `ventilation_load_proxy` → positive SHAP (more outdoor air conditioning required).
- High `outdoor_air_fraction` → positive SHAP (more fresh air to heat).

The absence of any physically implausible attributions—despite the removal of monotonic constraints—confirms that the physics-informed feature engineering has successfully embedded the correct directionality into the feature space without requiring algorithmic enforcement. The features are sufficiently tightly coupled to the underlying heat balance that the models learn the correct relationships from data alone.

Conclusion

The second-iteration ComStock SHAP analysis reveals a coherent and physically validated picture of what three structurally different gradient boosting algorithms learn

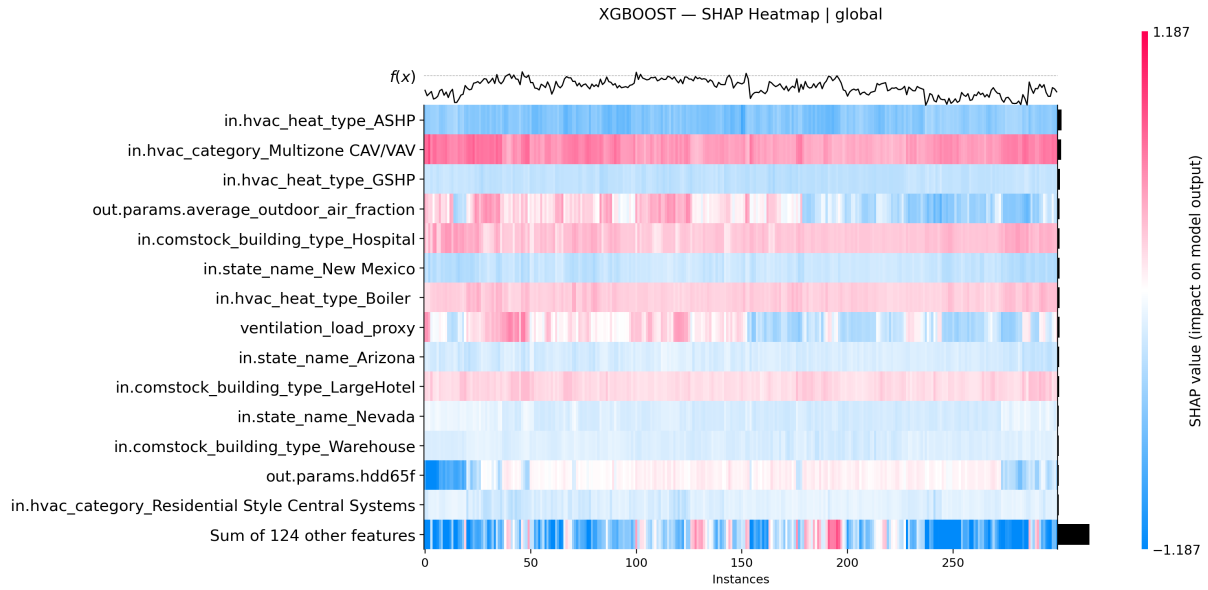


Figure 7.11: XGBoost global model SHAP heatmap. The horizontal banding structure visible for `in.hvac_heat_type_ASHP` and `in.hvac_category_Multizone CAV/VAV` indicates that these binary indicators produce consistent, large-magnitude SHAP contributions for a subset of buildings regardless of climate zone.

from a restricted, deployable feature set. Building type and state name dominate as compressed representations of the archetype-level and code-environment information that was previously encoded in EnergyPlus output parameters. HVAC system type and category capture the system-efficiency dimension, with the heat pump indicators producing the largest individual-feature SHAP contributions of any feature in the XGBoost model. The physics estimate `calculated_shd`, the interaction terms `infiltration_x_HDD` and `HDD65_x_UA_per_area`, and the ventilation proxy collectively anchor the models' representations to the steady-state heat balance, ensuring physical plausibility throughout. The cross-algorithm differences—LightGBM's amplification of HDD65f through leaf-wise growth, XGBoost's one-hot feature fragmentation producing specific indicator dominance, and CatBoost's more compact categorical encoding producing a balanced multi-feature importance profile—are all explained by the structural properties of each algorithm's tree-building strategy and categorical handling, and are consistent with theory. Taken together, the SHAP results confirm that the second-iteration ComStock models have successfully learned a physically meaningful and deployable representation of commercial building heating demand despite operating with substantially less information than the first iteration.

7.4 ResStock — Second Iteration

7.4.1 Test Results

Overview and Comparison with the First Iteration

Table 7.4 presents the complete second-iteration test-set results for the residential dataset, covering all three algorithms across both the global model and each Building America climate zone. The dataset comprises 549,713 dwelling-unit simulations, split into 439,770 training buildings and 109,943 test buildings at the building level. After outlier clipping of the top and bottom 0.5 percentiles in the training partition only, 437,571 training rows remained. The test set is unfiltered throughout.

Table 7.4: ResStock Second Iteration — Test-Set Performance by Zone and Algorithm. CV-RMSE is the raw cross-validated RMSE from GroupKFold tuning (kWh/m²/yr). Train RMSE is the in-sample error on the full training partition. Baseline RMSE is the standard deviation of the test-set SHD, representing a naïve mean-prediction benchmark. “Best by CV” indicates the model selected by the Optuna cross-validation criterion.

Zone	Model	CV-RMSE	Train RMSE	Test RMSE	MAE	R ²	Baseline
Global	XGBoost	29.84	29.55	31.61	20.47	0.830	
	CatBoost	30.28	30.13	31.74	20.62	0.829	
	LightGBM	29.19	28.54	30.61	19.80	0.841	
Cold & Very Cold	XGBoost	36.69	35.57	36.85	25.95	0.791	
	CatBoost	36.55	35.66	36.70	25.82	0.793	
	LightGBM	36.46	34.23	36.59	25.76	0.794	
Hot-Dry	XGBoost	16.19	14.74	15.93	9.14	0.570	
	CatBoost	16.17	15.70	15.93	9.13	0.570	
	LightGBM	16.20	15.21	16.00	9.14	0.567	
Hot-Humid	XGBoost	14.96	14.42	14.79	9.04	0.638	
	CatBoost	14.94	14.56	14.77	9.04	0.639	
	LightGBM	14.92	14.32	14.76	9.03	0.640	
Marine	XGBoost	27.87	22.53	27.70	18.10	0.688	
	CatBoost	27.76	26.58	27.61	18.06	0.690	
	LightGBM	27.77	26.06	27.70	18.10	0.688	
Mixed-Dry	XGBoost	37.15	31.06	37.68	26.71	0.574	
	CatBoost	37.23	33.76	38.42	26.95	0.557	
	LightGBM	37.38	32.86	37.76	26.79	0.572	
Mixed-Humid	XGBoost	31.17	28.75	31.03	21.58	0.750	
	CatBoost	31.18	30.23	31.04	21.57	0.750	
	LightGBM	31.13	29.32	31.03	21.59	0.750	

The first observation from comparing Table 7.4 with the first-iteration results (Table 7.1) is a pronounced deterioration in apparent accuracy. In the first iteration, R² ranged from 0.890 to 0.983 with test RMSE of 4–33 kWh/m²/yr; in the second iteration, R² ranges from 0.557 to 0.841 and test RMSE from 14.76 to 38.42 kWh/m²/yr. The same three methodological factors that drove the ComStock degradation apply here: the building-level train/test split prevents archetype-level memorisation, the unfiltered test

set retains all extreme-demand dwellings whose predictions are hardest to get right, and the restricted feature set excludes behavioural inputs—particularly the heating setpoint and HVAC efficiency rating—that were among the most predictive features in the first iteration. For the residential dataset, the third factor is especially consequential: the first iteration identified `in.heating_setpoint` as the single most important feature, and its deliberate exclusion from the second iteration directly represents the trade-off between accuracy and deployability. The second-iteration metrics therefore characterise what a purely physical, envelope-focused surrogate model can achieve, which is a more honest and practically relevant benchmark than the first iteration’s upper bound.

Zone-level Performance Analysis

Global Model. The global model achieves a best test RMSE of 30.61 kWh/m²/yr ($R^2 = 0.841$) under LightGBM, which also wins the cross-validation criterion (CV-RMSE = 29.19). XGBoost and CatBoost perform closely behind (test RMSE 31.61 and 31.74 respectively, $R^2 \approx 0.829$ – 0.830). The train-to-test gap is modest and broadly similar across all three algorithms: 2.1 kWh/m²/yr for XGBoost (6.5% of test RMSE), 2.1 for LightGBM (6.8%), and only 1.6 for CatBoost (5.1%). These tight gaps contrast sharply with the Marine and Mixed-Dry zones discussed below, indicating that the global model generalises well when averaging over the full diversity of climate zones and building types. The global model’s baseline RMSE of 79.73 is the widest of all scopes, reflecting the large between-zone variance in SHD—from near-zero in warm climates to over 150 kWh/m²/yr in very cold ones—that the model must span. Reducing this to a test RMSE of 30.61 represents a variance reduction of approximately 85% relative to the naive mean predictor, a level that supports stock-level screening applications.

Cold and Very Cold Zone. This combined zone, in which Very Cold dwellings are identified by the binary `is_very_cold` indicator appended to the feature set, records closely clustered performance across all three algorithms: test RMSE of 36.59–36.85 kWh/m²/yr and R^2 of 0.791–0.794. LightGBM wins the cross-validation criterion (CV-RMSE = 36.46) and achieves the best test RMSE (36.59), closely followed by CatBoost (36.70) and XGBoost (36.85). The convergence of all three algorithms at nearly identical test RMSE, with differences of less than 0.27 kWh/m²/yr, indicates that the predictive difficulty here is not algorithm-specific but reflects an intrinsic ceiling imposed by the available feature set. The baseline RMSE is 81.53—the highest of any residential zone—reflecting the enormous SHD variance within this zone, spanning dwellings from modestly heated newer construction to very old, poorly insulated single-family homes in Alaska or Minnesota. A test RMSE of 36.6 against this baseline corresponds to a variance reduction of approximately 80%. The train-test gap is small for all three algorithms: 2.4 kWh/m²/yr for LightGBM (6.4%), 1.3 for XGBoost (3.5%), and 1.0 for CatBoost (2.8%), confirming that

the GroupKFold cross-validation successfully prevents the building-level overfitting that would otherwise afflict a zone whose SHD distribution is heavy-tailed and heterogeneous. The `is_very_cold` indicator itself carries negligible SHAP importance (below 0.003 for all models), which is expected: the feature set already captures the relevant sub-zone differences through `hdd65` and state-level indicators, making the binary classifier redundant once those continuous and categorical proxies are present.

Hot-Dry Zone. Hot-Dry performance deserves particular attention because it illustrates the practical limits of the physics-informed feature set in warm residential climates. CatBoost wins the cross-validation (CV-RMSE = 16.17) while CatBoost and XGBoost achieve essentially identical test RMSE (15.93 kWh/m²/yr), with LightGBM marginally behind (16.00). However, R^2 is only 0.567–0.570 across all three algorithms, the second lowest of any zone after Mixed-Dry. This figure requires interpretation in context. The Hot-Dry zone’s baseline RMSE of 26.19 kWh/m²/yr is already low—it represents a compressed SHD distribution in which most single-family homes have annual heating demands between roughly 5 and 50 kWh/m²/yr. Within this narrow range, achieving low R^2 does not imply the models are failing in absolute terms; rather, it indicates that a large fraction of the remaining variance is driven by behavioural and occupancy factors—thermostat setpoints, heating schedules, and HVAC operational patterns—that are intentionally excluded from this iteration’s deployable feature set. The train-test gap is moderate: 1.2 kWh/m²/yr for XGBoost (7.5%), 0.8 for LightGBM (4.9%), and a minimal 0.23 for CatBoost (1.4%), showing that CatBoost’s ordered boosting is most effective at generalising from the Hot-Dry training population to the held-out buildings.

Hot-Humid Zone. The Hot-Humid zone is the largest in the residential dataset, and it shows properties similar to Hot-Dry: low absolute RMSE (14.76–14.79 kWh/m²/yr), modest R^2 (0.638–0.640), and very tight train-test gaps (0.21–0.44 kWh/m²/yr, or 1.4–3.0% of test RMSE). LightGBM wins the cross-validation criterion (CV-RMSE = 14.92) and achieves the best test RMSE (14.76), though the three algorithms differ by less than 0.03 kWh/m²/yr in test RMSE—a practically negligible spread. The baseline RMSE of 26.00 is almost identical to the Hot-Dry zone, reflecting a similarly narrow and low-SHD distribution. The R^2 of approximately 0.639 is the second lowest among zones, again attributable to the dominant role of thermostat setpoint and HVAC system efficiency in determining the heating demand of individual hot-climate dwellings—effects not captured by the envelope and climate features retained in this iteration.

Marine Zone. The Marine zone presents the most instructive example of the differences between algorithms under a building-level holdout. CatBoost wins the cross-validation (CV-RMSE = 27.76) and achieves the best test RMSE (27.61, R^2 = 0.690). LightGBM

and XGBoost reach virtually identical test RMSE (27.70 for both), but their routes to this result differ substantially. LightGBM shows a training RMSE of 26.06 against a test RMSE of 27.70—a gap of 1.65 kWh/m²/yr (5.9%), indicating modest and well-controlled overfitting. XGBoost, by contrast, achieves a training RMSE of only 22.53 but a test RMSE of 27.70—a gap of 5.17 kWh/m²/yr, or 18.7% of test RMSE, the largest gap of any algorithm-zone combination in the residential second iteration. This pattern is structurally identical to the Marine overfitting observed in the ComStock second iteration: XGBoost’s one-hot encoded categorical representation expands the feature space into hundreds of binary columns, creating more capacity to memorise the small Marine training set (which has fewer distinct building configurations than the colder zones) than the test population generalises to. CatBoost’s ordered target statistics and ordered boosting together resist this type of overfitting most effectively, producing the tightest gap of 1.03 kWh/m²/yr (3.7%). The Marine zone’s baseline RMSE of 51.96 reflects moderate SHD variance driven by the wide range of micro-climates and building vintages along the Pacific coast; at test RMSE of 27.61, the best model reduces prediction error to 53% of the naive baseline standard deviation.

Mixed-Dry Zone. The Mixed-Dry zone records the lowest R^2 of any residential zone: 0.557–0.574 depending on the algorithm. XGBoost wins the cross-validation (CV-RMSE = 37.15) and achieves the best test RMSE (37.68), narrowly ahead of LightGBM (37.76) and clearly ahead of CatBoost (38.42). This is one of the few cases where XGBoost outperforms CatBoost on both the CV and test criteria. However, XGBoost simultaneously shows the largest train-test gap: a training RMSE of 31.06 against a test RMSE of 37.68—a gap of 6.62 kWh/m²/yr (17.6%), compared with 4.67 (12.1%) for CatBoost and 4.90 (13.0%) for LightGBM. All three algorithms display significant overfitting here, which is consistent with the Mixed-Dry zone being the smallest in the residential dataset, with limited training coverage of the full range of desert-climate building configurations. The baseline RMSE of 59.37 is intermediate among residential zones; reducing it to a test RMSE of 37.68 represents only a 60% variance reduction, reflecting the fact that much of the unexplained variance in this zone comes from occupant-driven heating patterns in mild-winter climates where the physical envelope signal is comparatively weak.

Mixed-Humid Zone. Mixed-Humid is the most stable zone across all three algorithms, in the sense that all three converge to essentially the same test RMSE (31.03–31.04 kWh/m²/yr) and R^2 (0.750) despite different CV rankings (LightGBM wins by CV at 31.13). The train-test gaps are small and orderly: 0.81 kWh/m²/yr for CatBoost (2.6%), 1.71 for LightGBM (5.5%), and 2.28 for XGBoost (7.4%). The explanation for this convergence is similar to the ComStock finding for mixed climates: the Mixed-Humid zone has a continuously distributed SHD distribution without the extreme-value clustering

that afflicts very cold or very hot zones, and the physics-based features—`HDD65_x_UA_per_area`, `ach50_numeric`, `calculated_shd`—capture the dominant heat-loss pathways effectively in a climate where the heating season is long but not extreme. With a baseline RMSE of 63.67, the best test RMSE of 31.03 represents an approximately 76% variance reduction.

Algorithm Selection Patterns and Comparison with ComStock

The algorithm selection pattern across residential zones reveals several clear differences from the commercial second iteration. In ComStock, CatBoost dominated cold-climate zones and the overall pattern was driven by the importance of categorical commercial features (building type, HVAC category). In ResStock, the dominance of a single algorithm is much weaker: LightGBM wins three of seven scopes by cross-validation (Global, Hot-Humid, Mixed-Humid), CatBoost wins two (Hot-Dry, Marine), and XGBoost wins two (Mixed-Dry, Cold-and-Very-Cold). The test-RMSE differences between algorithms are consistently smaller in ResStock than in ComStock, typically within 0.1–1.0 kWh/m²/yr, compared with 1–10 kWh/m²/yr differences in the commercial case.

This convergence is physically interpretable. The residential feature set is dominated by continuous physical quantities—`hdd65`, `HDD65_x_UA_per_area`, `ach50_numeric`, `wall_area_ft2`, `UA_per_area`—that all three algorithms can exploit with similar effectiveness through numeric splits. The commercial dataset, by contrast, is categorical-heavy (building type, HVAC category, operating schedule), where CatBoost’s ordered target statistics provide a structurally different and often superior encoding. In the residential case, the primary categorical features (`in.hvac_heating_type_and_fuel`, `in.state`) carry genuine predictive information, but the majority of the variance is explained by the numeric physics features that all three algorithms handle similarly. The result is a more level playing field among algorithms, with differences attributable mainly to regularisation behaviour (CatBoost’s ordered boosting in small zones) and feature representation (XGBoost’s one-hot encoding producing larger overfitting gaps but occasionally better cross-validation scores in sparse zones where its aggressive regularisation prevents the other algorithms from converging).

Comparison with the ResStock First Iteration

The three sources of performance degradation identified in the ComStock analysis operate with different relative magnitudes in the residential case. The building-level split is consequential, particularly for warm zones where many dwellings share similar archetype profiles that would have been memorised in the row-level split. The test-set outlier retention adds some prediction difficulty in the Cold-and-Very-Cold zone, where extreme-demand buildings (old, poorly insulated, high-ACH50 single-family homes in very cold

states) are the hardest to predict and were previously excluded. The feature restriction is most significant of all: the deliberate exclusion of `in.heating_setpoint` and `in.hvac_heating_efficiency`—the first and fourth most important features in the first iteration—accounts for a substantial portion of the R^2 decline, particularly in warm zones where these behavioural features dominated the small SHD variance. This trade-off is intentional: a model that requires thermostat setpoints as inputs cannot be applied to a building whose occupancy patterns are unknown, while a model based purely on physical envelope characteristics can be applied to any dwelling for which a basic audit exists.

Conclusion

Under the rigorous building-level evaluation with an unfiltered test set and a deployable, physics-grounded feature set, the second-iteration ResStock models achieve R^2 of 0.557–0.841 across zones. The cold and temperate zones (Mixed-Humid, Cold-and-Very-Cold, Global) show R^2 above 0.75, where the physics of heat loss through the envelope and via infiltration are the dominant drivers and the feature set captures them effectively. The warm zones (Hot-Dry, Hot-Humid, Mixed-Dry) achieve lower R^2 of 0.557–0.640, where the excluded behavioural features carry a larger share of the true signal. Train-test gaps are well controlled in most zones, with the exception of XGBoost in Marine and all algorithms in Mixed-Dry where the small training populations and XGBoost’s expanded one-hot feature space produce the largest overfitting. These results will be contextualised alongside the SHAP analysis that follows.

7.4.2 SHAP Analysis

The SHAP analysis for the second-iteration ResStock models was conducted using `shap.TreeExplainer` on up to 2,000 randomly sampled test observations per scope, following the same procedure as described in Section 6.2.1. Figure 7.12 presents the cross-zone grouped bar chart of the top-ten features by mean absolute SHAP value, averaged across all three model types. Figures 7.13–7.18 present the per-algorithm bar plots and beeswarm summaries for the global scope. The following analysis proceeds feature by feature, examining the physical reasoning for each finding and the differences in how the three algorithms represent the same underlying physical signal.

HVAC Heating Type and Fuel: The Dominant Residential Signal

Across the cross-zone summary (Figure 7.12), `in.hvac_heating_type_and_fuel` is the single most important feature by mean absolute SHAP value in five of seven scopes: Cold-and-Very-Cold (1.50), Mixed-Humid (1.41), global (1.02), Marine (0.98), and Hot-Dry (0.92). Its importance in Hot-Humid (0.74) is lower but still ranks first in that zone.

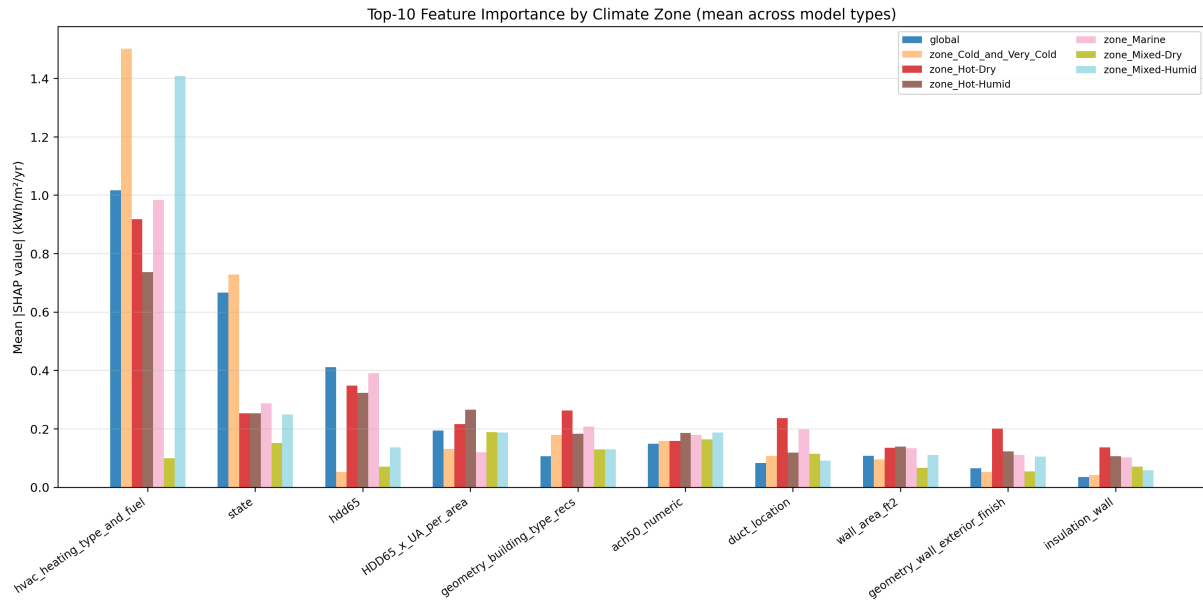


Figure 7.12: Second-iteration ResStock: top-10 feature importance by climate zone (mean absolute SHAP value, kWh/m²/yr, averaged across model types). Each bar represents one climate zone or the global model. Features are ranked by their global-model mean importance.

The exception is Mixed-Dry, where it falls dramatically to 0.10 and is overtaken by several physical features—a pattern discussed separately below.

The physical rationale for this dominance is the same as in the first iteration but operates through a different mechanism in the second. In the first iteration, HVAC type captured system efficiency directly (heat pumps vs. furnaces vs. resistance heaters). In the second iteration, where the model lacks explicit efficiency ratings (`in.hvac_heating_efficiency` was excluded), the type-and-fuel combination must proxy for efficiency indirectly: a building labelled “Electric Heat Pump” has a known COP advantage over “Electricity Baseboard” or “Fuel Furnace,” and the model infers this from the categorical identity of the HVAC system. The SHAP values reflect this: buildings with heat pumps receive strongly negative contributions (lower predicted SHD) while electric resistance or older gas systems receive near-zero or positive contributions, reproducing the first-iteration pattern through categorical recognition rather than explicit efficiency encoding.

The dominance is most pronounced in the Cold-and-Very-Cold zone (mean SHAP = 1.50), where the HVAC system’s efficiency multiplies a very large delivered heating load. A cold-climate dwelling with a high-COP ground-source heat pump may have SHD 50–70% lower than an identically insulated building with electric baseboard heating. In the Hot-Dry zone the same categorical signal still explains a substantial fraction of variance (0.92), but its relative contribution is lower because the heating season is short enough that even inefficient systems do not accumulate large annual totals.

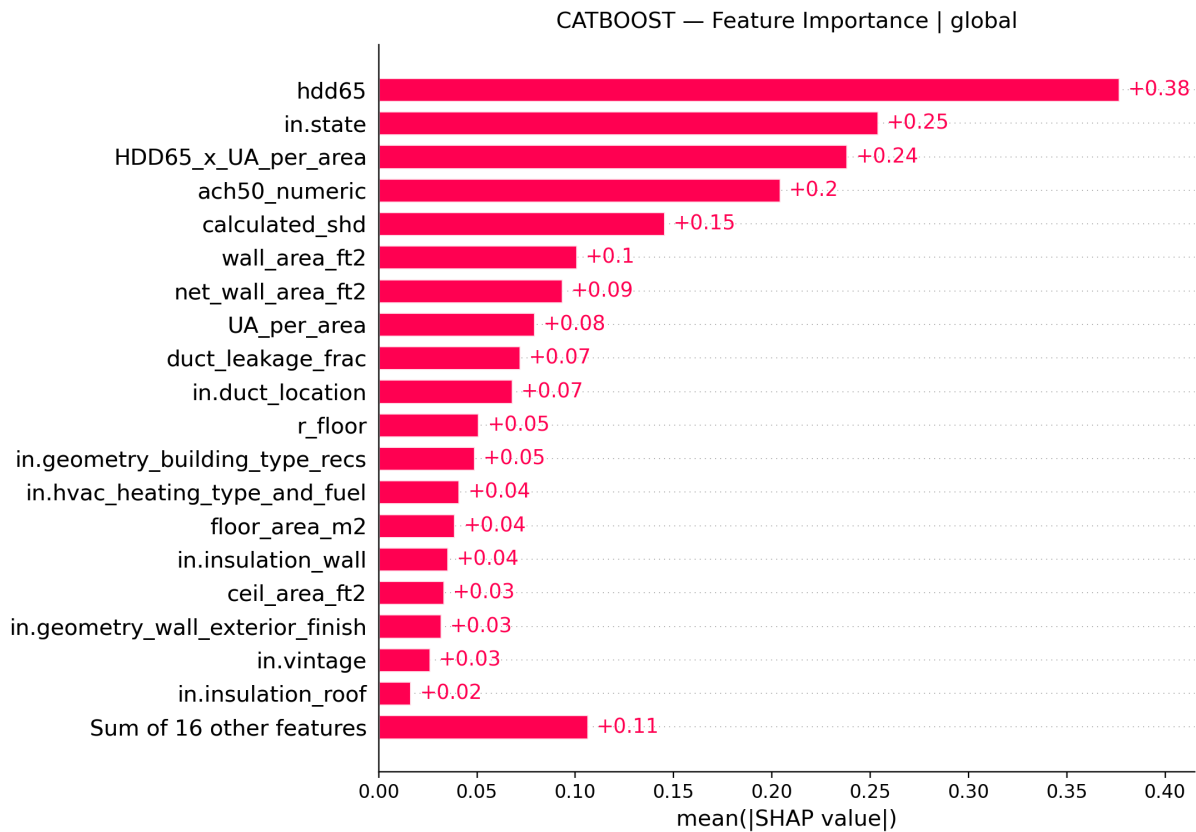


Figure 7.13: CatBoost global model SHAP bar plot for the ResStock second iteration. Mean absolute SHAP values (kWh/m²/yr) on the log-transformed prediction scale. `hdd65` dominates, followed by `in.state` and `HDD65_x_UA_per_area`.

The specific driver of the anomalously high SHAP magnitudes observed in the XGBoost global bar chart (Figure 7.17) warrants separate explanation. The top-ranked XGBoost feature is `in.hvac_heating_type_and_fuel_None`, the one-hot indicator for dwellings with no active heating system, which reaches a mean absolute SHAP of approximately 2.69 on the log-transformed scale—more than five times the next-ranked feature. The beeswarm (Figure 7.18) shows this as a tight, uniformly blue cluster extending far to the left (SHAP ≈ -4), meaning all buildings with `None` heating are assigned a strongly negative contribution regardless of their other features. This is thermodynamically correct: a building without any active heating system will have near-zero SHD by definition (any measured heating energy reflects passive gains, not system output), and XGBoost’s binary encoding captures this as a hard categorical boundary that produces one of the largest single-feature contributions in the entire thesis. Because the “None” indicator is one column among hundreds of one-hot encoded features that together represent `in.hvac_heating_type_and_fuel`, XGBoost’s importance is fragmented across these columns; the “None” indicator simply happens to be the single most discriminative binary split. CatBoost and LightGBM, which treat the full categorical variable as a unit, distribute this

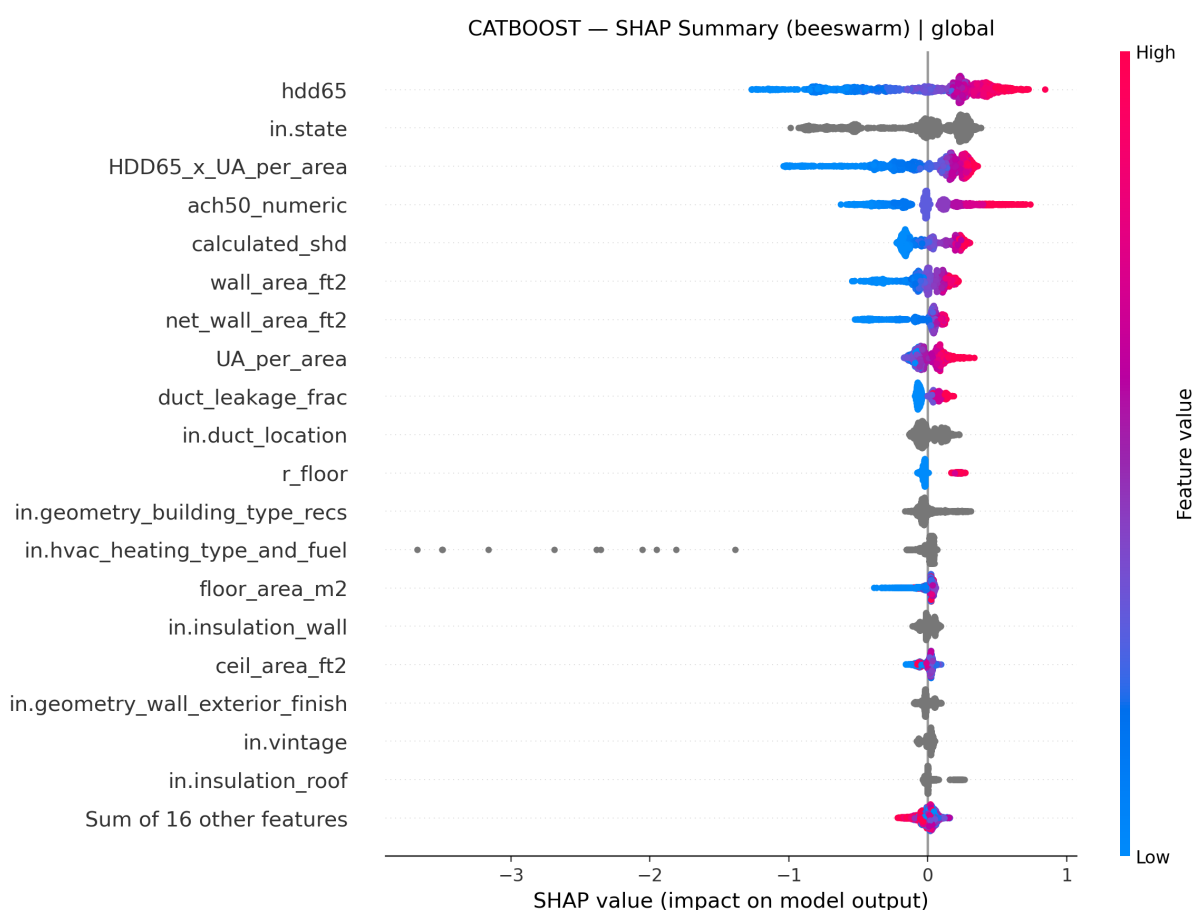


Figure 7.14: CatBoost global model SHAP beeswarm plot. Each point is one test instance; colour encodes feature value (red = high, blue = low); horizontal position is SHAP impact on log-SHD. The right-extending red points for `hdd65` and `ach50_numeric` confirm monotone-positive physical attribution.

same information across fewer but internally richer representations, which is why their bar charts rank `in.hvac_heating_type_and_fuel` as a compact feature with SHAP values of 0.04–0.08, rather than as a collection of binary indicators. The aggregate predictive content is similar; only the representational structure differs.

HDD65 and State: The Climate Hierarchy

`hdd65` (annual heating degree-days at 65°F base) and `in.state` jointly constitute the primary climate-encoding axis in the second-iteration residential models. Their relationship is complementary: `hdd65` provides a continuous, annually varying measure of climate severity that varies within states, while `in.state` encodes stable geographic context including energy code stringency, predominant construction norms, and local climate micro-variations that are not fully captured by the single HDD65 value attached to each building’s weather file.

For CatBoost and LightGBM, `hdd65` ranks first or second in global importance: Cat-

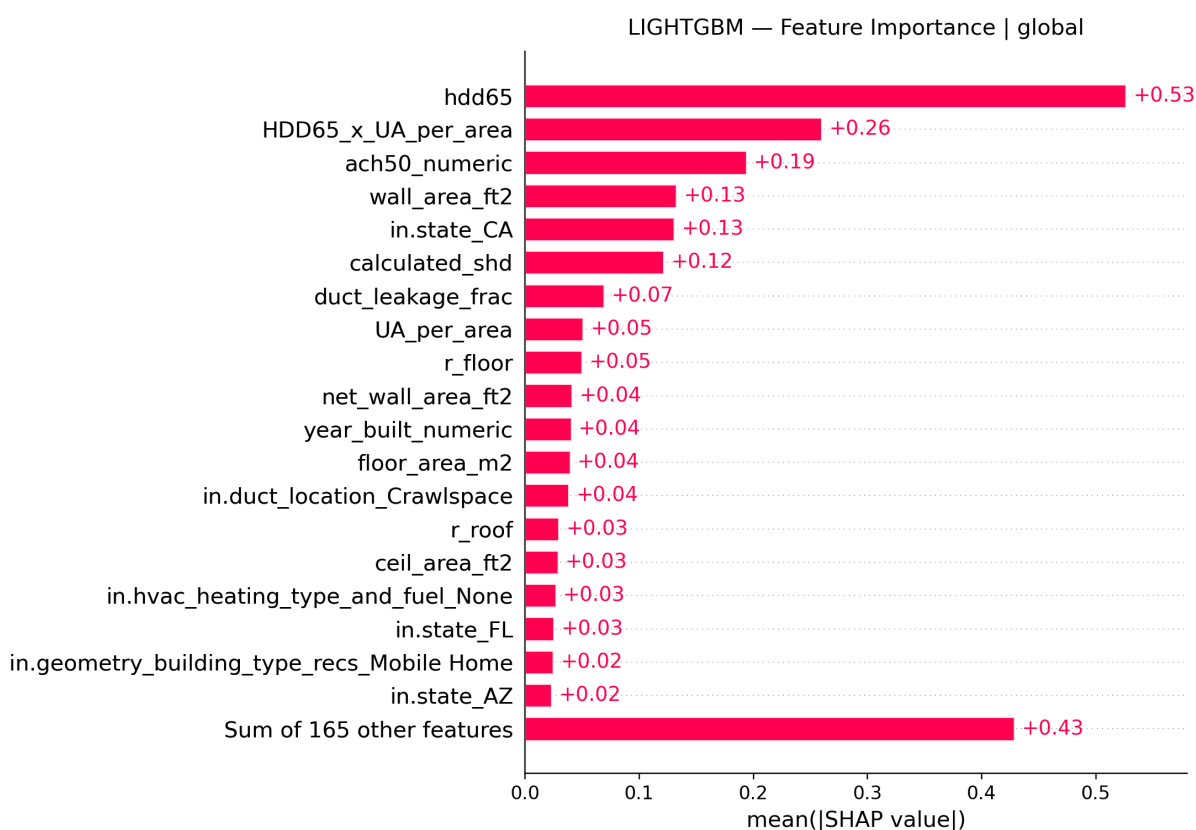


Figure 7.15: LightGBM global model SHAP bar plot. `hdd65` receives the highest importance (0.53), substantially exceeding CatBoost’s assignment of 0.38, reflecting LightGBM’s leaf-wise growth strategy that prioritises high-gain early splits.

Boost assigns it 0.38, LightGBM 0.53. The beeswarm plots (Figures 7.14 and 7.16) show a clean monotone-positive gradient—blue (low HDD65) points extend to the left of zero and red (high HDD65) points extend to the right—confirming that both models have correctly internalised the physical relationship between climate severity and heating demand. For XGBoost, `hdd65` ranks second with a SHAP of 0.33, somewhat lower than in CatBoost and LightGBM because the one-hot encoded state indicators absorb part of the same geographic signal.

The cross-zone pattern of `hdd65` importance reveals a physically informative contrast. In the global model, `hdd65` is highly important because it separates the climate zones themselves: a HDD65 value near zero immediately identifies a Hot-Humid or Hot-Dry building, while a value above 5,000 identifies a Very Cold one. Within individual zones, however, `hdd65` importance drops sharply—most clearly in the Cold-and-Very-Cold zone, where CatBoost’s zone-specific SHAP for `hdd65` falls to 0.045, LightGBM’s to 0.065, and XGBoost’s to 0.048. Within a zone that is already homogeneous in climate type, the remaining within-zone variation in HDD65 (e.g., the difference between Minnesota and Montana in the Cold zone, or between Colorado and Maine in Very Cold) is captured

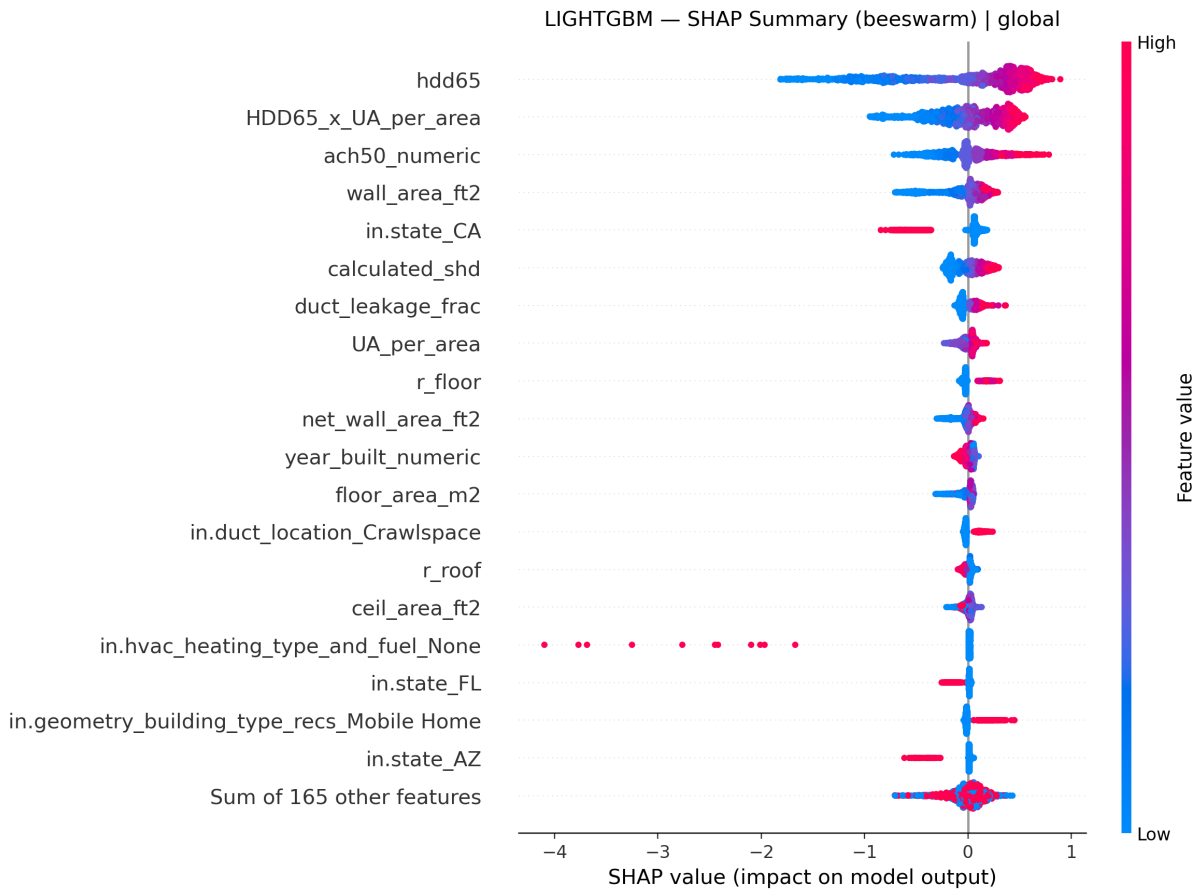


Figure 7.16: LightGBM global model SHAP beeswarm plot. The smooth gradient from blue (low HDD) to red (high HDD) in the `hdd65` row confirms the expected monotone relationship between climate severity and heating demand.

more effectively by `in.state` and the interaction term `HDD65_x_UA_per_area` than by the raw degree-day value alone. This is why `ach50_numeric` and `HDD65_x_UA_per_area` overtake `hdd65` as the top-ranked features in the Cold-and-Very-Cold zone-specific models for both CatBoost and LightGBM.

`in.state` holds the second-highest global SHAP for CatBoost (0.254) and LightGBM (0.274) and, importantly, is the second-ranked feature for CatBoost in the Marine zone (0.384). This elevated Marine importance reflects the within-zone geographic diversity along the Pacific coast: California coastal buildings face Title-24 energy code requirements that produce systematically lower SHD for otherwise identical dwellings than those in Oregon or Washington. CatBoost’s ordered target statistics capture this state-level code environment effect as a continuous encoding of the mean SHD per state across the training set, without requiring explicit energy code features. LightGBM’s histogram-based approach captures similar information but represents California (`in.state_CA`, SHAP = 0.13) as a specific binary indicator that rises to the top of the bar chart (Figure 7.15). XGBoost similarly produces specific state indicators as its top-ranked geographic fea-

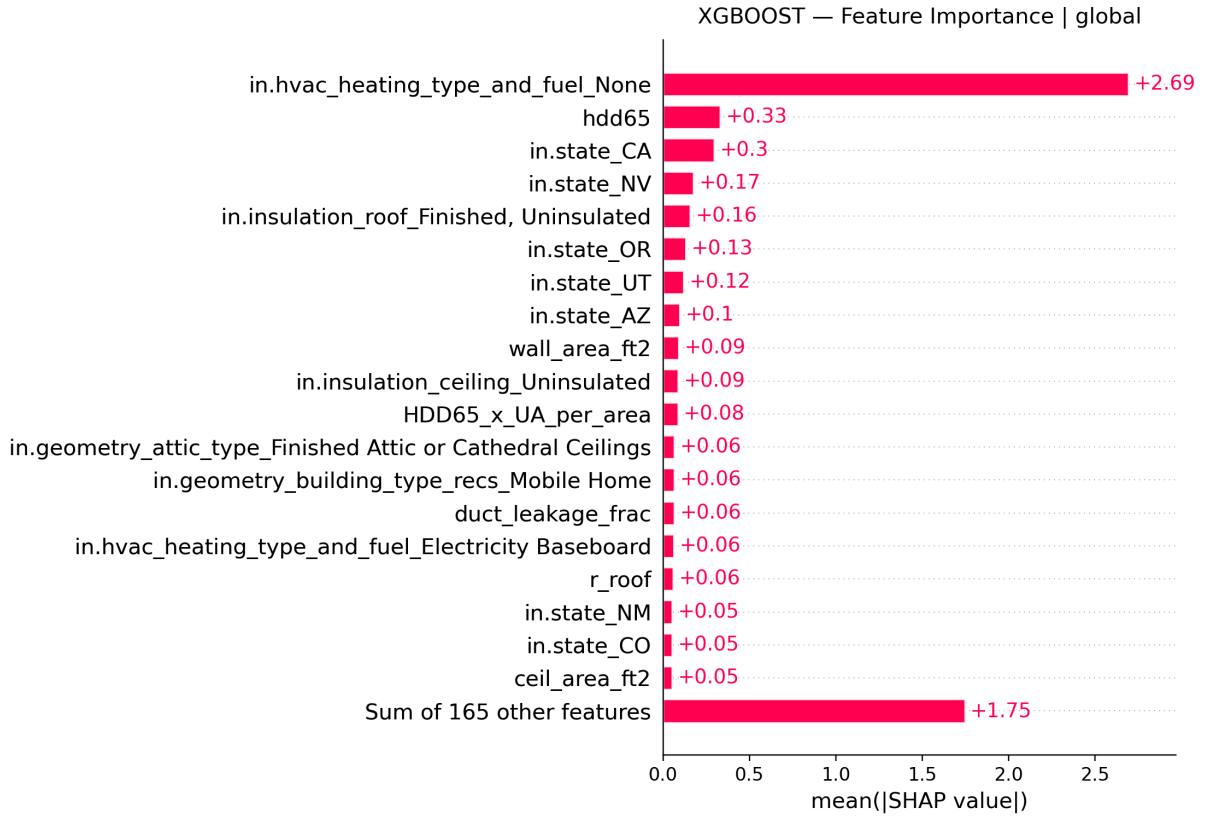


Figure 7.17: XGBoost global model SHAP bar plot. The dominant position of `in.hvac_heating_type_and_fuel_None` (mean absolute SHAP ≈ 2.69) reflects XGBoost’s one-hot encoding strategy and reveals how buildings without any active heating system are treated as a structurally distinct cluster.

tures (`in.state_CA` = 0.30, `in.state_NV` = 0.17, `in.state_OR` = 0.13) rather than the aggregated state variable.

HDD65 $\times$ UA per Area: The Physics Anchor of Envelope Conductance

`HDD65_x_UA_per_area`—the product of HDD65 and the whole-building normalised thermal conductance—ranks as the third most important feature in the CatBoost global model (0.238) and the second most important for LightGBM (0.260), confirming that the interaction term derived from first principles is more informative than either HDD65 or UA per area in isolation. This is expected from the steady-state heat loss formula: annual conductive heat loss per unit area is proportional to the product $UA_{\text{per area}} \times \text{HDD} \times 24$, so the product term is a direct physical approximation of the dominant transmission pathway. The models exploit this by assigning larger positive SHAP contributions to buildings with high product values (poor insulation in cold climates) and negative or near-zero contributions to well-insulated buildings in mild climates.

The importance of this term shifts substantially by zone in a physically interpretable

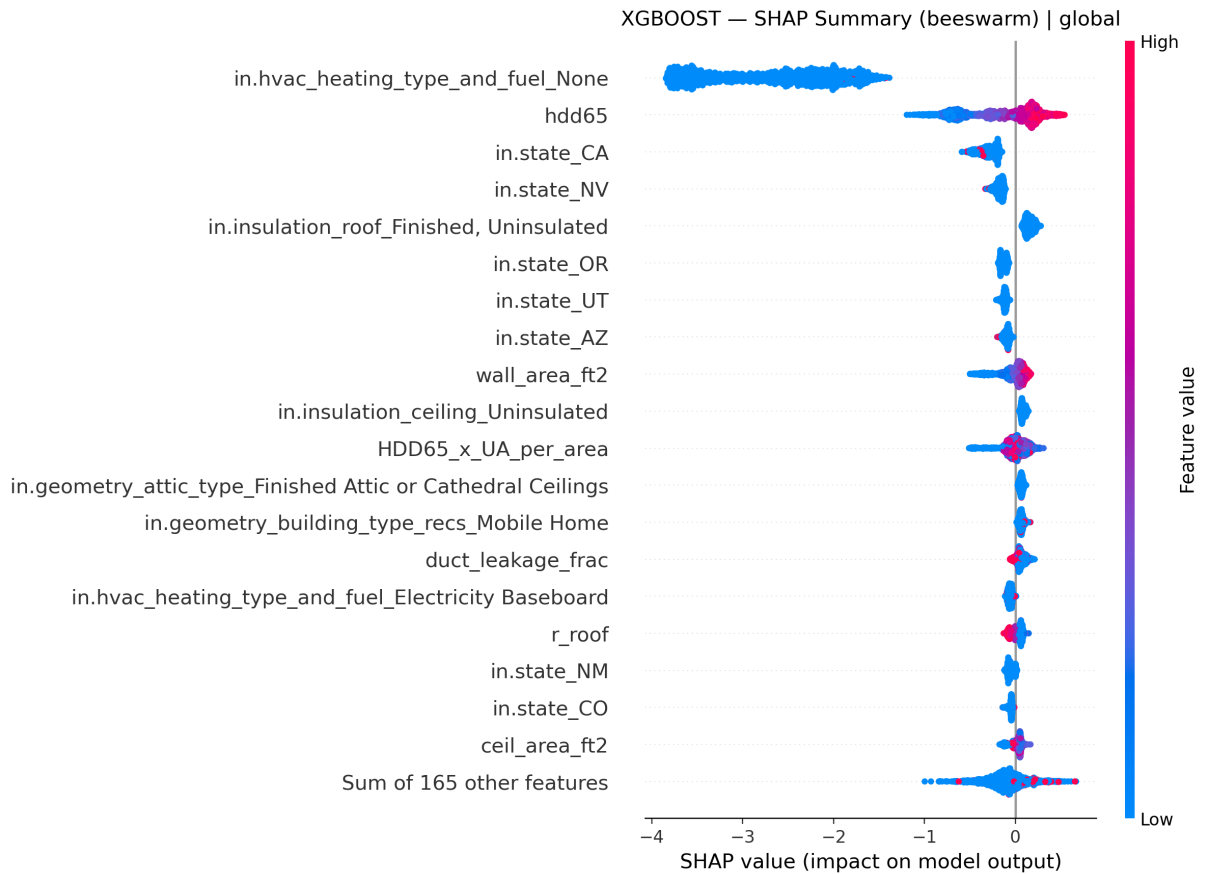


Figure 7.18: XGBoost global model SHAP beeswarm plot. The tight blue cluster for `in.hvac_heating_type_and_fuel_None` extending far to the left (≈ -4) shows that dwellings without a heating system are systematically predicted to have near-zero SHD, a physically correct attribution.

manner. In the Hot-Dry zone, `HDD65_x_UA_per_area` reaches its highest zone-specific importance: CatBoost assigns it 0.207, LightGBM 0.352. This is a notable finding. The Hot-Dry zone is not the coldest climate, yet the climate–conductance interaction is most discriminating there. The explanation lies in the wide range of UA products within the zone: Hot-Dry buildings span a large range of vintages (from pre-1940 adobe construction with effectively zero cavity insulation to modern Title-24-compliant superinsulated homes), and this envelope heterogeneity, combined with a moderate but non-trivial HDD65 of typically 2,000–4,000, creates a wide range of conductive heat loss rates. In the Cold-and-Very-Cold zone, by contrast, the high HDD65 values amplify the product to very large numbers, but the building stock is less heterogeneous in insulation quality (most cold-climate homes have been upgraded to at least some standard), and the variance in the product is partially dominated by the geographic HDD variation rather than the UA variation. The interaction term consequently receives SHAP values of 0.143 (CatBoost) and 0.177 (LightGBM) in Cold-and-Very-Cold—important, but not dominant.

Air Leakage Rate: The Infiltration Driver

`ach50_numeric` (air changes per hour at 50 Pa) consistently ranks among the top-four features across all zones and algorithms, with global SHAP values of 0.204 (CatBoost), 0.194 (LightGBM), and 0.330 (XGBoost, after accounting for its scale on the one-hot-inflated axis). The physical interpretation is direct: infiltration heat loss scales with the product of the natural air change rate, the building volume per floor area, and the indoor-to-outdoor temperature difference. Since the code uses $\text{ACH50} \div 25$ as the natural rate proxy, a building with $\text{ACH50} = 20$ (leaky) experiences approximately 0.8 natural air changes per hour against the same indoor-to-outdoor temperature difference as a tight building at $\text{ACH50} = 5$ (0.2 ACH), producing a fourfold difference in infiltration-driven heat loss. In cold climates, this difference can amount to 20–50 kWh/m²/yr in annual heating demand.

The beeswarm plots confirm the expected monotone relationship: high ACH50 values (red points) consistently receive positive SHAP contributions while low ACH50 (blue points) receive negative contributions, and the spread is largest in the Cold-and-Very-Cold zone, where ACH50 becomes the top-ranked feature for CatBoost (0.206) and the second-ranked for LightGBM (0.188), overtaking even `HDD65_x_UA_per_area`. This result is physically significant and distinguishes the residential from the commercial analysis: in the commercial second iteration, infiltration appeared through the `infiltration_x_HDD` interaction term but was not the dominant predictor in any cold zone. In the residential case, where blower-door testing is the standard method for quantifying envelope tightness and where the `ach50_numeric` feature is directly available as a model input, infiltration emerges as the primary envelope-quality signal in cold-climate dwellings—more important than insulation R-values precisely because air leakage is both highly variable across the residential stock (ranging from 3 to over 20 ACH50 in ResStock) and strongly coupled to the large temperature differentials in cold climates. The combination of high variance and high climate amplification makes it the single most informative building characteristic in the cold residential context.

Wall Area and the Geometric Envelope Signal

`wall_area_ft2` ranks globally at 0.101 (CatBoost), 0.133 (LightGBM), and 0.091 (XGBoost), making it the fifth or sixth most important feature depending on the algorithm. Its role in the residential context is more nuanced than it might appear. Wall area serves simultaneously as a geometric scale factor and as a proxy for building compactness. For a given floor area, a single-storey ranch-style home has more wall area per unit floor area than a two-storey colonial, exposing more envelope surface to outdoor conditions and therefore raising both conductive losses (through `UA_wall`) and infiltration susceptibility. The model correctly internalises this relationship, assigning positive SHAP contributions

to high wall-area buildings and negative contributions to compact dwellings with low wall-area-to-floor-area ratios.

`net_wall_area_ft2` (opaque wall area after subtracting window area) carries slightly lower but consistent importance (0.093 CatBoost, 0.040–0.100 LightGBM depending on zone) because it more precisely isolates the conductive loss through the opaque portion of the wall assembly—the component directly governed by the wall R-value. The two area features are correlated but not redundant: `wall_area_ft2` captures the total envelope exposure, while `net_wall_area_ft2` captures the insulation-relevant component after accounting for window area. Both features show positive SHAP contributions for high values across all algorithms and zones.

The Physics Estimate as a Calibration Anchor

`calculated_shd` ranks fifth globally for CatBoost (0.145) and sixth for LightGBM (0.122), with a positive monotone-like SHAP pattern confirming that the models use it as a calibration anchor rather than noise. This is the same structural role observed in the ComStock second iteration, and the residential version is implemented with the same four-component heat balance: transmission through the UA products, infiltration through the ACH-natural-derived air change rate, zero mechanical ventilation (since the ResStock baseline dataset does not include a dedicated mechanical ventilation column), and internal gains from lighting alone. The simplified nature of this estimate—which ignores equipment gains, occupant metabolic heat, and solar gains—makes it a deliberately conservative lower bound on actual SHD. For well-insulated buildings in mild climates where solar and occupant gains reduce the net heating requirement significantly, `calculated_shd` will over-predict the true EnergyPlus-simulated SHD, and the model corrects for this over-prediction using the envelope and HVAC features. For leaky buildings in very cold climates where infiltration dominates and internal gains are a small fraction of total heat loss, the estimate is more accurate and the model relies on it more heavily.

Notably, `calculated_shd` is substantially less important in the Cold-and-Very-Cold zone-specific models (0.047 CatBoost, not visible in top-ten for LightGBM within that zone) than in the global model or warm zones. This is consistent with the explanation above: in cold climates, the simplified energy balance captures the dominant heat loss pathways (transmission and infiltration) reasonably well on average, but the wide scatter of `ach50_numeric` and the strong geographic HDD variation make the direct physical features more discriminating than the aggregate estimate. The models essentially learn that in the cold zone, they are better served by using `ach50_numeric` and `HDD65_x_UA_per_area` directly than by using the estimate that packages them together less precisely.

Duct Location and Leakage: The Distribution System Penalty

`in.duct_location` and `duct_leakage_frac` together represent the thermal penalty of distributing conditioned air through ductwork that is partially or fully located outside the conditioned envelope. `in.duct_location` is a categorical feature encoding where the ducts are routed (attic, crawlspace, basement, conditioned space, or no ducts), while `duct_leakage_frac` encodes the percentage of supply air lost to the outside through duct joints and gaps. Together these features capture a heat-loss pathway that has no direct analogue in the commercial second iteration, where the commercial dataset’s mechanical ventilation systems are treated as a single aggregate flow-rate parameter.

In the global model, `in.duct_location` receives SHAP values of 0.068 (CatBoost), 0.075 (LightGBM), and 0.109 (XGBoost aggregated across one-hot indicators). `duct_leakage_frac` receives 0.072 (CatBoost), 0.069 (LightGBM), and 0.063 (XGBoost). The physical mechanism is well understood: ducts routed through an unconditioned attic lose heat through the duct walls to the outdoor-temperature-influenced attic space, and any air that leaks out is heated but never delivered to the living space. In cold climates, an attic with ducts running at 120°F through a space at −10°F represents a substantial, continuous heat drain. The CatBoost beeswarm shows clearly differentiated SHAP clusters for `in.duct_location`: the “Crawlspace” indicator (visible as the LightGBM-specific one-hot feature `in.duct_location_Crawlspace` in Figure 7.15 with SHAP = 0.04) assigns positive contributions to buildings with ducts in crawlspaces, reflecting the moderate but significant heat loss through the poorly sealed crawlspace floor. The “Conditioned Space” category receives near-zero or negative contributions, correctly identifying that ducts within the conditioned envelope do not incur distribution losses. `duct_leakage_frac` shows a monotone-positive pattern—higher fractional leakage produces higher positive SHAP contributions—most pronounced in the cold zones where each percentage point of leakage represents a larger absolute energy loss.

R-Floor: The Foundation Penalty

`r_floor` (the effective R-value of the floor assembly) ranks eleventh globally for CatBoost (0.051) and ninth for LightGBM (0.050), consistently appearing in the top twelve across all zone scopes. This is notable because `r_floor` is the insulation feature with the highest individual importance among all the R-value features (`r_wall`, `r_ceiling`, `r_roof`, `r_slab`, `r_foundation_wall`, `r_rim_joist`), which individually rank lower. The reason is that floor insulation is the most heterogeneous envelope component in the residential building stock: slab-on-grade homes have no floor insulation except a slab edge strip, homes over unconditioned crawlspaces have highly variable floor insulation from none to R-30, and homes over conditioned basements effectively have no meaningful floor-to-grade heat loss. This wide variability, combined with the significant thermal penalty of an unin-

insulated crawlspace-floor assembly in cold climates (where the ground below the crawlspace can fall to near-outdoor temperatures), makes `r_floor` the most discriminating of the individual R-value features. The CatBoost beeswarm shows the expected negative-SHAP tail for very low R-floor values (red points to the left, indicating low R-floor = high SHD contribution) with a smaller right-tail for high R-floor (blue points slightly right of zero, indicating diminishing marginal returns on insulation beyond approximately R-19).

In contrast, `r_ceiling`, `r_wall`, and `r_roof` receive lower individual SHAP values in the second iteration than might be expected from first principles, because much of their aggregate effect is already captured by the constructed features `UA_per_area` and `HDD65_x_UA_per_area`, which combine all envelope U-values and areas into a single conductance product. The individual R-value features provide the model with marginal discriminating power over and above the aggregate UA term—specifically in identifying which envelope component is the weakest link—but their standalone importance is naturally lower once the aggregate is already in the feature set. This is the deliberate consequence of including both individual R-values and aggregate UA products: the UA terms anchor the global heat balance, while the individual R-values provide residual correction for buildings where a single poorly insulated component dominates the total conductance.

Building Type, Vintage, and the Geometry Features

`in.geometry_building_type_recs` (dwelling type: single-family detached, single-family attached, multi-family 2–4 units, multi-family 5+ units, mobile home) ranks twelfth globally for CatBoost (0.049) but rises substantially in the Hot-Dry zone (0.092) and Hot-Humid zone (0.080). This zone-specific elevation reflects the fact that in warm climates, where the absolute SHD is low and physical features provide less discriminating power, the dwelling typology captures an important geometric determinant of per-unit heat loss: a unit in a mid-rise multi-family building has far less exposed envelope area per floor area than a detached single-family home of the same size, because shared party walls and floors are internal surfaces that do not lose heat to the outdoors. In cold climates this effect is subsumed by the dominant infiltration and conductance signals, but in warm climates it becomes proportionally more important.

`in.vintage`, despite encoding construction era (and thus the implicit insulation standards of the time), receives modest SHAP values across all zones (0.026 CatBoost, 0.040 LightGBM). This lower-than-expected importance is a direct consequence of the explicit R-value features in the second-iteration feature set: because insulation values are decoded from the descriptive strings and provided as continuous features (`r_wall`, `r_ceiling`, etc.), the model does not need to infer insulation quality from vintage as a proxy. The vintage feature provides residual information—primarily about construction quality aspects not captured by insulation alone, such as window airtightness, thermal bridging through structural framing, and the prevalent HVAC technology of the era—but its role

is secondary once the explicit physical parameters are present. `year_built_numeric`, which encodes vintage as a continuous year rather than a decade-band string, shows similar importance levels (0.016 CatBoost, 0.040 LightGBM), confirming that the ordinal continuous representation captures roughly the same information as the categorical string with neither a clear advantage.

Cross-Algorithm Interpretability Summary and Physical Validation

The most fundamental cross-algorithm contrast in the second-iteration ResStock SHAP analysis is the consequence of how each algorithm handles `in.hvac_heating_type_and_fuel`. CatBoost and LightGBM treat it as a single categorical variable and assign it compact, bounded SHAP values (0.04 and 0.08 respectively). XGBoost one-hot encodes it, producing the dominant `in.hvac_heating_type_and_fuel_None` indicator with an outsized mean absolute SHAP of 2.69, while the remaining fuel-type indicators contribute smaller individual amounts. The aggregate predictive content is equivalent—all three algorithms correctly identify HVAC type as a primary driver of residential SHD—but the representational structure differs in a way that makes XGBoost’s per-feature SHAP chart difficult to interpret without aggregation.

For the purely numeric features—`hdd65`, `HDD65_x_UA_per_area`, `ach50_numeric`—all three algorithms show consistent SHAP magnitudes and directions, with the quantitative differences explicable by LightGBM’s leaf-wise growth concentrating more early-split capacity on the highest-gain feature (`hdd65` = 0.53 for LightGBM vs. 0.38 for CatBoost vs. 0.33 for XGBoost). The beeswarm plots uniformly show monotone-positive gradients for climate and infiltration features, and monotone-negative gradients for insulation features—precisely what would be predicted from the steady-state heat balance and confirming that the second-iteration models have internalised physically correct relationships without requiring explicit monotonic constraint enforcement.

The zone-specific patterns collectively validate a consistent physical narrative: in cold climates, infiltration (ACH50) and climate-conductance interaction (`HDD65_x_UA_per_area`) dominate over HVAC type because the physics of heat loss through the building shell drives a much larger fraction of the total SHD than system efficiency; in warm climates, HVAC type and state-level code environment dominate because the total annual heat loss is small enough that system efficiency and behavioural patterns (partially encoded in state-level averages) account for a larger relative share of the residual variance. The Mixed-Dry zone’s anomalously low HVAC-type importance (0.099 vs. 0.74–1.50 in other zones) is consistent with the observation that Mixed-Dry is a transition climate where moderate HDD values make conductive and infiltration losses and system efficiency roughly equally important, and the small sample size prevents the model from resolving the HVAC signal clearly through the categorical features.

Conclusion

The second-iteration ResStock SHAP analysis demonstrates that all three gradient boosting algorithms have learned a physically coherent representation of residential heating demand from a deployable, envelope-focused feature set. The dominant feature hierarchy—HVAC type and fuel, climate severity (HDD65), state-level code environment, envelope conductance–climate interaction, and air leakage—corresponds precisely to the major terms of the residential heat balance. The cold-climate elevation of infiltration (`ach50_numeric`) to the top position within zone-specific models is a distinctive finding that differentiates the residential from the commercial analysis, reflecting the greater variability of residential envelope airtightness and the direct availability of ACH50 data in the ResStock feature set. The cross-algorithm differences—LightGBM’s amplification of `hdd65` through leaf-wise growth, XGBoost’s fragmentation of the HVAC categorical into binary indicators, and CatBoost’s compact categorical encoding—are all consistent with the structural properties of each algorithm and are reconciled by the convergent physical interpretation that the same underlying relationships are being learned through different representational pathways. The absence of monotonic constraint enforcement does not compromise physical plausibility, confirming that the physics-informed feature engineering is sufficient to guide the models toward thermodynamically valid representations of residential SHD across all climate zones of the U.S. building stock.

Chapter 8

Global Applicability and Methodology Transfer

The ML models developed in this research are linked to the architectural and operational norms of the North American building sector. While the computational framework for predicting SHD is shown to be effective, the trained models themselves cannot be directly applied to buildings that deviate significantly from typical U.S. building characteristics without substantial recalibration.

8.1 The Constraint of U.S.-Centric Archetypes

The ComStock and ResStock datasets are designed to represent the approximately 6 million commercial and 130 million residential buildings in the United States. They cannot be reliably used to represent buildings globally.

Several factors contribute to this geographic specificity:

- **Calibration to national surveys:** The models are calibrated to match the energy intensities and floor area distributions of CBECS and RECS. These surveys reflect U.S. economic activities and residential habits that may not mirror those in Europe, Asia, or the Global South.
- **Prototype geometry:** All building models in this framework are variations of rectangular prisms with flat roofs. This simplified geometry allows for large-scale simulation but fails to account for diverse global architectural styles, such as high-thermal-mass traditional stone structures or complex non-rectangular urban footprints common in older international cities.
- **HVAC and code dependencies:** The models rely heavily on the vintage of a building to assign HVAC system types and efficiencies based on the historical adoption of ASHRAE 90.1 and California’s Title 24 energy codes. Construction practices

in other regions—such as the prevalence of hydronic radiator systems in Northern Europe versus the forced-air Rooftop Units dominant in U.S. commercial stock—would render the current feature mappings inaccurate.

8.2 Methodology for International Adaptation

To apply this thesis’s methodology outside of the U.S., a systematic process of data localisation and feature alignment is required.

8.2.1 Localised Data Acquisition

The first step is gathering heating consumption data representative of the target region. Researchers should follow the “Stock Estimation” approach by identifying local building counts and types through government infrastructure databases. In the absence of metered data, a synthetic dataset should be generated using EnergyPlus (or any other reliable, industry-standard alternative), localised with regional weather files in EnergyPlus Weather (EPW) format and regional construction baselines.

8.2.2 Classification and Alignment

Buildings must be reclassified according to the ASHRAE Standard 169 climate zones. Global classifications are already provided online for all countries [23]. Input features must also be mapped to local equivalents. For example, U.S. “vintages” must be replaced with local construction eras, and primary heating fuels must reflect regional availability, such as the high prevalence of district heating in certain Eastern European regions.

International Climate Zone Definitions

Zone Number	Zone Name	Thermal Criteria (I-P Units)	Thermal Criteria (SI Units)
1A and 1B	Very Hot –Humid (1A) Dry (1B)	$9000 < CDD50^{\circ}F$	$5000 < CDD10^{\circ}C$
2A and 2B	Hot-Humid (2A) Dry (2B)	$6300 < CDD50^{\circ}F \leq 9000$	$3500 < CDD10^{\circ}C \leq 5000$
3A and 3B	Warm – Humid (3A) Dry (3B)	$4500 < CDD50^{\circ}F \leq 6300$	$2500 < CDD10^{\circ}C < 3500$
3C	Warm – Marine (3C)	$CDD50^{\circ}F \leq 4500$ AND $HDD65^{\circ}F \leq 3600$	$CDD10^{\circ}C \leq 2500$ AND $HDD18^{\circ}C \leq 2000$
4A and 4B	Mixed-Humid (4A) Dry (4B)	$CDD50^{\circ}F \leq 4500$ AND $3600 < HDD65^{\circ}F \leq 5400$	$CDD10^{\circ}C \leq 2500$ AND $HDD18^{\circ}C \leq 3000$
4C	Mixed – Marine (4C)	$3600 < HDD65^{\circ}F \leq 5400$	$2000 < HDD18^{\circ}C \leq 3000$
5A, 5B, and 5C	Cool-Humid (5A) Dry (5B) Marine (5C)	$5400 < HDD65^{\circ}F \leq 7200$	$3000 < HDD18^{\circ}C \leq 4000$
6A and 6B	Cold – Humid (6A) Dry (6B)	$7200 < HDD65^{\circ}F \leq 9000$	$4000 < HDD18^{\circ}C \leq 5000$
7	Very Cold	$9000 < HDD65^{\circ}F \leq 12600$	$5000 < HDD18^{\circ}C \leq 7000$
8	Subarctic	$12600 < HDD65^{\circ}F$	$7000 < HDD18^{\circ}C$

Figure 8.1: International Climate Zone Definitions, ANSI/ASHRAE/IESNA Standard 90.1-2007.

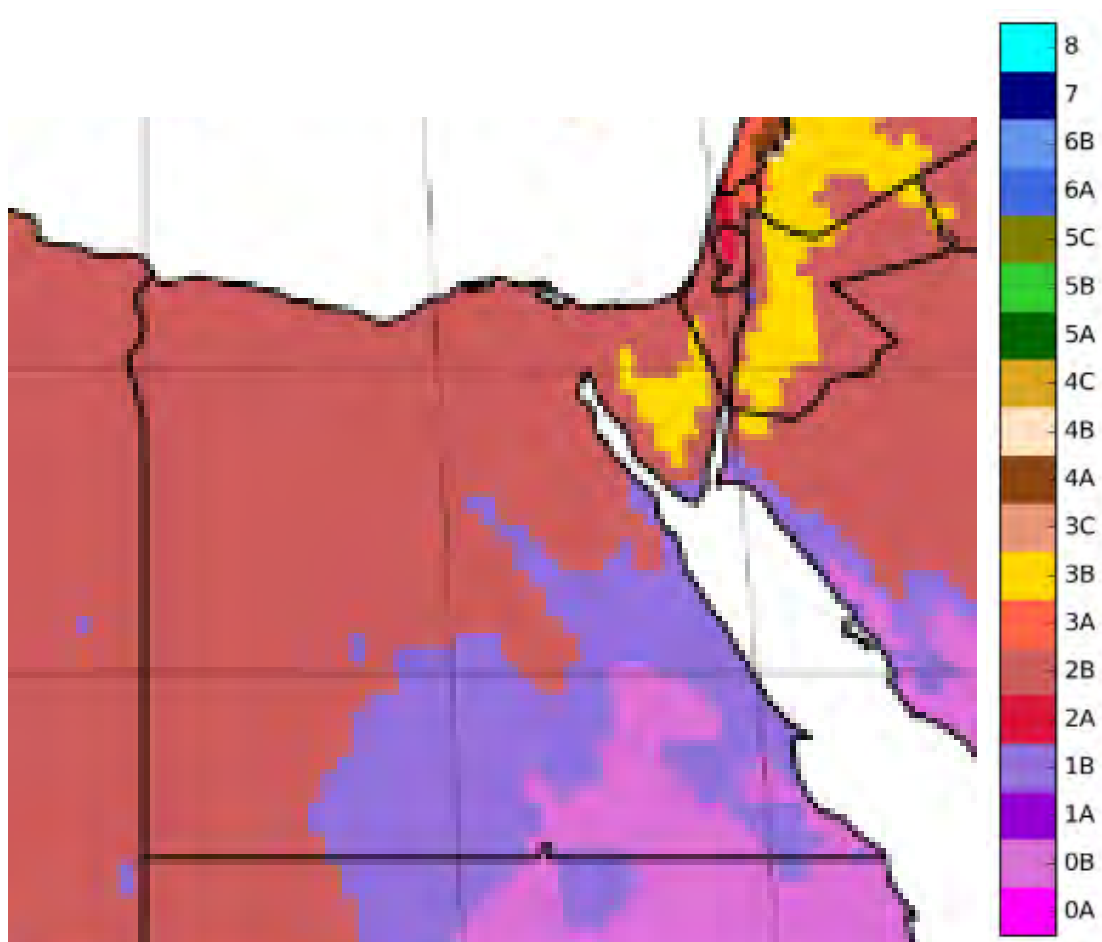


Figure 8.2: ASHRAE climate zone classifications in Egypt.

8.3 Methodological and Structural Limitations

Despite the high predictive accuracy achieved in this thesis, several limitations must be considered when attempting to scale or transfer the models.

8.3.1 Building Type Coverage Gaps

Even within the U.S. building stock, the current models have notable omissions. Approximately 36% of commercial energy use is not currently represented in ComStock. Key building types excluded from the model include grocery stores, religious worship centres, mixed-use offices, and universities. Predicting SHD for these complex or specialised structures using the current ensemble would likely result in significant error due to a lack of representative training samples.

8.3.2 Data Sparsity and Quality

The accuracy of the model is highly dependent on the number of samples available per climate zone. For instance, in sparsely populated regions, the number of simulation models may be very low, potentially leading to unrealistic load profiles. In this research, climate zones with fewer than 250–500 samples were excluded to maintain statistical relevance.

8.3.3 Assumptions in Physical Parameters

Due to the lack of available national data on the internal composition of walls and roofs, thermal performance values (U-values) are often estimated based on the energy code in force at the time of construction or last replacement. While this code-compliance-based framework is a necessary surrogate for missing data, it may not accurately capture the performance gap found in buildings where real-world insulation levels lag behind theoretical code requirements.

8.3.4 Operational Complexity

Commercial buildings exhibit significantly more diversity than residential units, and many intra-building characteristics must be inferred from whole-building metadata. The decision to use point estimates for certain parameters, such as lighting power density across various space types, simplifies the model but excludes the stochastic variability present in real-world operations. Additionally, the reliance on rectangular geometry and core-and-perimeter thermal zoning can result in unrealistically sized HVAC systems for buildings with complex internal layouts.

8.3.5 Validation

A fundamental limitation that runs beneath all results presented in this thesis is the nature of the validation chain on which they rest. The models are surrogates for ResStock and ComStock, and ResStock and ComStock are themselves surrogates for real buildings. Every accuracy figure reported — R^2 , RMSE, CV-RMSE — measures fidelity to the simulation engine, not to measured energy consumption. The distinction matters because the models inherit, without correction, whatever systematic biases exist in the underlying datasets. ResStock and ComStock are calibrated to national surveys (RECS and CBECS respectively) and validated against Advanced Metering Infrastructure data and end-use sub-metered sources, as discussed in Chapter 5. This is a meaningful and serious validation effort, and the resulting datasets are among the most rigorous national building stock simulations available. However, calibration to a population-level statistical distribution is categorically different from validation against individual building measurements. A model that perfectly reproduces ResStock’s SHD distribution will still inherit the performance gap between EnergyPlus’s deterministic occupancy assumptions and the stochastic behaviour of real occupants — a gap documented in the literature at 15–50% for individual buildings [6, 22]. The cold-climate models developed here, where envelope physics dominate and behavioural variance is proportionally smaller, are likely to transfer more faithfully to real-world conditions than the warm-climate models, where occupant behaviour drives a large share of the residual variance. But this remains a hypothesis rather than a demonstrated fact, because no validation against sub-metered or utility-billed real buildings was conducted. For these models to be considered production-ready in any practical deployment — supporting retrofit prioritisation decisions, utility demand forecasting, or regulatory compliance screening — this validation gap would need to be systematically closed. At minimum, this would require assembling a geographically and archetypally diverse sample of real buildings for which both the audit-grade input features used here and independent ground-truth annual heating consumption records are available. The models would then need to be evaluated against those real records, with performance disaggregated by climate zone, building type, and vintage, to establish where the simulation-to-reality transfer holds and where it breaks down. Discrepancies between simulated and real performance would need to be characterised not merely as error but as evidence for specific model components — envelope physics, infiltration estimation, occupancy assumptions — that require recalibration against empirical data. This is a substantially larger undertaking than the modelling conducted here, requiring sustained data collection effort and close collaboration with utilities, building owners, or energy agencies that hold metered consumption records at the building level. The absence of this validation is not a flaw unique to this thesis; it reflects a structural challenge in the building energy ML literature more broadly, where the scarcity of labelled real-world

datasets — the same scarcity that motivated the shift from BDG2 to simulation data in Chapter 4 — prevents the closure of the simulation-reality gap at scale. What this thesis demonstrates is that the surrogate modelling approach is technically viable and physically coherent within the simulation domain. Demonstrating that it is viable in the real world requires the next step that this work does not take.

Chapter 9

Conclusion

This thesis investigated using gradient boosting machine learning models as scalable surrogates for physics-based simulation in predicting the specific heating demand of buildings. The work proceeded across two datasets — ResStock for the residential sector and ComStock for the commercial sector — and two modelling iterations designed to probe the accuracy ceiling achievable with a broad feature set and the deployable floor achievable with audit-grade physical inputs alone.

A preliminary investigation using the Building Data Genome Project 2 dataset established that real-world sub-metered energy records cannot support the SHD prediction task at stock scale. CV-RMSE values ranged from 12% to over 310% across sites, and the apparent accuracy of better-performing sites was attributable to building-level data leakage rather than genuine generalisation. Four structural limitations — the absence of electrical heating records, the inability to disaggregate space heating from total fuel consumption, the impossibility of constructing clean annual labels from incomplete hourly series, and the narrow archetype coverage — were identified as irresolvable by algorithmic means. This finding motivated the shift to NREL’s nationally representative simulation repositories, which resolve all four issues by construction.

The first modelling iteration, employing a row-level train/test partition and a broad feature set including EnergyPlus output parameters, established an optimistic upper bound. ResStock models achieved R^2 between 0.890 and 0.983 across climate zones; ComStock models achieved R^2 between 0.845 and 0.973. These figures confirm that the combined metadata and simulation-output feature space contains sufficient information to reconstruct nearly all variance in SHD when building archetypes are freely shared across the partition boundary, but they overstate the accuracy that can be expected in genuine deployment.

The second iteration replaced the row-level partition with a building-level holdout, evaluated models on an unfiltered test set retaining all extreme-demand buildings, and restricted inputs to features derivable from building audits or public climate data without a prior simulation run. Under these more rigorous and operationally realistic conditions,

ResStock models achieved R^2 between 0.557 and 0.841 and ComStock models between 0.707 and 0.906, with CV-RMSE values ranging from 13.8% to 52.3% across all zones, datasets, and iterations. Performance was strongest in cold and mixed climates where envelope conductance and infiltration dominate the heat balance, and weakest in warm zones where thermostat setpoint and operational behaviour — deliberately excluded from the deployable feature set — carry a disproportionate share of the residual variance. The gap between the two iterations is therefore not primarily a loss of model quality but a measurement of how much of the first iteration’s apparent accuracy was attributable to archetype memorisation, outlier suppression, and non-deployable inputs.

Across both datasets and iterations, CatBoost demonstrated the most reliable generalisation under building-level evaluation, consistently exhibiting the smallest train-to-test RMSE gap and winning the cross-validation criterion in a majority of zones, with a tendency to avoid potential overfitting. Its ordered boosting and native categorical handling were particularly valuable in smaller zones and in the categorical-heavy commercial data. LightGBM was most competitive in global and diverse-population models, while XGBoost was most robust in data-scarce zones where its aggressive regularisation framework prevents overfitting on small training populations.

SHAP analysis conducted across all scopes confirmed that the models learned physically coherent representations rather than statistical artefacts. In the residential models, air leakage rate rose to the dominant predictor in cold-climate zones, reflecting the amplification of infiltration heat loss by large indoor-to-outdoor temperature differences. In the commercial models, building type and state name emerged as the primary features once simulation output parameters were removed, acting as compressed surrogates for archetype-level occupancy profiles and energy code environments. The engineered interaction terms — `HDD65_x_UA_per_area` and `infiltration_x_HDD` — received SHAP attributions in the physically expected directions across all algorithms, and this physical plausibility was achieved without monotonic constraint enforcement, confirming that physics-informed feature engineering is sufficient to guide the models toward thermodynamically valid representations.

There are two principal limitations of the present work, the first being its geographic specificity: the trained models are calibrated to U.S. construction norms and archetypes as encoded in ResStock and ComStock, and cannot be transferred to other building stocks without retraining on locally representative data. The second limitation being the lack of much needed rigorous validation needed to make those models production-ready and a true replacement to using simulation programs. The methodology, however, is mostly transferable. Given a nationally representative simulation dataset generated with locally calibrated weather files, construction baselines, and energy code histories — and following the same pipeline of physics-informed feature engineering, climate-zone stratification, and building-level cross-validation — the approach developed here can be reproduced for most

other regions. The resulting models could serve as efficient, interpretable alternatives to EnergyPlus for stock-scale screening, retrofit prioritisation, and energy policy analysis, delivering predictions in milliseconds rather than the minutes to hours required by full physics-based simulation.

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